

From Forecast Accuracy to Control Utility: Evaluating Quadrotor Dynamics Calibration at the Decision Interface

Extended Research Record

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This supplementary research record provides extended methods, complete retained results, and traceability for the accompanying paper. The main paper organizes forecast fidelity, shared-state action quality, feedback-conditioned consequences, and complete-flight utility as distinct evidence checks. This record separates independently confirmed findings from reused development evidence and retains unfavorable and null comparisons. Table values are generated from accepted JSON outputs; displayed rounding does not change the underlying records. The main argument follows S3 (calibration), S4 (shared-state choices and feedback continuation), S9 (adapter diagnosis), S5–S7 (development flights), and S13 (prospectively locked model selection on new instances). S10–S12 bound observation, model-form, and native-model assumptions. S8 retains the earlier representation study as an archival record; it is not a premise of the current contribution. S14 adds the source-matched nonlinear-readout sensitivity reported in the current main paper; it is a development analysis, separate from the earlier confirmation families.

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S1 Evidence Stages, Denominators, and Traceability

The scientific question is whether a calibrated predictor’s trajectory accuracy transfers to action ordering and full-flight utility. These are separate endpoints. The new main-paper confirmation (C1) and the earlier teacher/compensation confirmation (S8) are distinct locked experiments. D1–D34 are subsequent or diagnostic development records. Reusing their tasks for diagnosis or fitting decisions prevents calling them fresh confirmation. S9 adds adapter, response and exact cost-difference diagnostics without new independent data; S10 adds an observation audit and frozen motor-form transfer on the same development tasks. S11 adds the original recurrent-model comparator; S12 adds rotor-hub drag and a same-input speed-law observer, also on reused tasks.

The C1 lock was recorded before its new data collection at 2026-09-08 02:47:53 UTC and covers 182 source/model files. Five calibration campaigns per regime, two budgets, all registered comparators, validation selection and all eight primary contrasts are retained. Its bootstrap resamples independent campaigns, encoder seeds and geometry-stratified scene clusters; both branch realizations remain within a scene cluster. The 99.375% intervals adjust the eight-comparison family; descriptive 95% intervals do not. A null interval does not demonstrate equivalence.

Development bootstrap intervals instead condition on the specified calibration pool or campaign. Repeated snapshot times remain in their scene cluster. Five encoder seeds, five subsets from a shared pool, and five independent calibration campaigns are not interchangeable replication units. The final MPC bootstrap averages old scalar subsets as a fixed control and resamples scene clusters and learned seeds. Its 48 comparisons are unadjusted. D24 additionally resamples independent calibration campaigns on reused controller states; its 56 comparisons remain exploratory. D26 retains the same predictor choices under two shared continuations, adding 16 paired exploratory contrasts without refitting. D28–D31 add the native-model scope audit and a fixed development comparison with 36 descriptive screen comparisons. D32–D34 add 88 rotor-drag intervals and 14 descriptive components. The historical D1–D34 development record contains 402 unadjusted interval comparisons and 62 screen components (12 earlier, 36 native and 14 rotor-drag), separate from both confirmation families.

Table S1: Stage sizes and validity. M/L: mass/lag. Counts are per regime unless stated otherwise.

Stage	Design	Validity / retention
C1	736 four-action tasks	M: 286 prefixes, 118 all-valid; L: 659, 593
C1	5 calibration campaigns; 32/64 episodes	530 total accepted configurations
D1–D2	108 nine-action snapshots	M: 99 prefixes, 98 all-valid; L: 108, 106
D3–D4	Initial closed loop	94 configurations; 3,384 total flights
D13	Shared altitude intervention	8 configurations; 288 total flights
D5–D6	Corrected closed loop	94 configurations; 3,384 total flights
D7–D8	Response-preserving drift	140 accepted heads; 16 contrasts
D9–D10	Paired-data fitting / snapshot evaluation	84 fits; 28 selected heads; 30 evaluations
D11–D12	Final paired-data closed loop	30 configurations; 1,080 total flights
D14	Fixed affine-head audit	28 heads; 204 eligible unique states
D15–D17	Response capacity / history probe	179/192 states; 10 selected response heads
D18–D19	Response choice / cost-difference audit	12 combinations; 204/216 eligible states
D20	Observer assumption audit	207/216 initially alive snapshots
D21–D23	Frozen actuator-form transfer	30 configurations; 1,080 new flights
D24–D25	5 campaigns; 190 snapshot predictors	Same 98/106 all-valid states
D26–D27	One-interval action continuation	190 configs; 3 modes; 16 contrasts
D28–D31	Native recurrent calibration / audit	100 fits; 110 models; same 98/106 states
D32–D34	Rotor-hub drag / same-input observers	34 new configurations; 1,224 flights

S13 adds a separate prospectively locked selection experiment, P1: three new validation panels, a fixed eight-predictor pool and 20 new test scenes per regime with three winds each. Its two primary contrasts use 97.5% intervals; 40 secondary contrasts are exploratory. These counts are separate from C1, S8 and the historical development ledger. P1 does not refit models or convert the earlier development tasks into confirmation. Its mass primary is adverse and its lag primary does not establish an advantage for action-regret selection.

S1.1 Validity and units

Contact is absorbing. Unless explicitly marked Full, candidate position RMSE includes eligible pre-contact candidate positions; accuracy and regret require all candidates in a task or snapshot to remain valid. Main Figure 3 uses the cross-campaign Shared RMSE and regret in S4, aligned on the same all-nine-valid snapshots. The original-pool Full column in S4 retains its separate, likewise aligned comparison. Selected contact uses initially alive histories, a larger denominator. The severe mass exclusions in C1 therefore condition its prediction finding on survival. In the nine-action development setting, accuracy is agreement with minimum true tracking cost and regret is excess true cost, with squared-position and command penalties as defined in the main paper. C1 and S8 instead use final goal-distance cost over four candidates. Their regret values should not be compared directly.

Closed-loop RMSE uses valid tracking observations after the common two-second warm-up. It is accompanied by 12-second success, contact, truncated duration and terminal error. Common-time sensitivity restricts paired RMSE to the intersection of valid observations. This is critical when contact differs: a shorter valid flight can look easier. There are no MPC collision constraints, so the task is tracking rather than collision-aware navigation.

S1.2 Manuscript project and research evidence archive

Source IDs refer to exact archived result files. The manuscript project contains the complete typesetting sources, printed tables, figures, and `evidence_manifest.json`; paths in that manifest are research-repository paths, not dependencies needed to compile this document. The separate research evidence archive retains machine-readable strata, per-fit summaries, and verification records. C1 includes all 540 stratum summaries and 9,900 per-fit entries; D3, D5 and D11 also retain latency and tracking-availability fields. D24 includes per-state outcomes and protocol verification; D26 includes continuation traces and fixed-choice outcomes. The research archive lists 700 large checkpoint/readout, calibration, and forecast/context binaries held separately, with exact paths and hashes. A separate rotor-drag trace archive contains 238 configuration files (34 summaries and 204 NPZ traces), with archive and member hashes in the manifest. The manuscript project is self-contained for editing and compilation; it is not a standalone package for rerunning the experiments.

Table S2: Source index within the portable bundle; all paths are relative to the repository root.

ID	Result artifact
C1	runs/icra_calibration_confirmation_20260908/summary/summary.json
D1	runs/icra_snapshot_prediction_20260908/descriptive_summary.json
D1a	runs/icra_snapshot_common_support_20260909/attempt01/summary.json
D1b	runs/icra_snapshot_common_support_20260909/attempt01/verification.json
D2	runs/icra_snapshot_prediction_20260908/accepted_intervals.json
D3	runs/icra_mpc_matrix_20260908/summary/summary.json
D4	runs/icra_mpc_matrix_20260908/summary/exploratory_intervals.json
D5	runs/icra_mpc_authority_matrix_20260908/summary/summary.json
D6	runs/icra_mpc_authority_matrix_20260908/summary/exploratory_intervals.json
D7	runs/icra_structured_drift_20260908/summary/summary.json
D8	runs/icra_structured_drift_20260908/summary/exploratory_intervals.json
D9	runs/icra_decision_evaluation_20260908/descriptive_summary.json
D10	runs/icra_decision_evaluation_20260908/accepted_intervals.json
D11	runs/icra_paired_mpc_20260908/summary/summary.json
D12	runs/icra_paired_mpc_20260908/summary/exploratory_intervals.json
D13	runs/icra_vertical_authority_20260908/summary.json
D14	runs/icra_readout_structure_20260908/attempt01/summary.json
D15	runs/icra_response_capacity_20260908/attempt01/summary.json
D16	runs/icra_response_capacity_20260908/readout_gap01/summary.json
D17	runs/icra_history_response_20260908/attempt01/summary.json
D18	runs/icra_response_choice_20260908/attempt01/summary.json
D19	runs/icra_cost_difference_20260908/attempt01/summary.json
D20	runs/icra_observer_assumptions_20260909/attempt01/summary.json
D21	runs/icra_motor_form_transfer_20260909/summary/summary.json
D22	runs/icra_motor_form_transfer_20260909/summary/exploratory_intervals.json
D23	reports/motor_form_transfer_20260909/all_model_form_shifts.json
D24	runs/icra_campaign_snapshot_20260909/attempt01/summary.json

Table S2 continued

ID	Result artifact
D25	runs/icra_campaign_snapshot_20260909/attempt01/verification.json
D26	runs/icra_action_continuation_20260909/attempt02/analysis/summary.json
D27	runs/icra_action_continuation_20260909/attempt02/acceptance.json
D28	runs/icra_native_path_audit_20260909/attempt01/summary.json
D29	runs/icra_native_recurrent_20260909/fit_acceptance.json
D30	runs/icra_native_recurrent_20260909/outcomes_attempt01/summary.json
D31	runs/icra_native_recurrent_20260909/outcomes_attempt01/acceptance.json
D32	runs/icra_rotor_drag_20260909/acceptance.json
D33	runs/icra_rotor_drag_20260909/summary_attempt01/summary.json
D34	reports/rotor_drag_outcomes_20260909/verification.json

S2 Wind Environment, Model, Fit, and Controller Details

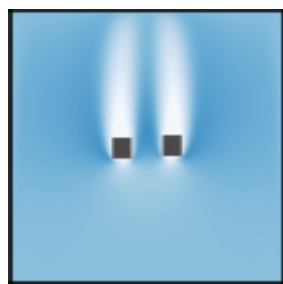
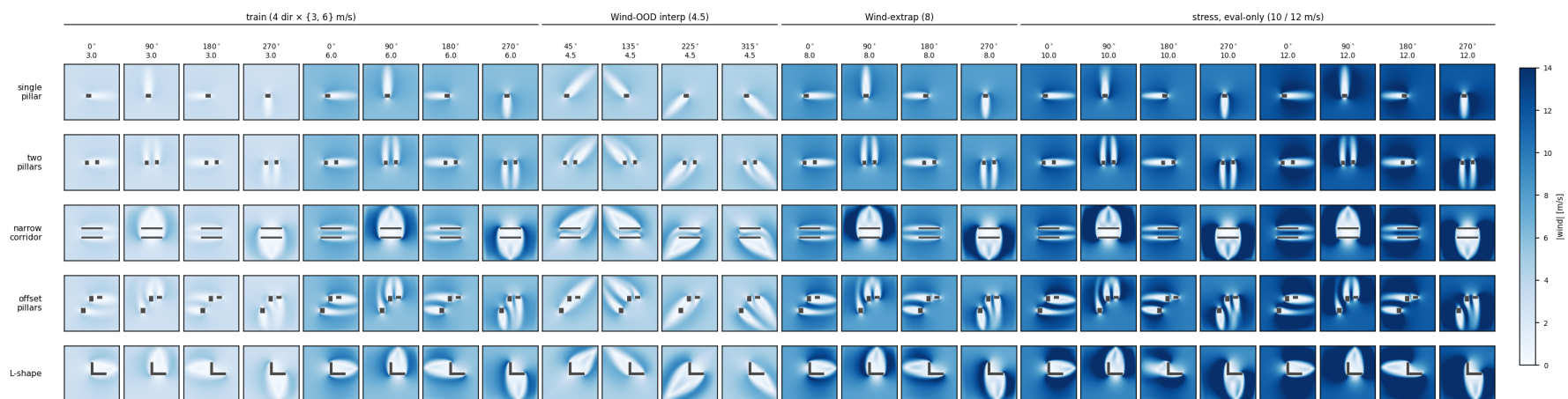
S2.1 Spatial wind and the evaluated task conditions

Obstacles make the disturbance encountered by a moving quadrotor depend on its position. The environment provides a controlled setting for asking whether a calibrated dynamics model helps choose commands while tracking a reference. It contains five obstacle families—single pillar, two pillars, narrow corridor, offset pillars, and L-shape—with six seeded instances each. Every instance occupies a 30×30 m horizontal domain. Obstacle footprints at the reference altitude $z_{\text{ref}} = 3$ m define the planar flow geometry.

A D2Q9 lattice-Boltzmann solver produces a horizontal velocity field on a 120×120 grid. The stored field averages the final 2,000 of 8,000 solver steps. During flight it remains frozen: bilinear lookup at the vehicle’s horizontal position supplies the local wind, with zero vertical component. Motion through the field therefore changes the encountered disturbance, even though the field itself is time invariant. This is a quasi-steady planar mean-flow model, without resolved three-dimensional turbulence. Local true wind affects the simulated dynamics but is excluded from the principal predictor’s inference inputs.

Figure S1 retains the original gallery for one instance per family. Its 24 columns comprise eight training conditions (four cardinal directions at 3 and 6 m/s), four new-direction conditions (diagonals at 4.5 m/s), four speed-extrapolation conditions (cardinals at 8 m/s), and eight historical evaluation-only stress conditions (cardinals at 10 and 12 m/s). These speeds specify far-field conditions; local values vary around obstacles. The gallery’s “Wind-OOD interp” label refers specifically to the 4.5 m/s group.

The current nine-action comparisons and complete flights use three strata: training winds on training scene instances (ID), 8 m/s winds on training instances (wind extrapolation), and 8 m/s winds on held-out instances (joint extrapolation). Each is evaluated under either mass multiplied by 1.4 or actuator time constant multiplied by 3; these vehicle changes are separate from the wind-condition axis. The 4.5 and 10/12 m/s columns provide broader gallery context, not additional conditions for these current action/flight results. C1 retains its separately specified calibration and evaluation strata. The gallery clarifies the environment underlying the study; it contributes no new experiment or evidence of real-flight transfer.



(a) Two pillars: 90°, 6 m/s.



(b) Narrow corridor: 90°, 6 m/s.

Figure S1: **Spatial wind environment and original condition coverage.** The full gallery is unchanged; the lower panels enlarge two existing cells. Color encodes horizontal wind-speed magnitude, not direction, on the original 0–14 m/s scale; higher values saturate. Gray regions are obstacle footprints. Column headings specify far-field direction and speed. The complete 24-condition gallery is broader than the current nine-action and complete-flight evaluation.

S2.2 Observation and deployment interface

The context uses 12 historical observations and commands and, when applicable, two depth frames. The observed state includes body-frame linear/angular velocity, gravity direction, altitude and nose-heading sine/cosine. Future commands are known candidates; future measured states, true wind, scene identifiers, and randomized plant parameters are excluded. World position supports candidate generation, displacement transformation and tracking costs. The wind observer estimates residual motion under nominal physical parameters and does not use true randomized mass or thrust.

In main Figure 1, h_t denotes the encoder’s feature history and $z_t = E_\theta(h_t)$ its shared context. Separately observed world position p_t^W and velocity v_t^W initialize candidate generation and nominal response; absolute world pose is not an encoder input. Each $u^a \in \mathbb{R}^4$ contains world-frame velocity and yaw rate; U^a repeats it over 30 steps. The index set $\mathcal{A} = \{1, \dots, 9\}$ refers to XY offsets before speed clipping. The fixed affine head D_ϕ includes FIT normalization, with $\phi = (W, b, \mu, \sigma)$; λ is its selected ridge penalty. Nominal displacement d_{nom}^a , residual $\hat{\delta}^a$ and their sum $\hat{d}^a$ share the anchor yaw frame. World positions satisfy $\hat{p}_k^{a,W} = p_t^W + R_z(\psi_t)\hat{d}_k^a$. The cost uses the future reference $r_{1:K}^W$, while candidate generation uses current r_t^W and $\dot{r}_t^W$. The previous command u^- is already activated; the newly selected command activates 20 ms later. Local feedback x_t contains observed pose and velocities for geometric control, not true wind, randomized parameters or rotor state.

The principal frozen representation in the new calibration study is supervised dynamics. Privileged-teacher variants are additional comparators, not the principal result. The following architecture/training details apply to the original representation training and original frozen residual readout. The original four-candidate and waypoint-segment metrics in this block differ from the later nine-candidate MPC cost and timed-reference tracking metric described below.

S2.3 Objectives and Training Details

The context representation has dimension 128, recurrent hidden state 256, and action embedding 64. The dynamics target encoder uses an exponential moving average with decay 0.996. For joint-embedding variants, the dynamics objective includes prediction against the moving-average target, a live-target matching term with coefficient 0.5 and a detached predictor target, and variance regularization on predicted, context, and live-target embeddings. The physical-state probe receives detached predictor features in these variants. Thus its supervised loss does not train the predictive backbone.

In PA-JEPA, a training-only vector encoder maps the local two-dimensional wind to a 64-dimensional target. A reconstruction loss constrains this target, while a context-side predictor matches its stopped-gradient embedding. Variance regularization also acts on the wind target. The direct variant replaces this teacher path with normalized wind-vector regression. Supervised variants train the physical-state predictor end to end; wind-only sets the physical-state loss weight to zero and has no joint-embedding dynamics loss. Although the implementation shares a model container, unused prediction modules do not make wind-only a dynamics-trained model.

All variants use AdamW with initial learning rate 3×10^{-4} , weight decay 10^{-5} , cosine scheduling, batch size 256, gradient-norm clipping at 5, and 50 epochs. Target normalization uses only training data. Checkpoint selection uses validation position RMSE, except wind-only, which uses validation wind RMSE. Joint-embedding state readouts are then reinitialized and trained for 20 epochs with the selected backbone frozen, using AdamW at 10^{-3} and validation-only readout selection. Supervised models retain their jointly trained state heads. These readout differences motivate the additional common frozen residual comparison.

S2.4 Residual Fit and Metric Definitions

The residual decoder uses the same fixed response time $\tau = 0.3\text{s}$ across representations. For a three-dimensional velocity command $\bar{u}_k$ in the anchor yaw frame and interval $\Delta t = 0.05\text{s}$,

$$v_{k+1} = \bar{u}_k + (v_k - \bar{u}_k)e^{-\Delta t/\tau}, \quad (1)$$

$$p_{k+1} = p_k + \Delta t \bar{u}_k + \tau(1 - e^{-\Delta t/\tau})(v_k - \bar{u}_k). \quad (2)$$

The initial velocity and heading are observed at the anchor. For nominal rollouts, velocity commands are rotated into the anchor yaw frame; the affine readout receives the original world-frame velocity/yaw-rate command sequence. A standardized ridge fit maps context and candidate commands to the stacked position residual; raw-history regression substitutes the flattened observed state history for the learned context. The penalty is selected from

$\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ on validation episodes. The confirmation runner loads the saved coefficients and never invokes training-data extraction or penalty selection.

Position RMSE pools the squared Euclidean error over eligible candidate positions before taking the square root, then averages model seeds. Action regret is the difference between the selected and minimum final squared goal-distance costs. These action statistics require all four candidates to be contact-free; their denominator and the separately reported selected contact frequency must accompany the scores.

Closed-loop contact is an absorbing event defined by the simulation’s clearance threshold or ground contact. The reported time is $\min(T_{\text{contact}}, 12\text{s})$. Tracking error averages the distance to the current waypoint segment only at valid observations after shared warm-up. Its observation duration can differ across methods. We therefore report contact, exposure and waypoint progress alongside tracking error; these quantities do not reduce to a single safety or task-success score.

S2.5 Feature-matched calibration confirmation

Physical-centered ridge adds the nominal physical trajectory to a linear correction whose features also include that physical trajectory. Zero-centered ridge uses the same expanded features. With an unpenalized intercept the two designs share a linear function class; the observed numerical audit supports no unique attribution to the additive prior. Raw-state controls include historical commands and raw-depth controls include both full observed frames. The scalar response time is selected using fit data. Both ridge centers, the retained affine adaptation head, budgets 32/64 and all methods remain in C1. Budgets include the 25% validation allocation.

Matched random/pretrained adaptation uses a linear 338-to-90 correction head and identical AdamW optimization at learning rate 3×10^{-4} , batch size at most 128, gradient clipping at 1, 100 epochs and weight decay $\{10^{-4}, 10^{-2}\}$. It minimizes trajectory residual MSE, selecting the weight decay and checkpoint by validation position RMSE, with epoch zero also evaluated. Observed-channel normalization is shared; only encoder initialization differs. After encoder selection, both ridge centers are refit on fit windows with validation-only ridge selection; the retained affine adaptation head is also reported. Random initialization receives nominal observed-channel statistics but no pretrained encoder weights. This is a comparison of fixed, budgeted procedures and does not claim optimal hyperparameter search for every alternative.

S2.6 Same-state branches and MPC

Snapshots occur at 3, 6 and 9 seconds on 36 tasks in each regime, with 12 each in ID, wind-extrapolation and joint-extrapolation strata. State cloning includes simulation clock, pose/velocity, rotors, randomized parameters and controller integrator/attitude state. Nine commands branch from a common history over 1.5 seconds. Existing thrust remains for 20 ms; the command becomes active at the next 50 Hz control tick. Repeat-center replay, parent-state immutability and complete instrumented-parent reproduction were verified. Invalid prefixes are saved rather than removed from collection logs.

MPC operates at 10 Hz with nine XY command offsets $\{-0.4, 0, 0.4\}^2$, horizontal speed cap 1.5 m/s, a 30-step 20 Hz forecast and the cost

$$J(a) = K^{-1} \sum_k \|p_k^a - r_k\|^2 + 0.02 \|u_v^a\|^2 + 0.05 \|u_v^a - u_v^-\|^2.$$

Here $u_v \in \mathbb{R}^3$ contains the velocity components of $u \in \mathbb{R}^4$; yaw rate is unpenalized. The timed waypoint-polyline reference moves at 0.6 m/s after a two-second stationary warm-up. All methods use the same task specification. Nominal feedback and observer feedback are direct closed-loop alternatives; physical, raw and frozen predictors use the same candidate MPC interface. Success requires surviving the full 12 seconds and terminal reference error below 0.5 m. The corrected shared vertical cap is 1.5 m/s after warm-up; the initial value was 0.5 m/s.

S2.7 Paired calibration and matched objective fitting

A new development campaign contains 32 parent flights per regime. The first 24 are fit and the last eight validation; all snapshots/branches of one parent remain together. Each regime uses 384 seconds of parent flight and 1,296 seconds of branches, plus 144 seconds of duplicate branches and 384 seconds of parent-replay checks. This supervision budget is not equivalent to 32 unbranched episodes in C1. Fit retains 62/72 all-nine-valid snapshots in mass and 69/72 in lag;

validation retains 24/24 each. Prefix-valid fit counts are 65/72 and 71/72. Selection never uses the reused development evaluation tasks.

Identical physical-feature, raw-history and five frozen-context representations are fit to two objectives: trajectory MSE or centered candidate-cost error. The linear physical correction has 90 outputs, fit-only feature standardization and zero-initialized weights. Both objectives use float64 Adam, learning rate 0.003, 500 steps and penalties $\lambda \in \{10^{-4}, 10^{-2}, 1\}$. Each objective is normalized by its initial fit loss; regularization is $\lambda \|W_{\text{nonintercept}}\|_F^2/90$. Both select by validation action regret, ties preferring larger λ . The new scalar fits $\tau \in \{0.1, 0.2, 0.3, 0.5, 0.7, 1, 1.5\}$ by validation regret, ties preferring smaller τ . No old readout coefficients initialize these fits. There are 84 fits, 28 selected heads, and two new scalar controls. Pure decision loss failed the retained task gate, so only the trajectory-loss readouts enter the final closed-loop study.

S3 Complete Calibration Confirmation Contrasts

All eight primary comparisons are shown together. F-Sup is frozen supervised context; Scalar-fit is calibrated scalar physics; Phys-feat is physical-feature ridge; Adapt-pre and Adapt-rand denote matched full-encoder adaptation. Negative prediction differences favor the left model. Higher accuracy and lower contact are favorable. All primary budgets are 32 with the prespecified readout; secondary tables do not create additional adjusted claims.

Table S3: All eight locked contrasts: 20,000 valid paired draws each; family-adjusted 99.375% and descriptive 95% intervals.

Regime	Comparison	Metric	Δ	99.375% interval	95% interval
Lag	F-Sup – Scalar-fit	RMSE (m)	-0.0590	[-0.0846, -0.0295]	[-0.0779, -0.0386]
Lag	F-Sup – Adapt-rand	RMSE (m)	-0.1653	[-0.2401, -0.1033]	[-0.2156, -0.1196]
Lag	F-Sup – Adapt-pre	RMSE (m)	-0.0668	[-0.1317, -0.0268]	[-0.1089, -0.0371]
Lag	Adapt-pre – Adapt-rand	RMSE (m)	-0.0985	[-0.1443, -0.0625]	[-0.1284, -0.0718]
Mass	F-Sup – Phys-feat	RMSE (m)	+0.1361	[+0.0546, +0.2557]	[+0.0756, +0.2110]
Lag	F-Sup – Scalar-fit	Accuracy (pp)	+1.2074	[-2.4798, +4.7067]	[-1.4214, +3.6014]
Lag	F-Sup – Scalar-fit	Contact (pp)	+0.0303	[-0.2219, +0.3135]	[-0.1072, +0.1997]
Mass	F-Sup – Phys-feat	Accuracy (pp)	-4.6102	[-14.9917, +3.1682]	[-11.4748, +1.0949]

S3.1 All pooled secondary means

F- prefixes identify frozen encoders: Blind, PA (privileged teacher), Direct (direct wind supervision), Sup (supervised dynamics), SupW (supervised plus wind) and Wind (wind-only). Raw-state and Raw-depth are unencoded controls. Z/P/N denote zero-centered ridge, physical-centered ridge and the retained neural prediction head, respectively; C denotes a physical control without a learned head. B is the total calibration budget and n the number of averaged fits. Absolute accuracy/contact are percentages; regret is final squared goal-distance excess. All five evaluation strata are retained in C1 in the bundle.

Table S4: Every pooled secondary method, budget and head. These are descriptive means.

Regime	Method	B	Head	n	RMSE (m)	Acc. (%)	Contact (%)	Regret
Mass	F-Blind	32	Z	25	0.3278	85.59	51.02	0.1931
Mass	F-Blind	32	P	25	0.3259	85.73	51.13	0.1773
Mass	F-Blind	64	Z	25	0.3418	85.32	51.30	0.1955
Mass	F-Blind	64	P	25	0.3537	84.37	51.38	0.2090
Mass	F-PA	32	Z	25	0.3195	86.58	51.36	0.1525
Mass	F-PA	32	P	25	0.3242	85.76	51.38	0.1624
Mass	F-PA	64	Z	25	0.3144	85.97	51.64	0.1535
Mass	F-PA	64	P	25	0.3180	85.66	51.68	0.1596
Mass	F-Direct	32	Z	25	0.3035	89.08	51.57	0.1071
Mass	F-Direct	32	P	25	0.3052	88.85	51.55	0.1066
Mass	F-Direct	64	Z	25	0.2945	89.46	51.71	0.1002
Mass	F-Direct	64	P	25	0.3000	89.12	51.76	0.1038

Table S4 continued

Regime	Method	B	Head	n	RMSE (m)	Acc. (%)	Contact (%)	Regret
Mass	F-Sup	32	Z	25	0.4007	83.86	51.66	0.2241
Mass	F-Sup	32	P	25	0.4154	83.36	51.59	0.2342
Mass	F-Sup	64	Z	25	0.4047	84.64	51.66	0.2027
Mass	F-Sup	64	P	25	0.4051	84.88	51.68	0.1920
Mass	F-SupW	32	Z	25	0.3391	86.41	51.30	0.1540
Mass	F-SupW	32	P	25	0.3495	85.53	51.24	0.1637
Mass	F-SupW	64	Z	25	0.3331	86.47	51.45	0.1584
Mass	F-SupW	64	P	25	0.3374	86.71	51.48	0.1522
Mass	F-Wind	32	Z	25	0.3520	87.49	51.37	0.1296
Mass	F-Wind	32	P	25	0.3630	86.75	51.37	0.1410
Mass	F-Wind	64	Z	25	0.3285	88.85	51.59	0.1056
Mass	F-Wind	64	P	25	0.3282	88.85	51.64	0.1050
Mass	Raw-state	32	Z	5	0.5978	78.81	51.40	0.3526
Mass	Raw-state	32	P	5	0.6201	74.75	51.47	0.4200
Mass	Raw-state	64	Z	5	0.4604	87.29	51.61	0.1286
Mass	Raw-state	64	P	5	0.4466	87.29	51.68	0.1241
Mass	Raw-depth	32	Z	5	0.6304	78.31	51.75	0.3837
Mass	Raw-depth	32	P	5	0.5252	78.31	51.33	0.3684
Mass	Raw-depth	64	Z	5	0.5565	86.10	51.33	0.1903
Mass	Raw-depth	64	P	5	0.5207	86.61	51.75	0.1778
Mass	Phys-feat	32	Z	5	0.2777	88.47	50.98	0.1316
Mass	Phys-feat	32	P	5	0.2793	87.97	50.98	0.1396
Mass	Phys-feat	64	Z	5	0.2823	87.12	51.19	0.1454
Mass	Phys-feat	64	P	5	0.2825	87.12	51.19	0.1454
Mass	Scalar-0.3	32	C	5	0.4225	90.68	50.00	0.0879
Mass	Scalar-1.0	32	C	5	0.3365	83.90	51.75	0.2290
Mass	Scalar-fit	32	C	5	0.4319	81.19	51.47	0.3019
Mass	Scalar-0.3	64	C	5	0.4225	90.68	50.00	0.0879
Mass	Scalar-1.0	64	C	5	0.3365	83.90	51.75	0.2290
Mass	Scalar-fit	64	C	5	0.4557	80.51	51.40	0.3201
Lag	F-Blind	32	Z	25	0.3669	86.63	5.05	0.1511
Lag	F-Blind	32	P	25	0.3699	86.78	5.00	0.1483
Lag	F-Blind	64	Z	25	0.3505	87.24	5.07	0.1346
Lag	F-Blind	64	P	25	0.3547	87.43	5.04	0.1312
Lag	F-PA	32	Z	25	0.3547	87.05	4.90	0.1385
Lag	F-PA	32	P	25	0.3580	86.81	4.89	0.1435
Lag	F-PA	64	Z	25	0.3313	88.53	4.90	0.1037
Lag	F-PA	64	P	25	0.3344	88.28	4.89	0.1080
Lag	F-Direct	32	Z	25	0.3094	89.55	5.01	0.0897
Lag	F-Direct	32	P	25	0.3118	89.38	4.98	0.0902
Lag	F-Direct	64	Z	25	0.2871	90.75	5.02	0.0671
Lag	F-Direct	64	P	25	0.2883	90.64	5.02	0.0679
Lag	F-Sup	32	Z	25	0.2872	89.10	5.14	0.1053
Lag	F-Sup	32	P	25	0.2899	89.03	5.10	0.1030
Lag	F-Sup	64	Z	25	0.2671	89.80	5.16	0.0912
Lag	F-Sup	64	P	25	0.2697	89.92	5.14	0.0883
Lag	F-SupW	32	Z	25	0.3271	88.37	5.04	0.1075
Lag	F-SupW	32	P	25	0.3287	88.24	5.03	0.1085
Lag	F-SupW	64	Z	25	0.3032	89.78	5.05	0.0794
Lag	F-SupW	64	P	25	0.3046	89.69	5.03	0.0801
Lag	F-Wind	32	Z	25	0.3576	87.58	4.94	0.1295
Lag	F-Wind	32	P	25	0.3591	87.41	4.93	0.1322
Lag	F-Wind	64	Z	25	0.3234	89.31	4.97	0.0874
Lag	F-Wind	64	P	25	0.3246	89.17	4.97	0.0892
Lag	Raw-state	32	Z	5	0.4240	84.65	4.80	0.2116
Lag	Raw-state	32	P	5	0.4327	84.15	4.83	0.2270
Lag	Raw-state	64	Z	5	0.3433	87.69	5.01	0.1246
Lag	Raw-state	64	P	5	0.3500	87.02	5.01	0.1363
Lag	Raw-depth	32	Z	5	0.4427	84.08	5.04	0.2125
Lag	Raw-depth	32	P	5	0.4466	84.45	4.67	0.2041

Table S4 continued

Regime	Method	B	Head	n	RMSE (m)	Acc. (%)	Contact (%)	Regret
Lag	Raw-depth	64	Z	5	0.3747	86.31	5.01	0.1559
Lag	Raw-depth	64	P	5	0.3851	86.04	5.01	0.1620
Lag	Phys-feat	32	Z	5	0.3820	86.58	4.98	0.1549
Lag	Phys-feat	32	P	5	0.3841	86.71	4.98	0.1515
Lag	Phys-feat	64	Z	5	0.3701	87.28	4.95	0.1386
Lag	Phys-feat	64	P	5	0.3699	87.28	4.95	0.1390
Lag	Scalar-0.3	32	C	5	0.4507	88.36	5.01	0.1135
Lag	Scalar-1.0	32	C	5	0.3966	84.32	5.16	0.2240
Lag	Scalar-fit	32	C	5	0.3488	87.82	5.07	0.1407
Lag	Scalar-0.3	64	C	5	0.4507	88.36	5.01	0.1135
Lag	Scalar-1.0	64	C	5	0.3966	84.32	5.16	0.2240
Lag	Scalar-fit	64	C	5	0.3372	88.77	5.04	0.1188
Mass	Adapt-rand	32	Z	25	0.4920	80.64	50.97	0.3061
Mass	Adapt-rand	32	P	25	0.5237	77.66	50.97	0.3980
Mass	Adapt-rand	32	N	25	0.4957	78.64	51.06	0.3474
Mass	Adapt-rand	64	Z	25	0.4238	86.78	51.50	0.1753
Mass	Adapt-rand	64	P	25	0.4136	86.20	51.44	0.1881
Mass	Adapt-rand	64	N	25	0.4307	85.08	51.51	0.2092
Lag	Adapt-rand	32	Z	25	0.4381	84.53	4.73	0.2137
Lag	Adapt-rand	32	P	25	0.4552	84.03	4.68	0.2300
Lag	Adapt-rand	32	N	25	0.4625	83.53	4.67	0.2428
Lag	Adapt-rand	64	Z	25	0.3646	86.46	4.88	0.1622
Lag	Adapt-rand	64	P	25	0.3725	85.92	4.81	0.1702
Lag	Adapt-rand	64	N	25	0.3770	85.69	4.78	0.1768
Mass	Adapt-pre	32	Z	25	0.4710	81.80	51.52	0.3078
Mass	Adapt-pre	32	P	25	0.4873	78.47	51.30	0.3774
Mass	Adapt-pre	32	N	25	0.5070	77.66	51.36	0.3947
Mass	Adapt-pre	64	Z	25	0.4049	84.68	51.65	0.2381
Mass	Adapt-pre	64	P	25	0.3975	84.34	51.75	0.2440
Mass	Adapt-pre	64	N	25	0.4148	82.98	51.71	0.2692
Lag	Adapt-pre	32	Z	25	0.3493	86.39	4.97	0.1651
Lag	Adapt-pre	32	P	25	0.3566	85.94	4.89	0.1697
Lag	Adapt-pre	32	N	25	0.3711	85.65	4.93	0.1768
Lag	Adapt-pre	64	Z	25	0.3004	88.32	5.08	0.1194
Lag	Adapt-pre	64	P	25	0.3069	87.80	5.03	0.1268
Lag	Adapt-pre	64	N	25	0.3135	87.60	5.06	0.1301

S4 Same-State Prediction and Decision Evidence

The initial development evaluations below use previously fit readouts at identical controller states and condition on the old shared calibration pool. The following cross-campaign subsection supplies the current main Figure 3. The frozen method averages five seeds and five old subsets; other methods average five subsets. These subsets are not independent calibration campaigns. The original Pre RMSE uses individually valid positions. The new Full RMSE uses exactly the all-nine-valid states used for accuracy and regret: 98 mass and 106 lag, with nine candidates and 30 times per state. Contact still uses initially alive snapshots.

D1a–D1b are a post-hoc denominator audit of saved D1 predictions, not new forecasts, fitting, rollouts or confirmation. Every original configuration and tier is retained in the bundle (360 configuration/tier rows, 40 method/tier means). Each configuration’s Full RMSE is the square root of mean squared Euclidean position error on shared support, followed by configuration averaging within method. Separate sum-of-squared-error aggregation agrees within 1.12×10^{-16} ; branch masks and saved common/contrast identities were checked. The 16 previous action intervals are unchanged and are not uncertainty intervals for the new RMSE.

Frozen-minus-scalar Full RMSE differences are -0.0966 m in mass and -0.0334 m in lag when pooled; regret differences remain $+0.0118$ and $+0.0140$. The reversal is not universal across tiers: lag ID instead has $+0.0267$ m prediction error and $+0.0096$ regret. All three mass tiers and both lag extrapolation tiers retain lower RMSE and higher regret. These are descriptive comparisons on reused development states.

Table S5: All five predictor classes. Pre: original pre-contact RMSE (m); Full: all-nine-valid RMSE (m), for the original calibration pool. C/Resp: candidate-mean/centered-action MSE (m^2) on full support, descriptive rather than causal. Both RMSE columns average configuration RMSEs.

Regime	Predictor	n	Pre	Full	Acc. (%)	Regret	Contact (%)	C	Resp
Mass	Scalar-old	5	0.3005	0.2989	76.73	0.0146	0.00	0.0836	0.0067
Mass	Axis	5	0.2839	0.2833	36.73	0.0325	0.00	0.0784	0.0020
Mass	Phys-feat	5	0.3375	0.3380	31.43	0.0372	0.00	0.1131	0.0023
Mass	Raw	5	0.2937	0.2945	38.98	0.0363	0.00	0.0845	0.0024
Mass	Frozen	25	0.2020	0.2023	50.29	0.0264	0.00	0.0391	0.0020
Lag	Scalar-old	5	0.3069	0.3056	63.96	0.0163	0.00	0.0905	0.0041
Lag	Axis	5	0.3370	0.3372	35.09	0.0390	0.00	0.1110	0.0030
Lag	Phys-feat	5	0.4111	0.4128	32.64	0.0490	0.00	0.1690	0.0029
Lag	Raw	5	0.3544	0.3566	42.26	0.0373	0.00	0.1248	0.0029
Lag	Frozen	25	0.2706	0.2722	53.43	0.0303	0.00	0.0717	0.0027

Table S6: All 16 registered exploratory contrasts; unadjusted 95% intervals. Differences follow the stated comparison.

Regime	Comparison	Metric	Δ	95% interval
Mass	Frozen – Scalar	Accuracy (pp)	-26.4490	[-35.6117, -17.1847]
Mass	Frozen – Scalar	Regret	+0.0118	[+0.0076, +0.0158]
Mass	Frozen – Axis	Accuracy (pp)	+13.5510	[+2.5841, +25.2395]
Mass	Frozen – Axis	Regret	-0.0061	[-0.0129, +0.0002]
Mass	Frozen – Phys-feat	Accuracy (pp)	+18.8571	[+8.1702, +30.7895]
Mass	Frozen – Phys-feat	Regret	-0.0108	[-0.0179, -0.0043]
Mass	Frozen – Raw	Accuracy (pp)	+11.3061	[+3.9574, +19.1808]
Mass	Frozen – Raw	Regret	-0.0099	[-0.0154, -0.0044]
Lag	Frozen – Scalar	Accuracy (pp)	-10.5283	[-21.4049, +2.9632]
Lag	Frozen – Scalar	Regret	+0.0140	[+0.0047, +0.0233]
Lag	Frozen – Axis	Accuracy (pp)	+18.3396	[+9.4653, +29.7290]
Lag	Frozen – Axis	Regret	-0.0087	[-0.0185, +0.0007]
Lag	Frozen – Phys-feat	Accuracy (pp)	+20.7925	[+12.1480, +32.5313]
Lag	Frozen – Phys-feat	Regret	-0.0187	[-0.0300, -0.0076]
Lag	Frozen – Raw	Accuracy (pp)	+11.1698	[+4.7586, +20.0844]
Lag	Frozen – Raw	Regret	-0.0070	[-0.0137, -0.0011]

S4.1 Independent calibration sources on reused controller states

D24–D25 cross-evaluate C1’s already fitted budget-32 physical-centered ridge heads on these same controller states. Protocol a947526 fixes five campaigns, both regimes, all five model seeds where applicable, and seven predictor groups before inference: 190 configurations and 20,520 state rows. The independent calibration campaigns are not independent task confirmation. No fitting or new rollout occurs; axis physics has no corresponding campaign fits and is not silently substituted from the old pool. Observer feedback remains a full-flight comparator.

Shared RMSE, regret and accuracy use the same 98 mass/106 lag all-nine-valid states; selected contact uses 99/108 initially alive states. RMSE is computed within each configuration and then averaged equally over seeds and campaigns. All 760 configuration/tier summaries, 280 campaign/method/tier means, 56 method/tier means and 240 campaign/contrast/tier differences are in the source bundle. Pre-contact RMSE remains a descriptive sensitivity. Raw-depth includes the full two-frame input, and both adaptation arms use their saved physical-centered ridge heads, not newly selected neural heads.

Frozen-minus-scalar pooled regret is positive in every campaign in both regimes. Lower prediction RMSE with higher regret occurs in all five mass campaigns and four of five lag campaigns; lag campaign 0 has both higher RMSE and regret. Across campaigns, mass RMSE difference is -0.0644 m, while lag difference is -0.0405 m with interval $[-0.1045, 0.0200]$. Thus the lag prediction advantage is not established by this exploratory interval. Lag ID retains higher RMSE and regret; lag joint-extrapolation accuracy can improve while regret rises. The regime boundary measured on C1’s different four-action tasks does not by itself determine rankings on these controller states.

All 56 pooled contrasts below use 20,000 paired bootstrap draws with fixed seed 30919007, resampling five campaigns, five model seeds and 21 scene clusters per regime stratified by geometry. Repeated snapshot times remain clustered; deterministic controls have one seed slot. All draws had positive denominators. Intervals are unadjusted 95% and exploratory. Zero observed selected-contact differences do not establish safety or equivalence.

All 285 recovered weight/checkpoint files match original hashes. The 105 unique encoders strict-load, retain saved nominal observation normalization, and pass CPU/CUDA context checks with TF32 off (maximum difference 1.44×10^{-6}). C1/deployed feature adapters and future/privileged-key isolation checks agree exactly. An independent verifier reconstructs every readout from saved contexts, nominal held-command displacement, world transform, original costs/masks/choices and all aggregates; maximum forecast difference is 2.23×10^{-14} . It checks interval point estimates, not an independent rerun of bootstrap quantiles. The first verifier omitted the deployed float32 conversion of known command features; that failed verifier/log is retained. Correcting its input conversion required no changed inference, result or tolerance.

Table S7: All seven groups on reused snapshots, equally averaged over independent calibration campaigns and model seeds. Shared/Pre RMSE in m; accuracy/contact in percent. All configurations and tiers are retained in D24.

Regime	Predictor	Shared	Pre	Regret	Acc. (%)	Contact (%)
Mass	Frozen	0.2091	0.2087	0.0326	40.0408	0.0000
Mass	Scalar	0.2734	0.2747	0.0147	75.9184	0.0000
Mass	Phys-feat	0.3241	0.3240	0.0377	29.7959	0.0000
Mass	Raw	0.2326	0.2321	0.0298	45.1020	0.0000
Mass	Raw-depth	0.3154	0.3150	0.0304	45.3061	0.0000
Mass	Adapt-rand	0.2982	0.2979	0.0367	38.7755	0.0000
Mass	Adapt-pre	0.2433	0.2430	0.0334	38.4082	0.0000
Lag	Frozen	0.2651	0.2635	0.0289	50.6415	0.0000
Lag	Scalar	0.3056	0.3069	0.0163	63.9623	0.0000
Lag	Phys-feat	0.3811	0.3800	0.0399	35.2830	0.0000
Lag	Raw	0.3072	0.3054	0.0294	49.4340	0.0000
Lag	Raw-depth	0.3512	0.3500	0.0319	50.3774	0.3704
Lag	Adapt-rand	0.3941	0.3926	0.0356	44.2642	0.0000
Lag	Adapt-pre	0.2922	0.2910	0.0272	51.8113	0.0000

Table S8: Every campaign: pooled frozen minus scalar. Lower RMSE and positive regret denote the descriptive reversal. Lag campaign 0 remains an explicit counterexample. Campaign-level differences are descriptive.

Regime	Campaign	Δ RMSE	Δ Regret	Δ Acc. (pp)	Δ Contact (pp)
Mass	0	-0.0815	0.0162	-34.8980	0.0000
Mass	1	-0.0329	0.0239	-43.2653	0.0000
Mass	2	-0.0454	0.0157	-32.8571	0.0000
Mass	3	-0.0281	0.0189	-35.7143	0.0000
Mass	4	-0.1339	0.0144	-32.6531	0.0000
Lag	0	0.0357	0.0168	-20.3774	0.0000
Lag	1	-0.0776	0.0161	-15.2830	0.0000
Lag	2	-0.0716	0.0132	-14.3396	0.0000
Lag	3	-0.0182	0.0086	-4.1509	0.0000
Lag	4	-0.0707	0.0087	-12.4528	0.0000

Table S9: All 56 fixed pooled comparisons; unadjusted exploratory 95% intervals. Calibration sources are independent; the evaluation tasks and question are reused development evidence.

Regime	Comparison	Metric	Δ	95% interval
Mass	Frozen – Scalar	RMSE (m)	-0.0644	[-0.1082, -0.0278]
Mass	Frozen – Phys-feat	RMSE (m)	-0.1150	[-0.1633, -0.0664]
Mass	Frozen – Raw	RMSE (m)	-0.0236	[-0.0713, +0.0241]
Mass	Frozen – Raw-depth	RMSE (m)	-0.1063	[-0.1590, -0.0556]
Mass	Frozen – Adapt-rand	RMSE (m)	-0.0891	[-0.1441, -0.0301]

Table S9 continued

Regime	Comparison	Metric	Δ	95% interval
Mass	Frozen – Adapt-pre	RMSE (m)	-0.0342	[-0.0743, +0.0064]
Mass	Adapt-pre – Adapt-rand	RMSE (m)	-0.0549	[-0.0873, -0.0180]
Mass	Frozen – Scalar	Regret	+0.0178	[+0.0117, +0.0238]
Mass	Frozen – Phys-feat	Regret	-0.0051	[-0.0143, +0.0026]
Mass	Frozen – Raw	Regret	+0.0028	[-0.0046, +0.0093]
Mass	Frozen – Raw-depth	Regret	+0.0021	[-0.0068, +0.0092]
Mass	Frozen – Adapt-rand	Regret	-0.0041	[-0.0134, +0.0045]
Mass	Frozen – Adapt-pre	Regret	-0.0008	[-0.0077, +0.0053]
Mass	Adapt-pre – Adapt-rand	Regret	-0.0033	[-0.0103, +0.0032]
Mass	Frozen – Scalar	Acc. (pp)	-35.8776	[-46.8475, -23.7140]
Mass	Frozen – Phys-feat	Acc. (pp)	+10.2449	[-3.9130, +25.1570]
Mass	Frozen – Raw	Acc. (pp)	-5.0612	[-17.3155, +7.6046]
Mass	Frozen – Raw-depth	Acc. (pp)	-5.2653	[-17.6444, +8.1923]
Mass	Frozen – Adapt-rand	Acc. (pp)	+1.2653	[-11.2407, +14.8042]
Mass	Frozen – Adapt-pre	Acc. (pp)	+1.6327	[-7.2762, +11.8751]
Mass	Adapt-pre – Adapt-rand	Acc. (pp)	-0.3673	[-10.9280, +10.4954]
Mass	Frozen – Scalar	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Phys-feat	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Raw	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Raw-depth	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Adapt-rand	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Adapt-pre	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Adapt-pre – Adapt-rand	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Scalar	RMSE (m)	-0.0405	[-0.1045, +0.0200]
Lag	Frozen – Phys-feat	RMSE (m)	-0.1159	[-0.1724, -0.0365]
Lag	Frozen – Raw	RMSE (m)	-0.0421	[-0.0814, +0.0082]
Lag	Frozen – Raw-depth	RMSE (m)	-0.0860	[-0.1412, -0.0211]
Lag	Frozen – Adapt-rand	RMSE (m)	-0.1290	[-0.2072, -0.0331]
Lag	Frozen – Adapt-pre	RMSE (m)	-0.0271	[-0.0690, +0.0102]
Lag	Adapt-pre – Adapt-rand	RMSE (m)	-0.1019	[-0.1599, -0.0321]
Lag	Frozen – Scalar	Regret	+0.0127	[+0.0060, +0.0189]
Lag	Frozen – Phys-feat	Regret	-0.0110	[-0.0207, -0.0029]
Lag	Frozen – Raw	Regret	-0.0005	[-0.0083, +0.0065]
Lag	Frozen – Raw-depth	Regret	-0.0030	[-0.0115, +0.0047]
Lag	Frozen – Adapt-rand	Regret	-0.0066	[-0.0186, +0.0025]
Lag	Frozen – Adapt-pre	Regret	+0.0018	[-0.0045, +0.0077]
Lag	Adapt-pre – Adapt-rand	Regret	-0.0084	[-0.0180, +0.0001]
Lag	Frozen – Scalar	Acc. (pp)	-13.3208	[-24.2912, +0.6667]
Lag	Frozen – Phys-feat	Acc. (pp)	+15.3585	[+3.3282, +29.5057]
Lag	Frozen – Raw	Acc. (pp)	+1.2075	[-7.9626, +11.4289]
Lag	Frozen – Raw-depth	Acc. (pp)	+0.2642	[-9.9829, +11.3726]
Lag	Frozen – Adapt-rand	Acc. (pp)	+6.3774	[-4.9311, +19.7740]
Lag	Frozen – Adapt-pre	Acc. (pp)	-1.1698	[-9.2500, +7.8589]
Lag	Adapt-pre – Adapt-rand	Acc. (pp)	+7.5472	[-2.6166, +18.2761]
Lag	Frozen – Scalar	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Phys-feat	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Raw	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Raw-depth	Contact (pp)	-0.3704	[-1.6667, +0.0000]
Lag	Frozen – Adapt-rand	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Adapt-pre	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Adapt-pre – Adapt-rand	Contact (pp)	+0.0000	[+0.0000, +0.0000]

S4.2 One-interval actions under shared continuation

D26–D27 test an evaluation-target assumption: the predictor and original branches hold a candidate for 1.5 s, whereas deployed MPC selects every 100 ms with 20 ms activation delay. Protocol c5e8e09 fixes the following intervention before simulation. Each original candidate starts at 20 ms, remains active for 100 ms, and at 120 ms switches to

either corrected nominal feedback or the same causal nominal-parameter observer feedback used in the complete-flight comparator. Both recompute commands at 50 Hz; the observer updates at 20 Hz using only past/current observed state. Control precedes observation at a coincident tick. This is a named shared continuation, not learned-MPC replanning.

All 190 C1 campaign configurations retain their previously saved choices: no inference, fitting, action reselection or task selection. The horizon, reference and initial-proposal command penalties are unchanged; subsequent feedback commands incur no extra penalty. We do not evaluate forecast RMSE against these altered future-command trajectories. Axis physics has no C1 campaign fits and is not substituted from another pool. All seven groups remain below. Regret and accuracy use the all-nine-valid intersection across all three continuations; own-support sensitivities and initially-alive selected contact are also retained. All 216 states and 61,560 configuration/state/mode records remain in D26, with explicit nulls outside the respective supports.

The shared intersection still contains 98 mass and 106 lag states. Frozen-minus-scalar regret falls from 0.01783 held to 0.00397 nominal/0.00485 observer in mass, and from 0.01266 to 0.00370/0.00599 in lag. All four new regret intervals remain positive. The descriptive reduction in the gap is 53–78%, depending on regime/continuation; it does not establish an improved controller. True argmin agreement with held branches is only 55.7–71.4%. Thus the magnitude and best candidate depend on continuation, while the frozen/scalar disadvantage persists in these two controls. This does not explain the entire complete-flight gap.

All six full parent replays, 18 snapshot inputs, old center branches and no-intervention nominal continuations match exactly. New center duplicates are exact; every new candidate matches both pre-switch position/mask records, and parent tensors/time remain unchanged. Independent acceptance reconstructs all 54 cost matrices, supports, every row/aggregate, candidate-order diagnostics and 16 interval point estimates (maximum difference 7.11×10^{-15}). All 60,264 causal observer estimates and nominal commands match exactly; feedforward reconstruction differs by at most 1.10×10^{-6} . The failed source-path and analysis-path preparations occurred before simulation and outcome calculation, respectively; source/logs are retained. There was no changed scientific protocol or tolerance.

The fixed family has 16 comparisons. Twelve compare frozen and scalar predictors in regret, accuracy and selected contact across two regimes and two continuations. Four compare each regret gap with its held-command counterpart. Each uses 20,000 paired draws, seed 30919008, five campaigns, five frozen-model seeds and 21 scene clusters stratified by geometry. The scalar has one seed slot; modes remain paired. All denominators were positive. These 16 intervals are unadjusted and exploratory; bootstrap quantiles and aerodynamics are not independently replicated. Zero contact differences establish neither equivalence nor safety.

Table S10: All continuation supports. Common: all nine candidates valid in every mode. Own: all nine valid in the indicated mode. Contact uses Alive.

Regime	Tier	States	Alive	Common	Held	Nominal	Observer
Mass	ID	36	36	36	36	36	36
Mass	W	36	34	34	34	34	34
Mass	J	36	29	28	28	29	28
Mass	All	108	99	98	98	99	98
Lag	ID	36	36	36	36	36	36
Lag	W	36	36	35	35	36	36
Lag	J	36	36	35	35	36	36
Lag	All	108	108	106	106	108	108

Table S11: All seven groups and all three continuations; equal model-seed/campaign means. Common-support regret/accuracy and own-support regret/accuracy; contact on original alive support. Acc./contact in percent.

Regime	Predictor	Continuation	Regret	Acc.	Own regret	Own acc.	Contact
Mass	Frozen	Held	0.03256	40.04	0.03256	40.04	0.00
Mass	Frozen	Nominal	0.00787	31.59	0.00858	31.27	0.00
Mass	Frozen	Observer	0.01036	23.76	0.01036	23.76	0.00
Mass	Scalar	Held	0.01474	75.92	0.01474	75.92	0.00
Mass	Scalar	Nominal	0.00390	64.08	0.00471	63.43	0.00
Mass	Scalar	Observer	0.00551	50.82	0.00551	50.82	0.00

Table S11 continued

Regime	Predictor	Continuation	Regret	Acc.	Own regret	Own acc.	Contact
Mass	Phys-feat	Held	0.03769	29.80	0.03769	29.80	0.00
Mass	Phys-feat	Nominal	0.00729	31.02	0.00797	30.71	0.00
Mass	Phys-feat	Observer	0.00784	31.22	0.00784	31.22	0.00
Mass	Raw	Held	0.02979	45.10	0.02979	45.10	0.00
Mass	Raw	Nominal	0.00660	40.41	0.00729	40.00	0.00
Mass	Raw	Observer	0.00836	34.29	0.00836	34.29	0.00
Mass	Raw-depth	Held	0.03042	45.31	0.03042	45.31	0.00
Mass	Raw-depth	Nominal	0.00599	46.33	0.00674	45.86	0.00
Mass	Raw-depth	Observer	0.00673	42.24	0.00673	42.24	0.00
Mass	Adapt-rand	Held	0.03669	38.78	0.03669	38.78	0.00
Mass	Adapt-rand	Nominal	0.00680	38.73	0.00746	38.34	0.00
Mass	Adapt-rand	Observer	0.00744	36.24	0.00744	36.24	0.00
Mass	Adapt-pre	Held	0.03336	38.41	0.03336	38.41	0.00
Mass	Adapt-pre	Nominal	0.00767	33.14	0.00835	32.81	0.00
Mass	Adapt-pre	Observer	0.00955	27.92	0.00955	27.92	0.00
Lag	Frozen	Held	0.02895	50.64	0.02895	50.64	0.00
Lag	Frozen	Nominal	0.00612	42.30	0.00602	42.59	0.00
Lag	Frozen	Observer	0.00889	31.51	0.00875	32.26	0.00
Lag	Scalar	Held	0.01629	63.96	0.01629	63.96	0.00
Lag	Scalar	Nominal	0.00242	67.17	0.00238	67.78	0.00
Lag	Scalar	Observer	0.00290	65.66	0.00289	65.37	0.00
Lag	Phys-feat	Held	0.03992	35.28	0.03992	35.28	0.00
Lag	Phys-feat	Nominal	0.00522	36.79	0.00517	37.22	0.00
Lag	Phys-feat	Observer	0.00522	39.81	0.00517	40.37	0.00
Lag	Raw	Held	0.02943	49.43	0.02943	49.43	0.00
Lag	Raw	Nominal	0.00528	44.72	0.00518	45.00	0.00
Lag	Raw	Observer	0.00704	35.85	0.00692	36.85	0.00
Lag	Raw-depth	Held	0.03192	50.38	0.03192	50.38	0.37
Lag	Raw-depth	Nominal	0.00504	47.55	0.00505	47.22	0.00
Lag	Raw-depth	Observer	0.00597	43.96	0.00597	44.26	0.00
Lag	Adapt-rand	Held	0.03559	44.26	0.03559	44.26	0.00
Lag	Adapt-rand	Nominal	0.00503	43.43	0.00495	43.63	0.00
Lag	Adapt-rand	Observer	0.00588	40.45	0.00580	41.07	0.00
Lag	Adapt-pre	Held	0.02719	51.81	0.02719	51.81	0.00
Lag	Adapt-pre	Nominal	0.00552	43.06	0.00543	43.22	0.00
Lag	Adapt-pre	Observer	0.00794	33.77	0.00784	34.22	0.00

Table S12: True candidate ordering relative to held-command branches on common support. All 36 pairs/state and first-index argmin ties; no near-tie filter. Pair/argmin agreement in percent; mean max-minus-min cost ranges. No exact pair ties occur. All tiers remain in D26.

Regime	Continuation	Pair agree	Argmin agree	Held range	New range
Mass	Nominal	86.17	71.43	0.19555	0.04224
Mass	Observer	82.60	59.18	0.19555	0.04218
Lag	Nominal	85.04	70.75	0.28560	0.04827
Lag	Observer	79.66	55.66	0.28560	0.04941

Table S13: All 16 fixed exploratory contrasts, with unadjusted 95% intervals. Regret/accuracy/contact: frozen minus scalar. Gap change: that difference under continuation minus held, on identical common support.

Regime	Continuation	Metric	Δ	95% interval
Mass	Nominal	Regret	+0.00397	[+0.00209, +0.00598]
Mass	Nominal	Gap change	-0.01386	[-0.01897, -0.00882]
Mass	Observer	Regret	+0.00485	[+0.00246, +0.00744]
Mass	Observer	Gap change	-0.01297	[-0.01812, -0.00791]

Table S13 continued

Regime	Continuation	Metric	Δ	95% interval
Mass	Nominal	Acc. (pp)	-32.48980	[-41.98060, -22.27116]
Mass	Observer	Acc. (pp)	-27.06122	[-36.72736, -16.83945]
Mass	Nominal	Contact (pp)	+0.00000	[+0.00000, +0.00000]
Mass	Observer	Contact (pp)	+0.00000	[+0.00000, +0.00000]
Lag	Nominal	Regret	+0.00370	[+0.00191, +0.00570]
Lag	Nominal	Gap change	-0.00896	[-0.01436, -0.00300]
Lag	Observer	Regret	+0.00599	[+0.00357, +0.00886]
Lag	Observer	Gap change	-0.00667	[-0.01245, -0.00007]
Lag	Nominal	Acc. (pp)	-24.86792	[-37.76484, -10.86402]
Lag	Observer	Acc. (pp)	-34.15094	[-45.93558, -23.68000]
Lag	Nominal	Contact (pp)	+0.00000	[+0.00000, +0.00000]
Lag	Observer	Contact (pp)	+0.00000	[+0.00000, +0.00000]

S5 Initial and Corrected Closed-Loop Matrices

The initial matrix used a 0.5 m/s vertical-command cap for all controllers. Under mass mismatch every controller group contacted in 18/36 tasks, with recorded poses at the ground threshold. A paired shared-cap intervention reduced feedback/observer contact to 4/36. The complete corrected matrix then applied the same 1.5 m/s post-warm-up cap to all 94 configurations. This is a shared controller correction, not a learned-model contribution. Both matrices are retained below. Without motor thrust logs, this evidence is not a direct identification of physical actuator saturation.

Each matrix compares seven controller groups: feedback and observer (one configuration each per regime), scalar/physical-feature/raw (five old subsets each), and frozen context and axis response as registered in the accepted matrix. The exact configuration counts are printed rather than interpreted as equal independent-seed counts. These are development tasks, not new confirmation. Common-time intervals address unequal contact truncation.

Table S14: All controller groups and tiers. RMSE and terminal error are in m; success and contact in %; truncated flight time in s. All: 36 tasks/configuration; ID/W/J: 12 each. Values average configurations within a group.

Regime	Controller	Tier	n	RMSE	Success	Contact	Time	Terminal
Mass	Feedback	All	1	1.663	27.78	50.00	8.941	2.657
Mass	Feedback	ID	1	1.361	33.33	33.33	10.077	2.081
Mass	Feedback	W	1	1.966	16.67	66.67	8.355	3.218
Mass	Feedback	J	1	1.661	33.33	50.00	8.391	2.671
Mass	Observer	All	1	1.637	27.78	50.00	8.747	2.696
Mass	Observer	ID	1	1.357	33.33	33.33	10.077	2.075
Mass	Observer	W	1	1.932	16.67	66.67	8.018	3.325
Mass	Observer	J	1	1.621	33.33	50.00	8.146	2.689
Lag	Feedback	All	1	0.582	83.33	2.78	11.854	0.532
Lag	Feedback	ID	1	0.156	100.00	0.00	12.000	0.094
Lag	Feedback	W	1	0.649	83.33	0.00	12.000	0.472
Lag	Feedback	J	1	0.942	66.67	8.33	11.561	1.030
Lag	Observer	All	1	0.421	91.67	2.78	11.773	0.364
Lag	Observer	ID	1	0.127	100.00	0.00	12.000	0.056
Lag	Observer	W	1	0.422	91.67	0.00	12.000	0.169
Lag	Observer	J	1	0.713	83.33	8.33	11.320	0.866
Mass	Scalar-old	All	5	1.646	27.78	50.00	8.939	2.635
Mass	Scalar-old	ID	5	1.360	33.33	33.33	10.068	2.088
Mass	Scalar-old	W	5	1.937	16.67	66.67	8.357	3.163
Mass	Scalar-old	J	5	1.642	33.33	50.00	8.391	2.655
Mass	Axis	All	5	1.665	25.00	50.00	8.932	2.686
Mass	Axis	ID	5	1.370	31.67	33.33	10.070	2.147
Mass	Axis	W	5	1.959	10.00	66.67	8.321	3.213
Mass	Axis	J	5	1.665	33.33	50.00	8.404	2.698
Mass	Phys-feat	All	5	1.660	25.00	50.00	8.935	2.670

Table S14 continued

Regime	Controller	Tier	n	RMSE	Success	Contact	Time	Terminal
Mass	Phys-feat	ID	5	1.364	31.67	33.33	10.071	2.114
Mass	Phys-feat	W	5	1.959	10.00	66.67	8.328	3.216
Mass	Phys-feat	J	5	1.657	33.33	50.00	8.407	2.680
Mass	Raw	All	5	1.669	24.44	50.00	8.934	2.681
Mass	Raw	ID	5	1.378	30.00	33.33	10.068	2.112
Mass	Raw	W	5	1.969	10.00	66.67	8.314	3.254
Mass	Raw	J	5	1.660	33.33	50.00	8.420	2.676
Lag	Scalar-old	All	5	0.534	88.33	2.78	11.843	0.487
Lag	Scalar-old	ID	5	0.146	100.00	0.00	12.000	0.080
Lag	Scalar-old	W	5	0.566	86.67	0.00	12.000	0.416
Lag	Scalar-old	J	5	0.889	78.33	8.33	11.528	0.967
Lag	Axis	All	5	0.579	86.67	2.78	11.844	0.573
Lag	Axis	ID	5	0.212	98.33	0.00	12.000	0.195
Lag	Axis	W	5	0.612	78.33	0.00	12.000	0.492
Lag	Axis	J	5	0.914	83.33	8.33	11.531	1.030
Lag	Phys-feat	All	5	0.613	85.00	2.78	11.849	0.604
Lag	Phys-feat	ID	5	0.231	98.33	0.00	12.000	0.202
Lag	Phys-feat	W	5	0.692	76.67	0.00	12.000	0.570
Lag	Phys-feat	J	5	0.916	80.00	8.33	11.547	1.040
Lag	Raw	All	5	0.580	85.56	2.78	11.845	0.579
Lag	Raw	ID	5	0.220	93.33	0.00	12.000	0.185
Lag	Raw	W	5	0.632	81.67	0.00	12.000	0.527
Lag	Raw	J	5	0.889	81.67	8.33	11.534	1.025
Mass	Frozen	All	25	1.654	26.33	50.00	8.945	2.665
Mass	Frozen	ID	25	1.366	33.00	33.33	10.069	2.075
Mass	Frozen	W	25	1.946	13.00	66.67	8.369	3.212
Mass	Frozen	J	25	1.650	33.00	50.00	8.395	2.707
Lag	Frozen	All	25	0.545	86.89	2.89	11.845	0.550
Lag	Frozen	ID	25	0.179	98.33	0.00	12.000	0.143
Lag	Frozen	W	25	0.590	83.00	0.00	12.000	0.508
Lag	Frozen	J	25	0.866	79.33	8.67	11.534	1.001

Table S15: All 48 registered exploratory contrasts; unadjusted 95% intervals. Differences follow the stated comparison.

Regime	Comparison	Metric	Δ	95% interval
Mass	Frozen – Feedback	RMSE (m)	-0.0085	[-0.0181, +0.0005]
Mass	Frozen – Feedback	Success (pp)	-1.4444	[-3.8750, +0.0000]
Mass	Frozen – Feedback	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Feedback	Common RMSE (m)	-0.0089	[-0.0199, +0.0017]
Mass	Frozen – Observer	RMSE (m)	+0.0176	[+0.0082, +0.0290]
Mass	Frozen – Observer	Success (pp)	-1.4444	[-3.8750, +0.0000]
Mass	Frozen – Observer	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Observer	Common RMSE (m)	-0.0170	[-0.0418, +0.0049]
Mass	Frozen – Scalar	RMSE (m)	+0.0077	[+0.0033, +0.0127]
Mass	Frozen – Scalar	Success (pp)	-1.4444	[-3.8750, +0.0000]
Mass	Frozen – Scalar	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Scalar	Common RMSE (m)	+0.0073	[+0.0022, +0.0126]
Mass	Frozen – Axis	RMSE (m)	-0.0105	[-0.0175, -0.0039]
Mass	Frozen – Axis	Success (pp)	+1.3333	[-0.7059, +4.2222]
Mass	Frozen – Axis	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Axis	Common RMSE (m)	-0.0126	[-0.0215, -0.0046]
Mass	Frozen – Phys-feat	RMSE (m)	-0.0061	[-0.0140, +0.0009]
Mass	Frozen – Phys-feat	Success (pp)	+1.3333	[-2.0000, +5.5556]
Mass	Frozen – Phys-feat	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Phys-feat	Common RMSE (m)	-0.0074	[-0.0165, +0.0004]
Mass	Frozen – Raw	RMSE (m)	-0.0146	[-0.0221, -0.0078]
Mass	Frozen – Raw	Success (pp)	+1.8889	[-1.7782, +6.1622]
Mass	Frozen – Raw	Contact (pp)	+0.0000	[+0.0000, +0.0000]

Table S15 continued

Regime	Comparison	Metric	Δ	95% interval
Mass	Frozen – Raw	Common RMSE (m)	-0.0149	[-0.0237, -0.0068]
Lag	Frozen – Feedback	RMSE (m)	-0.0375	[-0.0650, -0.0160]
Lag	Frozen – Feedback	Success (pp)	+3.5556	[-1.0256, +9.5136]
Lag	Frozen – Feedback	Contact (pp)	+0.1111	[+0.0000, +0.5854]
Lag	Frozen – Feedback	Common RMSE (m)	-0.0353	[-0.0633, -0.0140]
Lag	Frozen – Observer	RMSE (m)	+0.1244	[+0.0591, +0.1868]
Lag	Frozen – Observer	Success (pp)	-4.7778	[-12.1493, -0.2353]
Lag	Frozen – Observer	Contact (pp)	+0.1111	[+0.0000, +0.5854]
Lag	Frozen – Observer	Common RMSE (m)	+0.0981	[+0.0450, +0.1564]
Lag	Frozen – Scalar	RMSE (m)	+0.0112	[-0.0163, +0.0357]
Lag	Frozen – Scalar	Success (pp)	-1.4444	[-4.3243, +0.4571]
Lag	Frozen – Scalar	Contact (pp)	+0.1111	[+0.0000, +0.5854]
Lag	Frozen – Scalar	Common RMSE (m)	+0.0105	[-0.0169, +0.0352]
Lag	Frozen – Axis	RMSE (m)	-0.0346	[-0.0511, -0.0194]
Lag	Frozen – Axis	Success (pp)	+0.2222	[-3.8573, +3.7211]
Lag	Frozen – Axis	Contact (pp)	+0.1111	[+0.0000, +0.5854]
Lag	Frozen – Axis	Common RMSE (m)	-0.0349	[-0.0514, -0.0196]
Lag	Frozen – Phys-feat	RMSE (m)	-0.0679	[-0.0880, -0.0492]
Lag	Frozen – Phys-feat	Success (pp)	+1.8889	[-3.3333, +7.0476]
Lag	Frozen – Phys-feat	Contact (pp)	+0.1111	[+0.0000, +0.5854]
Lag	Frozen – Phys-feat	Common RMSE (m)	-0.0668	[-0.0872, -0.0482]
Lag	Frozen – Raw	RMSE (m)	-0.0352	[-0.0450, -0.0267]
Lag	Frozen – Raw	Success (pp)	+1.3333	[-1.6250, +5.4054]
Lag	Frozen – Raw	Contact (pp)	+0.1111	[+0.0000, +0.5854]
Lag	Frozen – Raw	Common RMSE (m)	-0.0352	[-0.0451, -0.0267]

S5.1 Shared altitude intervention

The intervention record D13 retains all eight configurations, four paired comparisons and exact baseline reproduction. Feedback/observer mass contact changes from 50.00% to 11.11%; the complete intervention outcomes are printed below. These are descriptive common-interface changes on reused tasks. The corrected matrix below is the source for learned-controller comparisons.

Table S16: All eight shared-cap intervention configurations. No new inference claim; RMSE in m and rates in %.

Regime	Controller	Cap (m/s)	RMSE	Success (%)	Contact (%)
Mass	Feedback	0.5	1.6626	27.78	50.00
Mass	Feedback	1.5	1.1205	30.56	11.11
Mass	Observer	0.5	1.6365	27.78	50.00
Mass	Observer	1.5	1.0333	36.11	11.11
Lag	Feedback	0.5	0.5824	83.33	2.78
Lag	Feedback	1.5	0.5759	83.33	0.00
Lag	Observer	0.5	0.4205	91.67	2.78
Lag	Observer	1.5	0.4091	94.44	0.00

S5.2 Corrected interface

Table S17: All controller groups and tiers. RMSE and terminal error are in m; success and contact in %; truncated flight time in s. All: 36 tasks/configuration; ID/W/J: 12 each. Values average configurations within a group.

Regime	Controller	Tier	n	RMSE	Success	Contact	Time	Terminal
Mass	Feedback	All	1	1.120	30.56	11.11	11.170	1.291

Table S17 continued

Regime	Controller	Tier	n	RMSE	Success	Contact	Time	Terminal
Mass	Feedback	ID	1	0.716	41.67	0.00	12.000	0.554
Mass	Feedback	W	1	1.359	16.67	8.33	11.467	1.606
Mass	Feedback	J	1	1.287	33.33	25.00	10.044	1.712
Mass	Observer	All	1	1.033	36.11	11.11	11.107	1.108
Mass	Observer	ID	1	0.709	41.67	0.00	12.000	0.545
Mass	Observer	W	1	1.170	25.00	16.67	10.920	1.430
Mass	Observer	J	1	1.222	41.67	16.67	10.400	1.349
Lag	Feedback	All	1	0.576	83.33	0.00	12.000	0.410
Lag	Feedback	ID	1	0.156	100.00	0.00	12.000	0.094
Lag	Feedback	W	1	0.650	83.33	0.00	12.000	0.474
Lag	Feedback	J	1	0.922	66.67	0.00	12.000	0.661
Lag	Observer	All	1	0.409	94.44	0.00	12.000	0.159
Lag	Observer	ID	1	0.127	100.00	0.00	12.000	0.056
Lag	Observer	W	1	0.395	100.00	0.00	12.000	0.097
Lag	Observer	J	1	0.705	83.33	0.00	12.000	0.326
Mass	Scalar-old	All	5	1.097	33.33	11.11	11.195	1.264
Mass	Scalar-old	ID	5	0.716	41.67	0.00	12.000	0.550
Mass	Scalar-old	W	5	1.313	25.00	8.33	11.481	1.566
Mass	Scalar-old	J	5	1.262	33.33	25.00	10.103	1.677
Mass	Axis	All	5	1.120	30.56	10.00	11.237	1.267
Mass	Axis	ID	5	0.734	40.00	0.00	12.000	0.594
Mass	Axis	W	5	1.337	18.33	8.33	11.500	1.590
Mass	Axis	J	5	1.290	33.33	21.67	10.210	1.615
Mass	Phys-feat	All	5	1.126	30.00	8.33	11.302	1.206
Mass	Phys-feat	ID	5	0.735	40.00	0.00	12.000	0.584
Mass	Phys-feat	W	5	1.373	16.67	8.33	11.503	1.627
Mass	Phys-feat	J	5	1.271	33.33	16.67	10.403	1.405
Mass	Raw	All	5	1.108	31.67	8.33	11.304	1.192
Mass	Raw	ID	5	0.734	38.33	0.00	12.000	0.584
Mass	Raw	W	5	1.346	16.67	8.33	11.509	1.602
Mass	Raw	J	5	1.245	40.00	16.67	10.403	1.390
Lag	Scalar-old	All	5	0.526	88.33	0.00	12.000	0.360
Lag	Scalar-old	ID	5	0.146	100.00	0.00	12.000	0.080
Lag	Scalar-old	W	5	0.568	86.67	0.00	12.000	0.420
Lag	Scalar-old	J	5	0.865	78.33	0.00	12.000	0.581
Lag	Axis	All	5	0.572	86.67	0.00	12.000	0.445
Lag	Axis	ID	5	0.212	98.33	0.00	12.000	0.195
Lag	Axis	W	5	0.613	78.33	0.00	12.000	0.497
Lag	Axis	J	5	0.889	83.33	0.00	12.000	0.643
Lag	Phys-feat	All	5	0.607	85.00	0.00	12.000	0.491
Lag	Phys-feat	ID	5	0.231	98.33	0.00	12.000	0.202
Lag	Phys-feat	W	5	0.694	76.67	0.00	12.000	0.575
Lag	Phys-feat	J	5	0.896	80.00	0.00	12.000	0.695
Lag	Raw	All	5	0.574	85.56	0.00	12.000	0.465
Lag	Raw	ID	5	0.220	93.33	0.00	12.000	0.185
Lag	Raw	W	5	0.633	81.67	0.00	12.000	0.531
Lag	Raw	J	5	0.868	81.67	0.00	12.000	0.678
Mass	Frozen	All	25	1.098	35.11	8.56	11.291	1.190
Mass	Frozen	ID	25	0.719	41.33	0.00	12.000	0.560
Mass	Frozen	W	25	1.335	21.67	8.33	11.498	1.596
Mass	Frozen	J	25	1.239	42.33	17.33	10.376	1.412
Lag	Frozen	All	25	0.542	86.89	0.00	12.000	0.452
Lag	Frozen	ID	25	0.179	98.33	0.00	12.000	0.143
Lag	Frozen	W	25	0.590	83.00	0.00	12.000	0.509
Lag	Frozen	J	25	0.859	79.33	0.00	12.000	0.704

Table S18: All 48 registered exploratory contrasts; unadjusted 95% intervals. Differences follow the stated comparison.

Regime	Comparison	Metric	Δ	95% interval
Mass	Frozen – Feedback	RMSE (m)	-0.0228	[-0.0449, -0.0066]
Mass	Frozen – Feedback	Success (pp)	+4.5556	[-1.4118, +11.8857]
Mass	Frozen – Feedback	Contact (pp)	-2.5556	[-8.1176, +0.0000]
Mass	Frozen – Feedback	Common RMSE (m)	-0.0201	[-0.0359, -0.0069]
Mass	Frozen – Observer	RMSE (m)	+0.0644	[+0.0073, +0.1391]
Mass	Frozen – Observer	Success (pp)	-1.0000	[-3.8668, +1.2903]
Mass	Frozen – Observer	Contact (pp)	-2.5556	[-7.6923, +0.7273]
Mass	Frozen – Observer	Common RMSE (m)	+0.0085	[-0.0087, +0.0275]
Mass	Frozen – Scalar	RMSE (m)	+0.0008	[-0.0220, +0.0171]
Mass	Frozen – Scalar	Success (pp)	+1.7778	[-2.5265, +8.0000]
Mass	Frozen – Scalar	Contact (pp)	-2.5556	[-8.1176, +0.0000]
Mass	Frozen – Scalar	Common RMSE (m)	+0.0044	[-0.0101, +0.0169]
Mass	Frozen – Axis	RMSE (m)	-0.0227	[-0.0423, -0.0095]
Mass	Frozen – Axis	Success (pp)	+4.5556	[-0.6002, +11.3333]
Mass	Frozen – Axis	Contact (pp)	-1.4444	[-4.5882, +0.0000]
Mass	Frozen – Axis	Common RMSE (m)	-0.0186	[-0.0305, -0.0090]
Mass	Frozen – Phys-feat	RMSE (m)	-0.0287	[-0.0413, -0.0172]
Mass	Frozen – Phys-feat	Success (pp)	+5.1111	[-1.3714, +13.0813]
Mass	Frozen – Phys-feat	Contact (pp)	+0.2222	[+0.0000, +0.9231]
Mass	Frozen – Phys-feat	Common RMSE (m)	-0.0285	[-0.0414, -0.0170]
Mass	Frozen – Raw	RMSE (m)	-0.0107	[-0.0177, -0.0015]
Mass	Frozen – Raw	Success (pp)	+3.4444	[-1.3714, +9.5000]
Mass	Frozen – Raw	Contact (pp)	+0.2222	[+0.0000, +0.9231]
Mass	Frozen – Raw	Common RMSE (m)	-0.0100	[-0.0171, -0.0012]
Lag	Frozen – Feedback	RMSE (m)	-0.0335	[-0.0632, -0.0095]
Lag	Frozen – Feedback	Success (pp)	+3.5556	[-1.0256, +9.5136]
Lag	Frozen – Feedback	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Feedback	Common RMSE (m)	-0.0335	[-0.0632, -0.0095]
Lag	Frozen – Observer	RMSE (m)	+0.1332	[+0.0794, +0.1962]
Lag	Frozen – Observer	Success (pp)	-7.5556	[-15.2824, -1.1425]
Lag	Frozen – Observer	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Observer	Common RMSE (m)	+0.1332	[+0.0794, +0.1962]
Lag	Frozen – Scalar	RMSE (m)	+0.0163	[-0.0133, +0.0416]
Lag	Frozen – Scalar	Success (pp)	-1.4444	[-4.3243, +0.4571]
Lag	Frozen – Scalar	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Scalar	Common RMSE (m)	+0.0163	[-0.0133, +0.0416]
Lag	Frozen – Axis	RMSE (m)	-0.0293	[-0.0483, -0.0125]
Lag	Frozen – Axis	Success (pp)	+0.2222	[-3.8573, +3.7211]
Lag	Frozen – Axis	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Axis	Common RMSE (m)	-0.0293	[-0.0483, -0.0125]
Lag	Frozen – Phys-feat	RMSE (m)	-0.0644	[-0.0868, -0.0445]
Lag	Frozen – Phys-feat	Success (pp)	+1.8889	[-3.3333, +7.0476]
Lag	Frozen – Phys-feat	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Phys-feat	Common RMSE (m)	-0.0644	[-0.0868, -0.0445]
Lag	Frozen – Raw	RMSE (m)	-0.0314	[-0.0408, -0.0230]
Lag	Frozen – Raw	Success (pp)	+1.3333	[-1.6250, +5.4054]
Lag	Frozen – Raw	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Raw	Common RMSE (m)	-0.0314	[-0.0408, -0.0230]

S6 Retained Remedial Ablations

S6.1 Response-preserving drift

The structured drift study restricts the correction to preserve a physical action response. It retains 140 heads and all methods at budgets 32/64. Its 12 task contrasts do not establish improvement, while all four prespecified-within-development prediction contrasts are adverse. These are exploratory comparisons after the main confirmation, not a

new independent confirmation. All strata remain in D7.

Table S19: All pooled drift and full-correction controls. Accuracy/contact in %; prediction RMSE in m.

Regime	Method	Budget	RMSE	Accuracy	Regret	Contact
Mass	Drift-frozen	32	0.4568	84.21	0.1745	48.05
Mass	Drift-raw	32	0.5298	81.87	0.2362	47.58
Mass	Drift-constant	32	0.3127	89.33	0.0784	47.45
Mass	Full-frozen	32	0.3539	82.56	0.1740	47.90
Mass	Full-raw	32	0.5094	77.87	0.3329	46.96
Mass	Full-phys	32	0.2793	85.87	0.0950	47.33
Mass	Scalar	32	0.3752	87.47	0.1165	47.45
Mass	Axis	32	0.3127	89.33	0.0784	47.45
Mass	Drift-frozen	64	0.4820	83.57	0.2014	48.20
Mass	Drift-raw	64	0.5325	80.53	0.2621	47.83
Mass	Drift-constant	64	0.3196	87.20	0.0907	47.58
Mass	Full-frozen	64	0.3734	83.36	0.1407	47.75
Mass	Full-raw	64	0.4211	83.20	0.1383	47.83
Mass	Full-phys	64	0.2584	85.87	0.0916	47.20
Mass	Scalar	64	0.4226	86.93	0.1446	47.70
Mass	Axis	64	0.3196	87.20	0.0907	47.58
Lag	Drift-frozen	32	0.3158	88.44	0.1172	6.16
Lag	Drift-raw	32	0.4367	81.02	0.2621	6.02
Lag	Drift-constant	32	0.3792	82.65	0.1889	6.27
Lag	Full-frozen	32	0.2899	87.82	0.1268	6.17
Lag	Full-raw	32	0.3790	81.90	0.2231	5.84
Lag	Full-phys	32	0.3982	83.54	0.1733	6.45
Lag	Scalar	32	0.3484	84.76	0.1580	6.33
Lag	Axis	32	0.3792	82.65	0.1889	6.27
Lag	Drift-frozen	64	0.3048	90.08	0.0944	6.24
Lag	Drift-raw	64	0.4013	81.50	0.2473	6.20
Lag	Drift-constant	64	0.3974	81.16	0.2366	6.14
Lag	Full-frozen	64	0.2729	89.10	0.1091	6.24
Lag	Full-raw	64	0.3601	82.18	0.2408	5.78
Lag	Full-phys	64	0.3846	83.06	0.1772	6.45
Lag	Scalar	64	0.3323	85.92	0.1339	6.33
Lag	Axis	64	0.3974	81.16	0.2366	6.14

Table S20: All 16 registered exploratory contrasts; unadjusted 95% intervals. Differences follow the stated comparison.

Regime	Comparison	Metric	Δ	95% interval
Mass	Drift-frozen – Full-frozen / B32	RMSE (m)	+0.1029	[+0.0655, +0.1380]
Mass	Drift-frozen – Full-frozen / B32	Accuracy (pp)	+1.6533	[-2.5317, +5.7534]
Mass	Drift-frozen – Full-frozen / B32	Regret	+0.0005	[-0.0757, +0.0857]
Mass	Drift-frozen – Full-frozen / B32	Contact (pp)	+0.1491	[-0.3099, +0.7143]
Mass	Drift-frozen – Full-frozen / B64	RMSE (m)	+0.1086	[+0.0571, +0.1667]
Mass	Drift-frozen – Full-frozen / B64	Accuracy (pp)	+0.2133	[-7.5072, +7.7041]
Mass	Drift-frozen – Full-frozen / B64	Regret	+0.0607	[-0.0606, +0.2156]
Mass	Drift-frozen – Full-frozen / B64	Contact (pp)	+0.4472	[-0.1334, +1.3892]
Lag	Drift-frozen – Full-frozen / B32	RMSE (m)	+0.0260	[+0.0147, +0.0375]
Lag	Drift-frozen – Full-frozen / B32	Accuracy (pp)	+0.6122	[-1.3334, +2.5653]
Lag	Drift-frozen – Full-frozen / B32	Regret	-0.0095	[-0.0371, +0.0193]
Lag	Drift-frozen – Full-frozen / B32	Contact (pp)	-0.0120	[-0.2477, +0.1939]
Lag	Drift-frozen – Full-frozen / B64	RMSE (m)	+0.0320	[+0.0179, +0.0476]
Lag	Drift-frozen – Full-frozen / B64	Accuracy (pp)	+0.9796	[-1.2917, +3.2764]
Lag	Drift-frozen – Full-frozen / B64	Regret	-0.0147	[-0.0530, +0.0198]
Lag	Drift-frozen – Full-frozen / B64	Contact (pp)	+0.0000	[-0.1554, +0.1667]

S6.2 Matched trajectory versus decision objective

The separate paired campaign and exact fitting protocol are specified in S2. Pure decision loss does not establish a task benefit for frozen context in either regime and substantially worsens its trajectory error. All 14 method/objective/regime groups and 18 exploratory objective contrasts are retained. The scalar row is a separate new-data physical control and has no matched learned-loss contrast. No fresh confirmation was launched for the rejected pure-cost candidate.

Table S21: All paired-data objective/control groups. RMSE in m; accuracy/contact in %.

Regime	Features	Objective	n	RMSE	Accuracy	Regret	Contact
Mass	Scalar-new	calibrated	1	0.7465	58.16	0.0179	0.00
Mass	Phys-feat	trajectory	1	0.2219	70.41	0.0149	0.00
Mass	Phys-feat	decision	1	0.5981	71.43	0.0133	0.00
Mass	Raw	trajectory	1	0.1827	68.37	0.0173	0.00
Mass	Raw	decision	1	0.8204	71.43	0.0151	1.01
Mass	Frozen	trajectory	5	0.1628	74.08	0.0150	0.00
Mass	Frozen	decision	5	0.8398	71.84	0.0142	0.20
Lag	Scalar-new	calibrated	1	0.2865	65.09	0.0147	0.00
Lag	Phys-feat	trajectory	1	0.2781	57.55	0.0190	0.00
Lag	Phys-feat	decision	1	0.4489	66.98	0.0148	0.00
Lag	Raw	trajectory	1	0.2337	66.04	0.0173	0.00
Lag	Raw	decision	1	1.0986	65.09	0.0184	0.93
Lag	Frozen	trajectory	5	0.2162	65.09	0.0189	0.00
Lag	Frozen	decision	5	1.2281	60.57	0.0198	0.00

Table S22: All 18 registered exploratory contrasts; unadjusted 95% intervals. Differences follow the stated comparison.

Regime	Comparison	Metric	Δ	95% interval
Mass	Phys-feat: decision – trajectory	Accuracy (pp)	+1.0204	[-4.7619, +5.7143]
Mass	Phys-feat: decision – trajectory	Regret	-0.0016	[-0.0030, -0.0001]
Mass	Phys-feat: decision – trajectory	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Raw: decision – trajectory	Accuracy (pp)	+3.0612	[-2.1978, +10.0000]
Mass	Raw: decision – trajectory	Regret	-0.0022	[-0.0064, +0.0019]
Mass	Raw: decision – trajectory	Contact (pp)	+1.0101	[+0.0000, +3.1915]
Mass	Frozen: decision – trajectory	Accuracy (pp)	-2.2449	[-10.1313, +3.7097]
Mass	Frozen: decision – trajectory	Regret	-0.0008	[-0.0035, +0.0020]
Mass	Frozen: decision – trajectory	Contact (pp)	+0.2020	[+0.0000, +1.0256]
Lag	Phys-feat: decision – trajectory	Accuracy (pp)	+9.4340	[+0.0000, +19.8452]
Lag	Phys-feat: decision – trajectory	Regret	-0.0042	[-0.0136, +0.0056]
Lag	Phys-feat: decision – trajectory	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Raw: decision – trajectory	Accuracy (pp)	-0.9434	[-9.6386, +7.6923]
Lag	Raw: decision – trajectory	Regret	+0.0010	[-0.0061, +0.0081]
Lag	Raw: decision – trajectory	Contact (pp)	+0.9259	[+0.0000, +2.9412]
Lag	Frozen: decision – trajectory	Accuracy (pp)	-4.5283	[-13.1429, +4.5455]
Lag	Frozen: decision – trajectory	Regret	+0.0009	[-0.0066, +0.0082]
Lag	Frozen: decision – trajectory	Contact (pp)	+0.0000	[+0.0000, +0.0000]

S7 Final Paired-Data Closed-Loop Evidence

Trajectory-loss heads from the new paired calibration enter the final gate. Per regime the 15 configurations are nominal feedback, observer feedback, new scalar, physical features, raw history, five old scalar subsets and five frozen-context seeds. Every configuration runs all 36 tasks for up to 12 seconds. There are 30 configurations and 1,080 rollouts total. Feedback/observer/old-scalar trajectories reproduce their previously accepted results.

The final data and fitting choices differ from earlier stages. Thus the local mass gains cannot be attributed to branching alone, and one paired calibration campaign does not support campaign-level uncertainty. All 48 exploratory contrasts

remain together. Frozen-context lag RMSE remains worse than observer feedback even when all methods have zero contact. The mass learned-minus-observer common-time interval includes zero; this does not establish equivalence.

Table S23: All controller groups and tiers. RMSE and terminal error are in m; success and contact in %; truncated flight time in s. All: 36 tasks/configuration; ID/W/J: 12 each. Values average configurations within a group.

Regime	Controller	Tier	n	RMSE	Success	Contact	Time	Terminal
Mass	Feedback	All	1	1.120	30.56	11.11	11.170	1.291
Mass	Feedback	ID	1	0.716	41.67	0.00	12.000	0.554
Mass	Feedback	W	1	1.359	16.67	8.33	11.467	1.606
Mass	Feedback	J	1	1.287	33.33	25.00	10.044	1.712
Mass	Observer	All	1	1.033	36.11	11.11	11.107	1.108
Mass	Observer	ID	1	0.709	41.67	0.00	12.000	0.545
Mass	Observer	W	1	1.170	25.00	16.67	10.920	1.430
Mass	Observer	J	1	1.222	41.67	16.67	10.400	1.349
Mass	Scalar-new	All	1	1.119	30.56	11.11	11.170	1.287
Mass	Scalar-new	ID	1	0.716	41.67	0.00	12.000	0.550
Mass	Scalar-new	W	1	1.353	16.67	8.33	11.468	1.599
Mass	Scalar-new	J	1	1.287	33.33	25.00	10.042	1.712
Mass	Phys-feat	All	1	1.095	33.33	11.11	11.169	1.268
Mass	Phys-feat	ID	1	0.715	41.67	0.00	12.000	0.550
Mass	Phys-feat	W	1	1.305	25.00	8.33	11.457	1.542
Mass	Phys-feat	J	1	1.265	33.33	25.00	10.049	1.712
Mass	Raw	All	1	1.140	30.56	11.11	11.168	1.325
Mass	Raw	ID	1	0.730	41.67	0.00	12.000	0.571
Mass	Raw	W	1	1.387	16.67	8.33	11.466	1.652
Mass	Raw	J	1	1.304	33.33	25.00	10.039	1.753
Mass	Scalar-old	All	5	1.097	33.33	11.11	11.195	1.264
Mass	Scalar-old	ID	5	0.716	41.67	0.00	12.000	0.550
Mass	Scalar-old	W	5	1.313	25.00	8.33	11.481	1.566
Mass	Scalar-old	J	5	1.262	33.33	25.00	10.103	1.677
Mass	Frozen	All	5	1.079	36.11	8.33	11.292	1.146
Mass	Frozen	ID	5	0.716	41.67	0.00	12.000	0.555
Mass	Frozen	W	5	1.280	25.00	8.33	11.472	1.522
Mass	Frozen	J	5	1.240	41.67	16.67	10.403	1.362
Lag	Feedback	All	1	0.576	83.33	0.00	12.000	0.410
Lag	Feedback	ID	1	0.156	100.00	0.00	12.000	0.094
Lag	Feedback	W	1	0.650	83.33	0.00	12.000	0.474
Lag	Feedback	J	1	0.922	66.67	0.00	12.000	0.661
Lag	Observer	All	1	0.409	94.44	0.00	12.000	0.159
Lag	Observer	ID	1	0.127	100.00	0.00	12.000	0.056
Lag	Observer	W	1	0.395	100.00	0.00	12.000	0.097
Lag	Observer	J	1	0.705	83.33	0.00	12.000	0.326
Lag	Scalar-new	All	1	0.520	91.67	0.00	12.000	0.358
Lag	Scalar-new	ID	1	0.145	100.00	0.00	12.000	0.080
Lag	Scalar-new	W	1	0.557	91.67	0.00	12.000	0.409
Lag	Scalar-new	J	1	0.856	83.33	0.00	12.000	0.585
Lag	Phys-feat	All	1	0.518	86.11	0.00	12.000	0.321
Lag	Phys-feat	ID	1	0.144	100.00	0.00	12.000	0.080
Lag	Phys-feat	W	1	0.549	83.33	0.00	12.000	0.338
Lag	Phys-feat	J	1	0.861	75.00	0.00	12.000	0.544
Lag	Raw	All	1	0.552	86.11	0.00	12.000	0.400
Lag	Raw	ID	1	0.166	100.00	0.00	12.000	0.103
Lag	Raw	W	1	0.606	83.33	0.00	12.000	0.509
Lag	Raw	J	1	0.883	75.00	0.00	12.000	0.588
Lag	Scalar-old	All	5	0.526	88.33	0.00	12.000	0.360
Lag	Scalar-old	ID	5	0.146	100.00	0.00	12.000	0.080
Lag	Scalar-old	W	5	0.568	86.67	0.00	12.000	0.420
Lag	Scalar-old	J	5	0.865	78.33	0.00	12.000	0.581
Lag	Frozen	All	5	0.506	87.78	0.00	12.000	0.346
Lag	Frozen	ID	5	0.154	100.00	0.00	12.000	0.092

Table S23 continued

Regime	Controller	Tier	n	RMSE	Success	Contact	Time	Terminal
Lag	Frozen	W	5	0.541	85.00	0.00	12.000	0.374
Lag	Frozen	J	5	0.823	78.33	0.00	12.000	0.571

Table S24: All 48 registered exploratory contrasts; unadjusted 95% intervals. Differences follow the stated comparison.

Regime	Comparison	Metric	Δ	95% interval
Mass	Frozen – Feedback	RMSE (m)	-0.0420	[-0.0739, -0.0147]
Mass	Frozen – Feedback	Success (pp)	+5.5556	[+0.0000, +13.3333]
Mass	Frozen – Feedback	Contact (pp)	-2.7778	[-8.8235, +0.0000]
Mass	Frozen – Feedback	Common RMSE (m)	-0.0395	[-0.0709, -0.0133]
Mass	Frozen – Observer	RMSE (m)	+0.0452	[+0.0003, +0.0957]
Mass	Frozen – Observer	Success (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Observer	Contact (pp)	-2.7778	[-7.8947, +0.0000]
Mass	Frozen – Observer	Common RMSE (m)	-0.0012	[-0.0294, +0.0303]
Mass	Frozen – Scalar	RMSE (m)	-0.0400	[-0.0691, -0.0144]
Mass	Frozen – Scalar	Success (pp)	+5.5556	[+0.0000, +13.3333]
Mass	Frozen – Scalar	Contact (pp)	-2.7778	[-8.8235, +0.0000]
Mass	Frozen – Scalar	Common RMSE (m)	-0.0372	[-0.0657, -0.0125]
Mass	Frozen – Scalar-old	RMSE (m)	-0.0184	[-0.0402, -0.0009]
Mass	Frozen – Scalar-old	Success (pp)	+2.7778	[+0.0000, +9.3750]
Mass	Frozen – Scalar-old	Contact (pp)	-2.7778	[-8.8235, +0.0000]
Mass	Frozen – Scalar-old	Common RMSE (m)	-0.0136	[-0.0341, +0.0002]
Mass	Frozen – Phys-feat	RMSE (m)	-0.0164	[-0.0275, -0.0067]
Mass	Frozen – Phys-feat	Success (pp)	+2.7778	[+0.0000, +9.3750]
Mass	Frozen – Phys-feat	Contact (pp)	-2.7778	[-8.8235, +0.0000]
Mass	Frozen – Phys-feat	Common RMSE (m)	-0.0155	[-0.0252, -0.0069]
Mass	Frozen – Raw	RMSE (m)	-0.0618	[-0.1052, -0.0270]
Mass	Frozen – Raw	Success (pp)	+5.5556	[+0.0000, +13.3333]
Mass	Frozen – Raw	Contact (pp)	-2.7778	[-8.8235, +0.0000]
Mass	Frozen – Raw	Common RMSE (m)	-0.0594	[-0.1023, -0.0261]
Lag	Frozen – Feedback	RMSE (m)	-0.0699	[-0.1043, -0.0379]
Lag	Frozen – Feedback	Success (pp)	+4.4444	[+0.0000, +10.9677]
Lag	Frozen – Feedback	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Feedback	Common RMSE (m)	-0.0699	[-0.1043, -0.0379]
Lag	Frozen – Observer	RMSE (m)	+0.0969	[+0.0562, +0.1389]
Lag	Frozen – Observer	Success (pp)	-6.6667	[-14.3750, +0.0000]
Lag	Frozen – Observer	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Observer	Common RMSE (m)	+0.0969	[+0.0562, +0.1389]
Lag	Frozen – Scalar	RMSE (m)	-0.0136	[-0.0367, +0.0117]
Lag	Frozen – Scalar	Success (pp)	-3.8889	[-11.0345, +0.0000]
Lag	Frozen – Scalar	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Scalar	Common RMSE (m)	-0.0136	[-0.0367, +0.0117]
Lag	Frozen – Scalar-old	RMSE (m)	-0.0200	[-0.0452, +0.0071]
Lag	Frozen – Scalar-old	Success (pp)	-0.5556	[-3.2432, +1.4286]
Lag	Frozen – Scalar-old	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Scalar-old	Common RMSE (m)	-0.0200	[-0.0452, +0.0071]
Lag	Frozen – Phys-feat	RMSE (m)	-0.0122	[-0.0333, +0.0063]
Lag	Frozen – Phys-feat	Success (pp)	+1.6667	[+0.0000, +5.6250]
Lag	Frozen – Phys-feat	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Phys-feat	Common RMSE (m)	-0.0122	[-0.0333, +0.0063]
Lag	Frozen – Raw	RMSE (m)	-0.0460	[-0.0942, -0.0030]
Lag	Frozen – Raw	Success (pp)	+1.6667	[+0.0000, +5.6250]
Lag	Frozen – Raw	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Raw	Common RMSE (m)	-0.0460	[-0.0942, -0.0030]

S8 Earlier Teacher, Readout, and Compensation Study

This earlier study is retained to distinguish the original teacher-based motivation from the final empirical claim. Its confirmation was independently locked before its own new trajectory realizations. It uses four-candidate final-goal ranking and waypoint-segment feedback-compensation metrics, not the later nine-candidate MPC protocol. The original tables below are preserved; their source text is included verbatim in the portable bundle. These results do not establish privileged-teacher necessity or an independent aerodynamic validation.

S8.1 Earlier Development Results

S8.1.1 Estimation versus Prediction

Table S25 shows that wind estimation and ordinary trajectory prediction rank methods differently. Wind-only estimation is competitive without a dynamics objective. PA-JEPA does not establish teacher necessity. The same-input observer’s offline ID wind RMSE is 1.825 m/s; this is a comparator, not a calibrated ceiling.

Table S25: ID development five-seed means. Wind uses the retained head; position is 30-step trajectory RMSE.

Method	Wind (m/s)	Position (m)
PA-JEPA	2.197	0.297
Direct wind JEPA	1.812	0.267
Supervised dynamics	–	0.219
Supervised + wind	1.608	0.229
Wind-only	1.647	–

S8.1.2 Changed Actions and Readout Choice

Original direct-wind JEPA obtains 0.658 m branch prediction RMSE, versus 1.017 m for supervised dynamics. But untuned physical response models obtain 0.484 m (0.3 s) and 0.415 m (1 s). Outperforming a learned comparator alone therefore does not establish additional capability.

The common residual readout changes this comparison (Fig. S2). Direct-wind JEPA reaches 0.380 m and 91.13% action accuracy; raw-history regression reaches 0.390 m and 89.40%. Their position difference is -0.0105 m with exploratory 95% interval $[-0.0274, 0.0053]$ m, which does not establish clear superiority over raw observations. Supervised dynamics has the highest mean residual-readout accuracy, 93.25%; PA-JEPA reaches 88.87%. All models and physical settings are retained in the results.

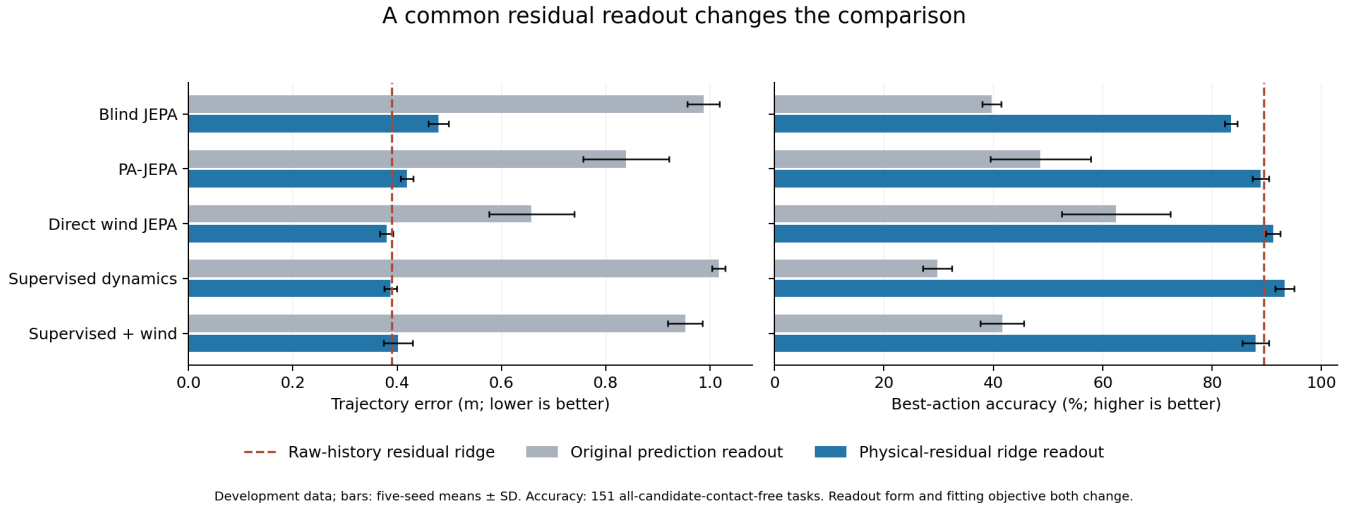


Figure S2: The common residual readout improves multiple frozen representations, while raw-history regression remains competitive. Error bars are seed standard deviations, not confidence intervals. Accuracy is conditional on all candidates being contact-free.

S8.1.3 Feedback Compensation

Table S26 reports pre-contact cross-track error. PA-JEPA improves over no compensation; its wind-extrapolation difference is -0.163 m, with exploratory 95% interval $[-0.210, -0.120]$ m. The observer remains numerically strongest in all tiers. PA-JEPA minus observer on wind extrapolation is $+0.139$ m, interval $[0.001, 0.393]$ m. Compensation benefit does not establish that a world-model objective was necessary.

Table S26: Development cross-track error (m). Interpret with contact rates and truncated flight time.

Source	ID	Wind extrap.	Joint extrap.
None	0.268	0.936	1.086
True-wind reference	0.218	0.671	0.865
Observer	0.212	0.634	0.801
PA-JEPA	0.220	0.773	0.858
Direct wind JEPA	0.221	0.773	0.850
Wind-only	0.223	0.790	0.896

ID contact rates are 2/48 throughout. Wind extrapolation has 2/48 except true wind, with 3/48. Joint extrapolation has 15% contact for no compensation and wind-only, and 12.5% for PA-JEPA and observer. These small counts do not establish safety superiority. Shared per-component 15 m/s clipping means true wind can have nonzero extrapolation estimation error.

S8.1.4 Consistency and Outstanding Validation

A ten-second nominal uniform-wind replay compares Torch with PhysX applied to the rigid asset. Across 200 positions, maximum discrepancy is 0.0234 m and last-record discrepancy is 0.0030 m. Both use the same force stack: this checks integration and coordinates, not independent aerodynamic validity. The complete trajectory was saved before an application-teardown timeout, which is retained in the record.

The complete development stress matrix motivated independent confirmation, reported next. The low-gain development contact advantage was also examined by replaying all 48 tasks for three sources at seed 0; common-time error comparisons showed that truncation alone did not explain the error ordering. Sensor noise is outside the locked confirmation scope; real-flight experiments remain unavailable.

S8.2 Independent Confirmation

Table S27: All eight locked primary comparisons. Differences are left minus right. Intervals are 99.375% familywise-adjusted; contact is in percentage points (pp).

Comparison	Difference	Adjusted interval
Residual direct – blind (m)	-0.0863	$[-0.1224, -0.0543]$
Residual direct – raw history (m)	-0.0111	$[-0.0309, +0.0089]$
Original PA-JEPA – direct (m)	+0.1855	$[+0.0568, +0.3143]$
Direct residual – original (m)	-0.3153	$[-0.4397, -0.2399]$
Low-gain wind-only – observer contact (pp)	+1.0417	$[-3.9122, +6.2341]$
Low-gain wind-only – observer error (m)	+0.1704	$[+0.0959, +0.2745]$
Low-gain PA-JEPA – wind-only contact (pp)	+0.6250	$[+0.0000, +3.3248]$
Nominal PA-JEPA – no compensation error (m)	-0.1713	$[-0.2440, -0.1101]$

Table S28: Complete confirmation stress matrix: mean pre-contact cross-track error (m) / contact (%). Learned results average five seeds; other sources are deterministic; task counts per source are 96/96/80 for ID/W/J. W: wind extrapolation; J: joint extrapolation.

Setting	Tier	None	True	Observer	PA-JEPA	Direct	Sup.+wind	Wind-only
Nominal	ID	0.270/3.12	0.225/1.04	0.216/1.04	0.228/2.08	0.226/1.67	0.228/2.08	0.228/2.08
Nominal	W	0.719/5.21	0.523/6.25	0.475/4.17	0.547/4.79	0.547/3.96	0.555/4.17	0.556/4.17
Nominal	J	0.862/12.50	0.685/7.50	0.663/3.75	0.672/8.50	0.675/7.50	0.680/8.50	0.684/9.00
Mass 1.4×	ID	0.279/4.17	0.223/1.04	0.216/2.08	0.251/2.29	0.244/2.29	0.241/2.08	0.242/2.08
Mass 1.4×	W	0.654/19.79	0.423/23.96	0.389/25.00	0.512/19.58	0.499/20.42	0.498/20.42	0.504/20.00
Mass 1.4×	J	0.751/22.50	0.523/23.75	0.510/23.75	0.615/23.75	0.600/24.50	0.598/25.25	0.609/25.00
Lag 3×	ID	0.266/2.08	0.217/1.04	0.211/0.00	0.222/0.83	0.221/1.04	0.222/0.00	0.222/0.00
Lag 3×	W	0.725/5.21	0.531/6.25	0.477/4.17	0.559/4.58	0.557/4.58	0.563/4.38	0.565/4.38
Lag 3×	J	0.869/11.25	0.700/8.75	0.677/5.00	0.689/8.75	0.694/8.25	0.699/8.75	0.703/8.75
Gain 0.6×	ID	0.355/2.08	0.234/1.04	0.215/0.00	0.238/0.00	0.237/0.00	0.234/0.00	0.235/0.00
Gain 0.6×	W	1.155/9.38	0.635/7.29	0.517/5.21	0.662/6.88	0.665/6.04	0.684/6.46	0.688/6.25
Gain 0.6×	J	1.155/3.75	0.723/13.75	0.629/8.75	0.775/10.25	0.778/10.00	0.779/10.00	0.781/9.25
Gain 1.4×	ID	0.241/2.08	0.215/2.08	0.210/2.08	0.215/2.08	0.214/2.08	0.217/2.08	0.217/2.08
Gain 1.4×	W	0.601/5.21	0.510/3.12	0.478/3.12	0.507/3.33	0.507/3.33	0.517/3.33	0.516/3.33
Gain 1.4×	J	0.738/11.25	0.651/5.00	0.638/3.75	0.644/5.25	0.642/6.00	0.646/5.00	0.647/5.00

After development, we froze models, residual coefficients, preprocessing, controllers, data-generation code and eight primary comparisons before collecting new trajectories. Two independent random seeds generate command, goal and vehicle realizations from the existing scene/field pool. This tests new realizations, not new procedural geometry families. No model or readout is refitted on confirmation data.

The prediction set has 736 shared-prefix tasks (2,944 candidate trajectories) across seen/held-out geometry and training/extrapolation wind strata. Of these, 664 have valid prefixes and 574 have all four candidates contact-free. Each closed-loop condition is tested on 96 ID, 96 wind-extrapolation and 80 joint-extrapolation tasks across the two batches. All five plant/controller settings and all seven compensation sources are retained.

The eight primary comparisons use 10,000 paired scene/seed bootstrap draws. Both batches remain together within each scene cluster; pooled prediction resamples within seen/held-out geometry strata. We report two-sided 99.375% intervals (Bonferroni familywise level 0.05 across eight comparisons). Prediction comparisons use 30 scene clusters; the prespecified wind-extrapolation control comparisons use 20. A null result does not establish equivalence.

S8.2.1 Prediction and Readout

Table S29 shows all original, residual and physical predictors. Direct wind supervision improves the residual comparison against blind JEPA, but superiority to raw-history regression is not established. PA-JEPA's original predictor is worse than the direct predictor in the prespecified comparison. Residual decoding improves direct prediction, but its combined changes to readout form and training objective prevent attributing this difference solely to the physical prior.

Table S29: Confirmation prediction. RMSE uses eligible candidate positions; accuracy uses 574 all-valid tasks. Means over five learned seeds; raw/physical controls deterministic.

Predictor	RMSE (m)	Accuracy (%)
<i>Original</i>		
Blind	1.009	35.99
PA-JEPA	0.865	46.38
Direct	0.679	63.34
Supervised	1.044	30.73
Supervised + wind	0.976	41.32
PA-JEPA, no depth	0.796	57.32
<i>Residual</i>		
Blind	0.450	85.30
PA-JEPA	0.404	86.48
Direct	0.364	89.97
Supervised	0.379	89.72
Supervised + wind	0.377	87.80
Wind-only	0.410	86.34
Raw history	0.375	87.98
<i>Physical</i>		
Response 1 s	0.405	84.67
Ideal command	0.866	79.09
Response 0.3 s	0.472	88.33
Velocity hold	0.948	17.42

S8.2.2 Control and Contact

Under nominal feedback and wind extrapolation, PA-JEPA reduces mean pre-contact error from 0.719 to 0.547 m compared with no compensation. The observer obtains 0.475 m. Under low feedback gain, the observer obtains 0.517 m and 5.21% contact, versus wind-only at 0.688 m and 6.25% contact. The development contact advantage is therefore not confirmed. Its locked wind-only-observer contact interval includes zero, while wind-only’s larger tracking error is confirmed (Table S27).

Descriptively, the low-gain observer and wind-only achieve truncated first-contact times of 11.753 and 11.717 s, and mean waypoint counts of 3.146 and 2.913, respectively. These secondary means are not additional adjusted superiority tests.

PA-JEPA does not establish a contact advantage over wind-only. These outcomes support compensation in the modeled setting but not the necessity of a world-model objective or privileged teacher. Table S28 retains all settings and tiers. Tracking error remains conditional on valid flight; contact, truncated duration and progress must guide interpretation.

S9 Adapter Structure, Response Probes, and Cost Differences

These post-hoc development diagnostics reuse the existing paired-calibration data and controller-state snapshots. They introduce no independent confirmation, new simulator setting or new full-flight experiment. The history probe fits small response heads on the prior calibration campaign; the later choice and cost audits freeze those heads. All methods and failed screens are retained. The elementary algebra below is an explanation of this deployed adapter and cost, not a new control algorithm or a causal decomposition of every prior failure.

S9.1 What shared context can change in a fixed affine head

After absorbing fixed normalization and the intercept, the deployed readout can be written

$$\text{vec}(\hat{d}^a) = W_z z_t + W_U \text{vec}(U^a) + (I + W_d) \text{vec}(d_{\text{nom}}^a) + b.$$

This expression uses flattened yaw-frame displacements and command sequences. For candidates sharing a history, subtracting two predictions removes $W_z z_t + b$; the common world-frame transform preserves this cancellation. Known heading, observed velocity, clipped candidate commands and nominal physical features may still affect response; fitting another head can also change W_U, W_d . Shared context can change squared-cost rankings through the common forecast. This restriction applies to the fixed affine adapter, not the original recurrent world-model forward pass or all possible learned representations. It does not imply that all history is useless.

D14 audits all 28 selected paired-data physical/raw/frozen heads under both trajectory and decision objectives. There are 3,024 state/head records from 216 unique states, of which 2,856 records from 204 states are all-nine-valid. Command/physical coefficient rows alone reconstruct saved centered response MSE within 2.284×10^{-10} m². Synthetic shared-feature changes and full-affine reconstruction agree within 4.2×10^{-15} m. These are algebra checks, not feasible encoder-manifold interventions; the audit does not reload encoder checkpoints. Physical/raw/frozen feature counts are 90/402/338, with 0/192/128 candidate-shared history columns.

Table S30: All fixed-head groups. MSE in m² on all-nine-valid states (98 mass, 106 lag); Frozen averages five heads. These total MSE values use a different denominator from pre-contact position RMSE.

Regime	Features	Objective	n	Common	Response	Total	Regret
Mass	Phys	trajectory	1	0.0450974	0.0024591	0.0475566	0.0149184
Mass	Phys	decision	1	0.3480035	0.0062465	0.3542500	0.0132800
Mass	Raw	trajectory	1	0.0297949	0.0013621	0.0311571	0.0173070
Mass	Raw	decision	1	0.6608283	0.0071237	0.6679519	0.0151003
Mass	Frozen	trajectory	5	0.0247666	0.0012631	0.0260296	0.0149934
Mass	Frozen	decision	5	0.6906387	0.0072312	0.6978699	0.0141948
Lag	Phys	trajectory	1	0.0699446	0.0058363	0.0757810	0.0189956
Lag	Phys	decision	1	0.1899074	0.0111874	0.2010948	0.0148144
Lag	Raw	trajectory	1	0.0530539	0.0021089	0.0551628	0.0173322
Lag	Raw	decision	1	1.1952765	0.0199216	1.2151981	0.0183565
Lag	Frozen	trajectory	5	0.0445302	0.0019538	0.0464840	0.0188928
Lag	Frozen	decision	5	1.4969430	0.0145862	1.5115293	0.0198378

S9.2 Oracle response capacity is not a deployable gain

D15 uses the unchanged 24 FIT and eight validation parents per regime, with three snapshots per parent. Of 192 states, 179 have all nine valid branches. A shared four-column affine command basis includes centered world-XY and heading-XY commands. A separate per-state affine oracle leaves less residual: it removes 98.76–99.30% of the shared oracle’s residual, but that shared residual is only 0.44–1.01% of centered true trajectory energy. The actual 28 heads lie in the known-heading affine class within 2.193×10^{-15} m. Global basis condition numbers are 272–1084; the nonredundant per-state command basis has condition below 1.726. Both oracles use their own split’s future labels. Their residuals are class floors on these finite candidates, not held-out generalization, physical Jacobians or achievable robot gains.

Table S31: Own-split oracle capacity. Energies/residuals in m^2 ; final column is removed fraction of shared residual, not of all response energy.

Regime	Split	States	True energy	Shared residual	State residual	Removed (%)
Mass	FIT	62	0.0540306	0.0003447	0.0000038	98.91
Mass	Val	24	0.0552201	0.0003774	0.0000029	99.24
Lag	FIT	69	0.0706726	0.0007116	0.0000050	99.30
Lag	Val	24	0.0763604	0.0003325	0.0000041	98.76

D16 then projects fitted-head error onto the corresponding class floor and excess within the class (heading-only physical features versus world-plus-heading raw/frozen). The identity is numerically verified; the floor fraction is not a causal share of control failure. All 24 head-group/split combinations follow.

Table S32: All fitted-head/split groups. Response MSE = own-split class floor + within-class excess, all in m^2 . Frozen averages five heads.

Regime	Split	Features	Objective	Response	Floor	Excess	Floor (%)
Mass	FIT	Phys	trajectory	0.0024019	0.0003831	0.0020188	15.95
Mass	Val	Phys	trajectory	0.0027018	0.0005649	0.0021369	20.91
Mass	FIT	Phys	decision	0.0055743	0.0003831	0.0051913	6.87
Mass	Val	Phys	decision	0.0058278	0.0005649	0.0052629	9.69
Mass	FIT	Raw	trajectory	0.0003794	0.0003447	0.0000348	90.84
Mass	Val	Raw	trajectory	0.0005750	0.0003774	0.0001976	65.64
Mass	FIT	Raw	decision	0.0065288	0.0003447	0.0061841	5.28
Mass	Val	Raw	decision	0.0067216	0.0003774	0.0063442	5.62
Mass	FIT	Frozen	trajectory	0.0004187	0.0003447	0.0000741	82.31
Mass	Val	Frozen	trajectory	0.0006245	0.0003774	0.0002470	60.44
Mass	FIT	Frozen	decision	0.0065795	0.0003447	0.0062348	5.24
Mass	Val	Frozen	decision	0.0067739	0.0003774	0.0063965	5.57
Lag	FIT	Phys	trajectory	0.0062142	0.0007284	0.0054859	11.72
Lag	Val	Phys	trajectory	0.0073114	0.0005960	0.0067154	8.15
Lag	FIT	Phys	decision	0.0109206	0.0007284	0.0101923	6.67
Lag	Val	Phys	decision	0.0116336	0.0005960	0.0110376	5.12
Lag	FIT	Raw	trajectory	0.0007264	0.0007116	0.0000148	97.96
Lag	Val	Raw	trajectory	0.0006772	0.0003325	0.0003447	49.10
Lag	FIT	Raw	decision	0.0196385	0.0007116	0.0189269	3.62
Lag	Val	Raw	decision	0.0210032	0.0003325	0.0206707	1.58
Lag	FIT	Frozen	trajectory	0.0008247	0.0007116	0.0001131	86.29
Lag	Val	Frozen	trajectory	0.0009054	0.0003325	0.0005729	36.72
Lag	FIT	Frozen	decision	0.0142089	0.0007116	0.0134972	5.01
Lag	Val	Frozen	decision	0.0148810	0.0003325	0.0145485	2.23

S9.3 Does motion history add response information?

D17 compares nominal physics, Shared, Current, Current+Q8, Current+Obs and Current+H8. The 21 Current features contain the latest 12-dimensional observed state, previous four-dimensional command, center-candidate XYZ command and candidate-mean XY command. H8 compresses 180 past-state/command differences into eight whitened principal components. Q8 uses eight whitened components of quadratic current features and matches H8's 16 added interaction columns; this does not prove equal function-class complexity. Obs supplies two clipped residual estimates from the existing nominal-parameter observer and the same 12 states; it is a feature control, not a new observer closed-loop evaluation. No depth, pretrained encoder, true wind or randomized plant parameters enter this probe.

Every method uses centered nominal physics at $\tau = 0.3$ s. Shared uses the four command columns; Current adds current-feature interactions with centered world-XY commands. Parents have equal fitting weight. Six folds group FIT parents by parent ID modulo six; normalization/PCA are refit inside each fold. Out-of-fold parent-macro response MSE selects $\lambda \in \{10^{-4}, 10^{-2}, 1\}$ (ties favor larger regularization), then the model is refit on all eligible FIT states. All 180 CV fits, 10 final heads and selections are retained; final weights/selection were saved before validation scoring.

The original split stays fixed. Mass has 62 FIT states from 21/24 parents, lag 69 from 24/24; three mass parents have no eligible state. Both validation sets have 24 states from eight parents. Validation was used in earlier development and is not a new confirmation set.

The score first averages candidate/time squared XYZ response error within each state, then states within a parent, then parents equally. It differs from the later controller-snapshot state average. The synthetic history-dependent response is recovered at $\text{MSE } 5.923 \times 10^{-18}$, while Current has 4.983×10^{-4} ; this verifies an intended code path, not empirical usefulness. Validation-label perturbation does not change fits, and forbidden-input perturbation does not change features. An independent script rebuilds all 10 final fits and all 1,152 state/method records exactly.

Table S33: All response probes and FIT-only selections. CV/FIT/validation values are parent-macro response MSE (m^2). No encoder is fitted or evaluated here.

Regime	Response	Columns	λ	FIT CV	FIT	Validation
Mass	Nominal	—	—	—	0.0056820	0.0058169
Mass	Shared	4	0.0001	0.0003966	0.0003598	0.0005673
Mass	Current	46	1	0.0014643	0.0007394	0.0007199
Mass	Current+Q8	62	1	0.0202877	0.0007194	0.0007125
Mass	Current+Obs	50	1	0.0013129	0.0007271	0.0007138
Mass	Current+H8	62	1	0.0014917	0.0007242	0.0007130
Lag	Nominal	—	—	—	0.0024364	0.0019012
Lag	Shared	4	0.01	0.0008210	0.0007566	0.0006747
Lag	Current	46	1	0.0006280	0.0004482	0.0003965
Lag	Current+Q8	62	1	0.0006154	0.0004247	0.0004047
Lag	Current+Obs	50	1	0.0006163	0.0004287	0.0003969
Lag	Current+H8	62	1	0.0006338	0.0004245	0.0003829

The lag history-support screen required at least 10% lower mean validation error and improvement on at least six of eight parents versus each of Shared, Current and Q8; the additional observer comparison used the same criterion. All fail. Current alone accounts for most lag mean improvement over Shared; mass favors Shared. The mass Q8 CV outlier is retained: fold 4 MSE is 0.138113, with parent 4 at 0.410905, yielding pooled CV 0.020288. It is finite and reproducible; its cause was not isolated, and no parent, rank or lambda was changed after seeing it. This fixed small-data probe does not exclude other histories or representations.

Table S34: All lag Current+H8 practical screens. Positive relative reduction favors H8. These are development screening rules, not significance tests.

Comparator	MSE reduction (%)	Parents improved	Pass
Shared	43.25	5/8	No
Current	3.42	4/8	No
Current+Q8	5.38	3/8	No
Current+Obs	3.52	4/8	No

S9.4 Action choice with a fixed deployable common forecast

D18 replaces only the centered candidate response while holding the same full predictor’s nine-candidate mean fixed. The four complete trajectory-objective physical-feature/raw heads come from D9; the four Shared/Current response heads come from D17. The predicted mean is never replaced by future truth. Native uses the full head’s original response and exactly reproduces the old forecast. The full heads previously used validation-regret selection; response heads used FIT CV on the same calibration campaign. Parameter counts and training/selection procedures therefore differ, even though this substitution performs no new fitting. H8/Q8 are not promoted after their failed screen.

All 216 prior controller-state snapshots and 1,296 state/configuration records remain. Action metrics use 98 mass/106 lag all-nine-valid states; selected contact uses 99/108 initially alive prefixes. Position RMSE uses 26,711/29,144 individually valid candidate positions. Predictions are invariant to future-truth/reference/validity/depth-array perturbation. The predicted common trajectory changes by at most 3.553×10^{-15} m. Independent acceptance replays all frozen

forecasts exactly and recomputes every row and aggregate; maximum metric difference is 8.527×10^{-14} . These short branches do not establish full-flight safety or observer superiority.

Table S35: All 12 fixed-common forecast combinations. RMSE in m; centered response MSE in m^2 . Contact is selected-branch count/live-prefix count.

Regime	Mean	Response	Valid	RMSE	Resp. MSE	Acc. (%)	Regret	Contact
Mass	Phys	Native	98	0.22186	0.0024591	70.41	0.0149184	0/99
Mass	Phys	Shared	98	0.21954	0.0013632	68.37	0.0162791	0/99
Mass	Phys	Current	98	0.22021	0.0016532	66.33	0.0161307	0/99
Mass	Raw	Native	98	0.18266	0.0013621	68.37	0.0173070	0/99
Mass	Raw	Shared	98	0.18266	0.0013632	68.37	0.0173070	0/99
Mass	Raw	Current	98	0.18346	0.0016532	68.37	0.0177105	0/99
Lag	Phys	Native	106	0.27807	0.0058363	57.55	0.0189956	0/108
Lag	Phys	Shared	106	0.27143	0.0021059	61.32	0.0209999	0/108
Lag	Phys	Current	106	0.27032	0.0014812	60.38	0.0204341	0/108
Lag	Raw	Native	106	0.23367	0.0021089	66.04	0.0173322	0/108
Lag	Raw	Shared	106	0.23366	0.0021059	66.04	0.0173322	0/108
Lag	Raw	Current	106	0.23234	0.0014812	66.04	0.0170700	0/108

Table S36: All 36 development tier combinations, 36 snapshots per row; no tier selected for a new claim. Complete common/total errors and valid-position counts remain in D18.

Regime	Mean	Response	Tier	Valid	RMSE	Resp. MSE	Acc. (%)	Regret	Contact
Mass	Phys	Native	ID	36	0.10138	0.0022007	88.89	0.0006780	0/36
Mass	Phys	Native	W	34	0.30799	0.0024986	61.76	0.0279567	0/34
Mass	Phys	Native	J	28	0.20988	0.0027435	57.14	0.0173951	0/29
Mass	Phys	Shared	ID	36	0.09159	0.0003104	86.11	0.0018041	0/36
Mass	Phys	Shared	W	34	0.30814	0.0025915	55.88	0.0312133	0/34
Mass	Phys	Shared	J	28	0.20684	0.0012253	60.71	0.0167553	0/29
Mass	Phys	Current	ID	36	0.09400	0.0007575	86.11	0.0018041	0/36
Mass	Phys	Current	W	34	0.30824	0.0026527	52.94	0.0307688	0/34
Mass	Phys	Current	J	28	0.20776	0.0015911	57.14	0.0167758	0/29
Mass	Raw	Native	ID	36	0.08466	0.0003104	77.78	0.0077217	0/36
Mass	Raw	Native	W	34	0.23598	0.0025743	58.82	0.0309475	0/34
Mass	Raw	Native	J	28	0.19932	0.0012424	67.86	0.0130676	0/29
Mass	Raw	Shared	ID	36	0.08466	0.0003104	77.78	0.0077217	0/36
Mass	Raw	Shared	W	34	0.23602	0.0025915	58.82	0.0309475	0/34
Mass	Raw	Shared	J	28	0.19928	0.0012253	67.86	0.0130676	0/29
Mass	Raw	Current	ID	36	0.08726	0.0007575	77.78	0.0054800	0/36
Mass	Raw	Current	W	34	0.23615	0.0026527	61.76	0.0333767	0/34
Mass	Raw	Current	J	28	0.20022	0.0015911	64.29	0.0144122	0/29
Lag	Phys	Native	ID	36	0.11195	0.0066065	83.33	0.0037640	0/36
Lag	Phys	Native	W	35	0.31534	0.0048090	45.71	0.0283048	0/36
Lag	Phys	Native	J	35	0.34650	0.0060714	42.86	0.0253532	0/36
Lag	Phys	Shared	ID	36	0.07957	0.0004038	86.11	0.0026225	0/36
Lag	Phys	Shared	W	35	0.31168	0.0025794	48.57	0.0285908	0/36
Lag	Phys	Shared	J	35	0.34296	0.0033832	48.57	0.0323116	0/36
Lag	Phys	Current	ID	36	0.07978	0.0004380	86.11	0.0026225	0/36
Lag	Phys	Current	W	35	0.31063	0.0019001	48.57	0.0285908	0/36
Lag	Phys	Current	J	35	0.34121	0.0021354	45.71	0.0305978	0/36
Lag	Raw	Native	ID	36	0.05882	0.0004084	86.11	0.0029286	0/36
Lag	Raw	Native	W	35	0.21555	0.0025811	60.00	0.0139076	0/36
Lag	Raw	Native	J	35	0.33759	0.0033858	51.43	0.0355720	0/36
Lag	Raw	Shared	ID	36	0.05878	0.0004038	86.11	0.0029286	0/36
Lag	Raw	Shared	W	35	0.21554	0.0025794	60.00	0.0139076	0/36
Lag	Raw	Shared	J	35	0.33758	0.0033832	51.43	0.0355720	0/36
Lag	Raw	Current	ID	36	0.05907	0.0004380	86.11	0.0025726	0/36
Lag	Raw	Current	W	35	0.21402	0.0019001	60.00	0.0144398	0/36
Lag	Raw	Current	J	35	0.33575	0.0021354	51.43	0.0346118	0/36

The prespecified descriptive action screen requires Current to reduce regret by at least 10% relative to both Native and Shared, with no lower accuracy or higher selected-contact rate. All four regime/common-source screens fail, as do the eight separate comparisons below. Raw Native/Shared choose identically on these snapshots; they are not independent supporting tests. In lag, Current can reduce response MSE by 74.6% under the physical-feature common forecast while increasing regret by 7.57%. These failed screens do not establish a control remedy.

Table S37: Current versus every fixed comparator. Accuracy differences are percentage points; final counts are wrong-to-correct / correct-to-wrong. All contact-rate differences are zero.

Regime	Mean	Baseline	Δ regret	Reduction (%)	Δ acc.	Changed	To/from	Pass
Mass	Phys	Native	+0.0012124	-8.13	-4.08	8	1/5	No
Mass	Phys	Shared	-0.0001483	+0.91	-2.04	6	0/2	No
Mass	Raw	Native	+0.0004035	-2.33	+0.00	4	1/1	No
Mass	Raw	Shared	+0.0004035	-2.33	+0.00	4	1/1	No
Lag	Phys	Native	+0.0014385	-7.57	+2.83	15	6/3	No
Lag	Phys	Shared	-0.0005659	+2.69	-0.94	2	0/1	No
Lag	Raw	Native	-0.0002622	+1.51	+0.00	5	1/1	No
Lag	Raw	Shared	-0.0002622	+1.51	+0.00	5	1/1	No

S9.5 Exact accounting for candidate cost differences

For the following cost decomposition only, d^a denotes a deviation from the candidate mean, rather than the anchor-frame displacement in Figure 1. At a fixed state, write true and predicted trajectories as $p^a = \mu + d^a$ and $\hat{p}^a = \hat{\mu} + q^a$, where $\mu, \hat{\mu}$ are their respective means across nine candidates. Set $e_0 = \hat{\mu} - \mu$, $\epsilon^a = q^a - d^a$, and $\delta_a = \hat{J}(a) - J(a)$. Let $\langle x, y \rangle = K^{-1} \sum_k x_k^\top y_k$ and $\|x\|_K^2 = \langle x, x \rangle$. Direct expansion gives

$$\begin{aligned}
\delta_a - \delta_b = & \underbrace{2\langle e_0, d^a - d^b \rangle}_{A_{ab}: \text{common-error interaction}} \\
& + \underbrace{2\langle \hat{\mu} - \mu, \epsilon^a - \epsilon^b \rangle}_{B_{ab}: \text{response alignment}} \\
& + \underbrace{(\|q^a\|_K^2 - \|d^a\|_K^2) - (\|q^b\|_K^2 - \|d^b\|_K^2)}_{C_{ab}: \text{response energy}}. \tag{3}
\end{aligned}$$

The known command penalties cancel between forecast and truth errors, but remain in predicted/true action choices. Expanding the two squared tracking costs and subtracting cancels the candidate-common squared-mean term and yields Eq. 3; no dynamical assumption is needed. For selected $\hat{a}$ and true-best a^* , the exact relation is

$$J(\hat{a}) - J(a^*) = \hat{J}(\hat{a}) - \hat{J}(a^*) - (A_{\hat{a}a^*} + B_{\hat{a}a^*} + C_{\hat{a}a^*}).$$

The first difference on the right is nonpositive. Common error can affect ordering even when response is exact, while small response MSE alone does not fix alignment with the reference or the decision margin. Even average pair-cost error is not a guarantee about the selected action. These identities use branch truth for diagnosis and do not define an online oracle correction.

D19 checks all 36 unordered pairs per state/configuration. The eligible 44,064 records are repeated comparisons from 204 unique states (1,224 state/configuration records), not 44,064 independent samples. All 1,296 state/configuration records, including invalid states, remain archived. Direct cost differences agree with the decomposition within 2.698×10^{-14} ; a separate per-pair formula recomputes the terms within 4.441×10^{-16} . Original choices and regrets reproduce exactly. No fit, rollout, p-value or new screen is added.

Table S38: All pair-cost error groups. RMS of signed A/B/C and their sum; component RMS values do not add. Regret is the original action metric, not a new controller result.

Regime	Mean	Response	Pairs	RMS A	RMS B	RMS C	RMS sum	Regret
Mass	Phys	Native	3528	0.058520	0.105809	0.016780	0.100695	0.0149184
Mass	Phys	Shared	3528	0.058520	0.127689	0.008334	0.121200	0.0162791
Mass	Phys	Current	3528	0.058520	0.109614	0.010923	0.102852	0.0161307
Mass	Raw	Native	3528	0.061416	0.149035	0.008349	0.154486	0.0173070
Mass	Raw	Shared	3528	0.061416	0.149149	0.008334	0.154613	0.0173070
Mass	Raw	Current	3528	0.061416	0.127806	0.010923	0.133931	0.0177105
Lag	Phys	Native	3816	0.091132	0.091704	0.025719	0.122790	0.0189956
Lag	Phys	Shared	3816	0.091132	0.163151	0.012344	0.175468	0.0209999
Lag	Phys	Current	3816	0.091132	0.105016	0.010778	0.123017	0.0204341
Lag	Raw	Native	3816	0.078288	0.172811	0.012363	0.180626	0.0173322
Lag	Raw	Shared	3816	0.078288	0.173005	0.012344	0.180698	0.0173322
Lag	Raw	Current	3816	0.078288	0.113462	0.010778	0.124952	0.0170700

The terms can cancel or reinforce. Their magnitudes must not be reported as independent causal fractions. With $M_{ij} = \mathbb{E}[T_i T_j]$ for $T = (A, B, C)$, total mean squared pair error equals $\sum_i M_{ii} + 2 \sum_{i < j} M_{ij}$. The table above supplies $\sqrt{M_{ii}}$; the complete signed off-diagonal terms follow. Raw/Current in mass has smaller overall pair-cost RMS than Raw/Native but larger selected-action regret, underscoring the distinction between an average ranking loss and the cost of the actual choice.

Table S39: All signed second-moment cross terms. Together with squared component RMS in the preceding table, these specify the full symmetric matrix. They are not statistical effect estimates.

Regime	Mean	Response	M_{AB}	M_{AC}	M_{BC}
Mass	Phys	Native	-0.00242572	+0.00001421	+0.00003038
Mass	Phys	Shared	-0.00263287	-0.00001644	+0.00009484
Mass	Phys	Current	-0.00251041	-0.00002022	+0.00004034
Mass	Raw	Native	-0.00117972	-0.00001291	+0.00009905
Mass	Raw	Shared	-0.00118087	-0.00001443	+0.00010455
Mass	Raw	Current	-0.00116722	-0.00001693	+0.00004016
Lag	Phys	Native	-0.00122877	+0.00007209	+0.00000727
Lag	Phys	Shared	-0.00228381	-0.00002507	+0.00016570
Lag	Phys	Current	-0.00211702	+0.00000630	-0.00004749
Lag	Raw	Native	-0.00195742	-0.00000067	+0.00019833
Lag	Raw	Shared	-0.00197556	-0.00000051	+0.00019593
Lag	Raw	Current	-0.00172547	+0.00001286	-0.00004026

S9.6 Reproduction scope and interpretation

The portable bundle retains D14–D19 summaries, all per-state records, source manifests, response weights, choice forecast components, pair terms/costs, protocols and audit scripts, indexed by hash. It does not include every raw simulator dataset or original encoder checkpoint; replaying raw-input inference requires the corresponding repository artifacts identified in those manifests. Acceptance verifies 143 source hashes for D19, with no change to previous scientific outputs.

This sequence narrows the adapter’s capabilities and records unsuccessful remedies. It supports a scoped empirical prediction-to-decision boundary, but does not resolve method novelty, observer-controlled robot utility or independent aerodynamic validity.

S10 Observation Assumptions and Actuator-Form Transfer

D20 audits existing observed histories; D21–D23 evaluate a single alternative motor state equation with all original controllers frozen. These are distinct development checks. The motor-form protocol was fixed before implementing the alternative, but uses the same earlier development tasks and old calibration fits. Neither check adds an independent calibration campaign, a new algorithm, or real-flight evidence.

S10.1 Executed observation and force interfaces

Table S40 distinguishes the current executed benchmark from properties merely listed in configuration files. These statements concern this revision’s data/control paths, not every historical project experiment. All physical coefficients remain unmeasured for this vehicle. In particular, configured IMU noise does not imply that the predictor or observer receives noisy observations.

Table S40: Implementation scope of the evaluated revision. Configuration entries are not evidence that an effect was active.

Component	Executed behavior and limitation
State inputs	12 channels from clean simulated pose, velocity and angular rate; 20 Hz histories. The configured noisy IMU channels do not enter these predictors.
Observer	Corrected causal state-history inverse using nominal parameters. No true wind, randomized mass, actual rotor thrust or future state input.
Wind	Spatial bilinear lookup of a frozen planar u/v field. Time is unused; vertical wind is zero. Motion still encounters spatial variations.
Aerodynamics	Unidentified anisotropic quadratic body drag, linear rotor drag and angular drag. A horizontal wind can still induce a world-vertical force component.
Original actuator	First-order thrust state with nominal time constant 0.0333 s, existing randomization and the lag multiplier. It is not a rotor-speed state.
Randomization	The executed function varies mass, inertia, body/rotor drag, thrust scale and motor time constant. Other YAML ranges are not all applied.
Timing	20 Hz wind estimate averages six backward-difference inverses and clips to $[-15, 15]$ m/s; the next 50 Hz controller tick uses the latest estimate.

S10.2 Nominal residual inversion does not prove true-force recovery

Let b be body z expressed in the world frame and a_{aero} the actual aerodynamic acceleration. For synchronous, unclipped continuous-time quantities,

$$a + ge_z = (T_{\text{actual}}/m_{\text{actual}})b + a_{\text{aero}} \implies r = a_{xy} - \frac{a_z + g}{b_z}b_{xy} = a_{\text{aero},xy} - \frac{a_{\text{aero},z}}{b_z}b_{xy}.$$

The projection removes actual thrust; nominal mass also cancels from this specific residual. It can therefore support useful compensation without separately identifying mass and thrust. Vertical aerodynamic force, sample timing and attitude variation limit the interpretation. The implemented inverse uses $\hat{T} = m_0 \text{clip}((a_z + g)/\max(b_z, 0.2), 0, 4g)$ and $r = a_{xy} - (\hat{T}/m_0)b_{xy}$.

Writing $A = \frac{1}{2}\rho \text{mean}(C_d A_{xy})$ and $B = k_{\text{rot}}\hat{T}/(m_0g)$, the inverse solves $|m_0r| = As^2 + Bs$ for relative speed. With $w_{\text{inst}} = v_{xy} + sr/|r|$ and the same nominal forward drag D , it follows that $D(v_{xy} - w_{\text{inst}}, \hat{T}/(m_0g)) = r$. This is an inverse/forward consistency identity, not an error bound against actual wind or aerodynamic force. All saved histories satisfy it within 2.665×10^{-15} m/s²; a separate 512-sample synthetic projection check agrees within 1.243×10^{-14} and is not flight evidence.

D20 retains all 216 old snapshots and reports 207 initially alive separately (99 mass, 108 lag). Future branch survival never selects this sample. The observer uses 11 backward differences over 12 clean states at 0.05 s intervals and averages the final six instantaneous wind estimates. No attitude-denominator or inferred-thrust clamp activates over the 2,277 live-history intervals, including the final 1,242 used intervals. The final wind clamp affects seven lag states, one component each. All dead prefixes remain in the archive and in the All rows below; they are not flight-performance observations.

In Table S41, V_r is within-six residual variability RMS. G is the norm of the difference between the nominal residual induced by the averaged wind and the latest instantaneous residual, with latest velocity and inferred thrust factor held fixed. G_c additionally includes float32 conversion and the final wind clamp. These are differences between inferred signals, not true-force or deployed-compensation errors. The actual controller uses the next tick’s velocity and previous commanded thrust sum, which cannot be reconstructed from these snapshot files. The maximum live lag G_c is 1.162196 m/s²; a small average does not make every transient negligible.

Table S41: All 16 population/tier summaries. Repeated task times are not independent flights. Clips counts wind components; all acceleration columns are state means in m/s². Each state supplies 11 differences, of which six enter averaging.

Regime	Tier	Mask	n	Clips	$ r $	V_r	G	G_c
Mass	All	Live	99	0/198	1.07276	0.01323	0.01921	0.01921
Mass	All	All	108	0/216	1.20692	0.01212	0.01761	0.01761
Mass	ID	Live	36	0/72	0.37766	0.00683	0.00948	0.00948
Mass	ID	All	36	0/72	0.37766	0.00683	0.00948	0.00948
Mass	W	Live	34	0/68	1.70322	0.01538	0.02291	0.02291
Mass	W	All	36	0/72	1.76574	0.01453	0.02164	0.02164
Mass	J	Live	29	0/58	1.19647	0.01863	0.02694	0.02694
Mass	J	All	36	0/72	1.47735	0.01501	0.02170	0.02170
Lag	All	Live	108	7/216	1.60641	0.04204	0.05509	0.08374
Lag	All	All	108	7/216	1.60641	0.04204	0.05509	0.08374
Lag	ID	Live	36	0/72	0.51029	0.04351	0.05495	0.05495
Lag	ID	All	36	0/72	0.51029	0.04351	0.05495	0.05495
Lag	W	Live	36	3/72	2.28056	0.03783	0.05107	0.12142
Lag	W	All	36	3/72	2.28056	0.03783	0.05107	0.12142
Lag	J	Live	36	4/72	2.02838	0.04477	0.05927	0.07486
Lag	J	All	36	4/72	2.02838	0.04477	0.05927	0.07486

Separate scalar reconstruction of rotation, quadratic inversion and every per-state metric verifies all 35 source hashes, 20 output hashes, 216 rows and 16 groups. Maximum independent array/metric differences are $8.438 \times 10^{-14}/4.483 \times 10^{-15}$; the original observer’s float32 output agrees exactly. The audit does not fit or change a controller. Filtering and measurement timing are established issues: Tal and Karaman’s INDI system uses filtered IMU/motor-speed measurements and motion-capture latency propagation. Its richer sensing interface and hardware-specific parameters cannot be transplanted here without a defined sensor setting.

S10.3 One fixed alternative motor state equation

The independently motivated model-form question is whether the observed control limits depend on first-order thrust dynamics. Eschmann, Albani and Loianno distinguish first-order rotor-speed dynamics and a static thrust curve (their Eqs. (6), (10)). We test the quadratic special case. We do not use their measured coefficients, reproduce their identification method, or claim a calibrated Starling motor.

Let c be each original thrust command clipped to $[0, T_{\max}]$, $\alpha = 1 - \exp(-\Delta t/\tau)$, and $s = \sqrt{T}$ a proportional-speed state in $\sqrt{N}$, not measured RPM. The sole replacement is

$$\text{original: } T^+ = T + \alpha(c - T), \quad \text{alternative: } s^+ = s + \alpha(\sqrt{c} - s), \quad T^+ = (s^+)^2.$$

Both preserve the same steady thrust, maximum thrust and small-signal time constant at every positive equilibrium. For a held step from T_0 to c , setting $e = \exp(-t/\tau)$ gives

$$T_{\text{original}}(t) - T_{\text{alternative}}(t) = e(1 - e)(\sqrt{c} - \sqrt{T_0})^2.$$

This elementary relation identifies the finite-transient difference without choosing a larger local delay. Nominal $\tau = 0.0333$ s, every old randomization multiplier, the mass/lag regimes, force/torque mapping, aero/wind equations and observations remain unchanged. The two plants share unmeasured aerodynamic assumptions.

All 30 D11 controllers stay frozen, including physical/raw/context heads, new and all five old scalar controls, and observer/feedback. Five encoder seeds remain five models, not five new campaigns. The same 36 tasks per regime, three tiers, control rates, 20 ms MPC activation, two-second warm-up and shared altitude cap produce 1,080 new

flights. This is transfer of old fits, not equal-budget recalibration in the new plant. An isolated wrapper preserves the old simulator and artifacts.

Before the matrix runs, CPU/GPU closed-form step checks cover 400 steps, five time constants, four rotors, upward/downward/clipped commands, reset subsets and identical wrench scaling. Errors are below 3.2×10^{-15} . Four old-model feedback/observer controls then reproduce 60 prior arrays exactly. All 32 Slurm entries (preflight, 30 transfer rows, acceptance) finish successfully; all 281 source/weight/task hashes are checked remotely and 187 recovered-file hashes agree. Five original encoder checkpoints are verified but not included in this portable bundle.

Table S42: All controller groups and tiers. RMSE and terminal error are in m; success and contact in %; truncated flight time in s. All: 36 tasks/configuration; ID/W/J: 12 each. Values average configurations within a group.

Regime	Controller	Tier	n	RMSE	Success	Contact	Time	Terminal
Mass	Feedback	All	1	1.121	30.56	11.11	11.170	1.291
Mass	Feedback	ID	1	0.716	41.67	0.00	12.000	0.554
Mass	Feedback	W	1	1.359	16.67	8.33	11.467	1.606
Mass	Feedback	J	1	1.287	33.33	25.00	10.044	1.712
Mass	Observer	All	1	1.033	36.11	11.11	11.107	1.108
Mass	Observer	ID	1	0.709	41.67	0.00	12.000	0.545
Mass	Observer	W	1	1.170	25.00	16.67	10.920	1.430
Mass	Observer	J	1	1.222	41.67	16.67	10.400	1.349
Mass	Scalar-new	All	1	1.119	30.56	11.11	11.170	1.287
Mass	Scalar-new	ID	1	0.716	41.67	0.00	12.000	0.550
Mass	Scalar-new	W	1	1.353	16.67	8.33	11.468	1.599
Mass	Scalar-new	J	1	1.287	33.33	25.00	10.042	1.712
Mass	Phys-feat	All	1	1.095	33.33	11.11	11.169	1.268
Mass	Phys-feat	ID	1	0.715	41.67	0.00	12.000	0.550
Mass	Phys-feat	W	1	1.305	25.00	8.33	11.457	1.542
Mass	Phys-feat	J	1	1.265	33.33	25.00	10.049	1.712
Mass	Raw	All	1	1.141	30.56	11.11	11.168	1.326
Mass	Raw	ID	1	0.731	41.67	0.00	12.000	0.572
Mass	Raw	W	1	1.387	16.67	8.33	11.466	1.652
Mass	Raw	J	1	1.304	33.33	25.00	10.039	1.753
Mass	Scalar-old	All	5	1.097	33.33	11.11	11.195	1.264
Mass	Scalar-old	ID	5	0.716	41.67	0.00	12.000	0.550
Mass	Scalar-old	W	5	1.313	25.00	8.33	11.481	1.566
Mass	Scalar-old	J	5	1.262	33.33	25.00	10.103	1.678
Mass	Frozen	All	5	1.079	36.11	8.33	11.292	1.146
Mass	Frozen	ID	5	0.716	41.67	0.00	12.000	0.555
Mass	Frozen	W	5	1.280	25.00	8.33	11.472	1.523
Mass	Frozen	J	5	1.240	41.67	16.67	10.403	1.362
Lag	Feedback	All	1	0.576	83.33	0.00	12.000	0.410
Lag	Feedback	ID	1	0.156	100.00	0.00	12.000	0.094
Lag	Feedback	W	1	0.650	83.33	0.00	12.000	0.474
Lag	Feedback	J	1	0.922	66.67	0.00	12.000	0.661
Lag	Observer	All	1	0.409	94.44	0.00	12.000	0.159
Lag	Observer	ID	1	0.127	100.00	0.00	12.000	0.056
Lag	Observer	W	1	0.396	100.00	0.00	12.000	0.097
Lag	Observer	J	1	0.706	83.33	0.00	12.000	0.326
Lag	Scalar-new	All	1	0.520	91.67	0.00	12.000	0.358
Lag	Scalar-new	ID	1	0.146	100.00	0.00	12.000	0.081
Lag	Scalar-new	W	1	0.557	91.67	0.00	12.000	0.407
Lag	Scalar-new	J	1	0.857	83.33	0.00	12.000	0.585
Lag	Phys-feat	All	1	0.518	86.11	0.00	12.000	0.321
Lag	Phys-feat	ID	1	0.144	100.00	0.00	12.000	0.081
Lag	Phys-feat	W	1	0.550	83.33	0.00	12.000	0.338
Lag	Phys-feat	J	1	0.862	75.00	0.00	12.000	0.544
Lag	Raw	All	1	0.554	86.11	0.00	12.000	0.402
Lag	Raw	ID	1	0.164	100.00	0.00	12.000	0.092
Lag	Raw	W	1	0.608	83.33	0.00	12.000	0.514
Lag	Raw	J	1	0.889	75.00	0.00	12.000	0.599
Lag	Scalar-old	All	5	0.526	88.33	0.00	12.000	0.360

Table S42 continued

Regime	Controller	Tier	n	RMSE	Success	Contact	Time	Terminal
Lag	Scalar-old	ID	5	0.146	100.00	0.00	12.000	0.080
Lag	Scalar-old	W	5	0.568	86.67	0.00	12.000	0.420
Lag	Scalar-old	J	5	0.865	78.33	0.00	12.000	0.581
Lag	Frozen	All	5	0.507	87.22	0.00	12.000	0.348
Lag	Frozen	ID	5	0.154	100.00	0.00	12.000	0.093
Lag	Frozen	W	5	0.541	85.00	0.00	12.000	0.378
Lag	Frozen	J	5	0.824	76.67	0.00	12.000	0.572

S10.4 Complete model-form sensitivity

The following differences subtract the original D11 result from the matched alternative result. Every group and tier remains, including unfavorable changes. At the pooled method level, the largest RMSE change is 0.001780 m (lag/Raw); across all tiers it is 0.006096 m. Pooled contact rates are unchanged. Lag/Frozen has one fewer successful model-task outcome out of 180, a 0.556 percentage-point mean reduction. These small changes restrict what this check establishes: it is a mild model-form sensitivity test, not broad physics or real-flight validation, and aggregate proximity does not prove individual trajectories equivalent.

Table S43: All 56 alternative-minus-original motor-form differences. RMSE/terminal in m, flight time in s, success/contact in percentage points. These are descriptive paired changes, not new inferential contrasts.

Regime	Controller	Tier	Δ RMSE	Δ success	Δ contact	Δ time	Δ terminal
Mass	Feedback	All	+0.000096	+0.000	+0.000	+0.000000	+0.000017
Mass	Feedback	ID	+0.000100	+0.000	+0.000	+0.000000	+0.000005
Mass	Feedback	W	+0.000095	+0.000	+0.000	+0.000000	+0.000021
Mass	Feedback	J	+0.000092	+0.000	+0.000	+0.000000	+0.000024
Mass	Observer	All	+0.000104	+0.000	+0.000	+0.000000	+0.000013
Mass	Observer	ID	+0.000102	+0.000	+0.000	+0.000000	+0.000005
Mass	Observer	W	+0.000106	+0.000	+0.000	+0.000000	+0.000004
Mass	Observer	J	+0.000103	+0.000	+0.000	+0.000000	+0.000031
Mass	Scalar-new	All	+0.000093	+0.000	+0.000	-0.000139	-0.000039
Mass	Scalar-new	ID	+0.000101	+0.000	+0.000	+0.000000	+0.000002
Mass	Scalar-new	W	+0.000085	+0.000	+0.000	-0.000417	-0.000144
Mass	Scalar-new	J	+0.000093	+0.000	+0.000	+0.000000	+0.000023
Mass	Phys-feat	All	+0.000121	+0.000	+0.000	-0.000139	-0.000026
Mass	Phys-feat	ID	+0.000127	+0.000	+0.000	+0.000000	+0.000001
Mass	Phys-feat	W	+0.000122	+0.000	+0.000	-0.000417	-0.000112
Mass	Phys-feat	J	+0.000113	+0.000	+0.000	+0.000000	+0.000033
Mass	Raw	All	+0.000308	+0.000	+0.000	+0.000000	+0.000363
Mass	Raw	ID	+0.000363	+0.000	+0.000	+0.000000	+0.000505
Mass	Raw	W	+0.000291	+0.000	+0.000	+0.000000	+0.000304
Mass	Raw	J	+0.000270	+0.000	+0.000	+0.000000	+0.000280
Mass	Scalar-old	All	+0.000123	+0.000	+0.000	-0.000139	+0.000098
Mass	Scalar-old	ID	+0.000138	+0.000	+0.000	+0.000000	+0.000007
Mass	Scalar-old	W	+0.000114	+0.000	+0.000	+0.000000	+0.000030
Mass	Scalar-old	J	+0.000118	+0.000	+0.000	-0.000417	+0.000258
Mass	Frozen	All	+0.000148	+0.000	+0.000	+0.000000	+0.000038
Mass	Frozen	ID	+0.000192	+0.000	+0.000	+0.000000	+0.000046
Mass	Frozen	W	+0.000121	+0.000	+0.000	+0.000000	+0.000009
Mass	Frozen	J	+0.000131	+0.000	+0.000	+0.000000	+0.000059
Lag	Feedback	All	+0.000134	+0.000	+0.000	+0.000000	+0.000109
Lag	Feedback	ID	-0.000025	+0.000	+0.000	+0.000000	+0.000009
Lag	Feedback	W	+0.000207	+0.000	+0.000	+0.000000	+0.000147
Lag	Feedback	J	+0.000219	+0.000	+0.000	+0.000000	+0.000171
Lag	Observer	All	+0.000211	+0.000	+0.000	+0.000000	+0.000096
Lag	Observer	ID	-0.000041	+0.000	+0.000	+0.000000	-0.000020
Lag	Observer	W	+0.000384	+0.000	+0.000	+0.000000	+0.000071

Table S43 continued

Regime	Controller	Tier	Δ RMSE	Δ success	Δ contact	Δ time	Δ terminal
Lag	Observer	J	+0.000291	+0.000	+0.000	+0.000000	+0.000238
Lag	Scalar-new	All	+0.000135	+0.000	+0.000	+0.000000	-0.000337
Lag	Scalar-new	ID	+0.000086	+0.000	+0.000	+0.000000	+0.000123
Lag	Scalar-new	W	+0.000091	+0.000	+0.000	+0.000000	-0.001016
Lag	Scalar-new	J	+0.000230	+0.000	+0.000	+0.000000	-0.000120
Lag	Phys-feat	All	+0.000248	+0.000	+0.000	+0.000000	+0.000269
Lag	Phys-feat	ID	-0.000103	+0.000	+0.000	+0.000000	+0.000157
Lag	Phys-feat	W	+0.000447	+0.000	+0.000	+0.000000	+0.000324
Lag	Phys-feat	J	+0.000400	+0.000	+0.000	+0.000000	+0.000327
Lag	Raw	All	+0.001780	+0.000	+0.000	+0.000000	+0.001445
Lag	Raw	ID	-0.002941	+0.000	+0.000	+0.000000	-0.010987
Lag	Raw	W	+0.002183	+0.000	+0.000	+0.000000	+0.004846
Lag	Raw	J	+0.006096	+0.000	+0.000	+0.000000	+0.010476
Lag	Scalar-old	All	+0.000160	+0.000	+0.000	+0.000000	+0.000033
Lag	Scalar-old	ID	+0.000043	+0.000	+0.000	+0.000000	+0.000111
Lag	Scalar-old	W	+0.000242	+0.000	+0.000	+0.000000	-0.000062
Lag	Scalar-old	J	+0.000195	+0.000	+0.000	+0.000000	+0.000050
Lag	Frozen	All	+0.000517	-0.556	+0.000	+0.000000	+0.002166
Lag	Frozen	ID	+0.000272	+0.000	+0.000	+0.000000	+0.001331
Lag	Frozen	W	+0.000742	+0.000	+0.000	+0.000000	+0.003713
Lag	Frozen	J	+0.000536	-1.667	+0.000	+0.000000	+0.001455

S10.5 Strong-control and common-time comparisons

All 48 unadjusted exploratory scene/encoder-seed comparisons below use the original resampling procedure (10,000 draws), conditional on the original paired calibration campaign. Old scalar subsets remain fixed-averaged. They are separate from the earlier 194 development comparisons and the two independently locked confirmation families.

Lag/Frozen RMSE is 0.506519 m versus observer 0.409345 m; their common-time difference is +0.097174 m with interval [0.056806, 0.138948]. In mass, the own-flight RMSE difference is +0.045263 m, but the common-time difference is -0.001164 m with interval [-0.029407, 0.030390]. Mass/Frozen has the same success and descriptively lower contact, whose interval reaches zero. These outcomes preserve the strong-observer boundary; they do not establish equivalence or a superior learned controller. All comparisons, rather than only the physical baselines with local improvements, are retained.

Table S44: All 48 registered exploratory contrasts; unadjusted 95% intervals. Differences follow the stated comparison.

Regime	Comparison	Metric	Δ	95% interval
Mass	Frozen – Feedback	RMSE (m)	-0.0419	[-0.0738, -0.0147]
Mass	Frozen – Feedback	Success (pp)	+5.5556	[+0.0000, +13.3333]
Mass	Frozen – Feedback	Contact (pp)	-2.7778	[-8.8235, +0.0000]
Mass	Frozen – Feedback	Common RMSE (m)	-0.0394	[-0.0708, -0.0132]
Mass	Frozen – Observer	RMSE (m)	+0.0453	[+0.0004, +0.0958]
Mass	Frozen – Observer	Success (pp)	+0.0000	[+0.0000, +0.0000]
Mass	Frozen – Observer	Contact (pp)	-2.7778	[-7.8947, +0.0000]
Mass	Frozen – Observer	Common RMSE (m)	-0.0012	[-0.0294, +0.0304]
Mass	Frozen – Scalar	RMSE (m)	-0.0400	[-0.0690, -0.0144]
Mass	Frozen – Scalar	Success (pp)	+5.5556	[+0.0000, +13.3333]
Mass	Frozen – Scalar	Contact (pp)	-2.7778	[-8.8235, +0.0000]
Mass	Frozen – Scalar	Common RMSE (m)	-0.0371	[-0.0657, -0.0124]
Mass	Frozen – Scalar-old	RMSE (m)	-0.0184	[-0.0402, -0.0009]
Mass	Frozen – Scalar-old	Success (pp)	+2.7778	[+0.0000, +9.3750]
Mass	Frozen – Scalar-old	Contact (pp)	-2.7778	[-8.8235, +0.0000]
Mass	Frozen – Scalar-old	Common RMSE (m)	-0.0136	[-0.0341, +0.0003]
Mass	Frozen – Phys-feat	RMSE (m)	-0.0164	[-0.0275, -0.0066]
Mass	Frozen – Phys-feat	Success (pp)	+2.7778	[+0.0000, +9.3750]

Table S44 continued

Regime	Comparison	Metric	Δ	95% interval
Mass	Frozen – Phys-feat	Contact (pp)	-2.7778	[-8.8235, +0.0000]
Mass	Frozen – Phys-feat	Common RMSE (m)	-0.0155	[-0.0252, -0.0069]
Mass	Frozen – Raw	RMSE (m)	-0.0619	[-0.1054, -0.0272]
Mass	Frozen – Raw	Success (pp)	+5.5556	[+0.0000, +13.3333]
Mass	Frozen – Raw	Contact (pp)	-2.7778	[-8.8235, +0.0000]
Mass	Frozen – Raw	Common RMSE (m)	-0.0596	[-0.1025, -0.0263]
Lag	Frozen – Feedback	RMSE (m)	-0.0695	[-0.1039, -0.0376]
Lag	Frozen – Feedback	Success (pp)	+3.8889	[+0.0000, +10.3226]
Lag	Frozen – Feedback	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Feedback	Common RMSE (m)	-0.0695	[-0.1039, -0.0376]
Lag	Frozen – Observer	RMSE (m)	+0.0972	[+0.0568, +0.1389]
Lag	Frozen – Observer	Success (pp)	-7.2222	[-15.5556, +0.0000]
Lag	Frozen – Observer	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Observer	Common RMSE (m)	+0.0972	[+0.0568, +0.1389]
Lag	Frozen – Scalar	RMSE (m)	-0.0132	[-0.0362, +0.0123]
Lag	Frozen – Scalar	Success (pp)	-4.4444	[-12.2222, +0.0000]
Lag	Frozen – Scalar	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Scalar	Common RMSE (m)	-0.0132	[-0.0362, +0.0123]
Lag	Frozen – Scalar-old	RMSE (m)	-0.0197	[-0.0448, +0.0076]
Lag	Frozen – Scalar-old	Success (pp)	-1.1111	[-4.1379, +0.6061]
Lag	Frozen – Scalar-old	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Scalar-old	Common RMSE (m)	-0.0197	[-0.0448, +0.0076]
Lag	Frozen – Phys-feat	RMSE (m)	-0.0120	[-0.0332, +0.0068]
Lag	Frozen – Phys-feat	Success (pp)	+1.1111	[+0.0000, +4.2857]
Lag	Frozen – Phys-feat	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Phys-feat	Common RMSE (m)	-0.0120	[-0.0332, +0.0068]
Lag	Frozen – Raw	RMSE (m)	-0.0472	[-0.0957, -0.0042]
Lag	Frozen – Raw	Success (pp)	+1.1111	[+0.0000, +4.2857]
Lag	Frozen – Raw	Contact (pp)	+0.0000	[+0.0000, +0.0000]
Lag	Frozen – Raw	Common RMSE (m)	-0.0472	[-0.0957, -0.0042]

S10.6 Independent local metric checks and reproducibility scope

The local check reconstructs all 1,080 tracking-error sequences from recovered positions and independent distance-based reference interpolation, agreeing within 3.553×10^{-15} m. It rebuilds all 120 configuration/tier rows, 56 method/tier means and all 48 paired point estimates (maximum difference 6.158×10^{-17}). Intervals use the source-verified original bootstrap implementation, not a second result-selected procedure. Repository regression checks pass all 72 tests, with nine existing warnings.

The expanded bundle preserves the observed-signal arrays/rows, complete transfer positions/actions/validity masks, summaries, protocols, hashes, preflight receipt, audit scripts and full model-form shifts. Original calibration datasets, all snapshot inputs and encoder checkpoints are not all bundled; exact paths and source hashes identify them. Earlier result bundles and source versions remain separate. Neither this audit nor this model-form transfer establishes necessary history/depth information, a new predictive mechanism or observer-controlled task benefit. No additional confirmation campaign or hardware test follows from these results.

S11 Original Recurrent Predictor: Scope and Budgeted Comparison

S11.1 The retained adaptation head is affine

D28 audits all 100 C1 primary-budget encoder-adaptation checkpoints. Their 338-to-90 heads are affine; all context encoders changed during fitting, while all 16 tensors in the original action embedding, prediction initialization, prediction GRU, probe and latent head exactly equal seeded initialization. The saved observed-only statistics also omit native target normalization. Label-free gradient checks reach context and the adaptation head, not those original prediction modules. Thus C1’s correctly named full-encoder adaptation and retained affine head are not full native-model adaptation. S9’s cancellation argument remains limited to the affine adapter.

S11.2 One fixed native-model comparison

The original supervised model propagates future commands through a recurrent state initialized by observed history. The simulated actuator carries unobserved rotor thrust through a first-order recurrence, which motivates checking this existing forecast path; it does not prove recoverable memory or a useful response interaction. Protocol dbabfbc fixes random and complete-pretrained initialization, both regimes, five C1 campaigns and five seeds: 100 configurations at budget 32, using exactly the old 24 FIT/eight validation episodes. The full original nominal checkpoint initializes the pretrained arm; unused C1 prediction modules are never treated as trained.

Both native arms share their seed’s nominal observed and target normalizers, the original architecture and a physical-unit displacement-MSE objective over 30 steps. AdamW uses learning rate 3×10^{-4} , batches at most 128, clipping 1, 100 epochs and weight decay $\{10^{-4}, 10^{-2}\}$. Validation position RMSE selects among all 202 epoch/decay candidates, including epoch 0; exact ties retain the earlier candidate. No branch outcome selects weights. Twenty selected models use epoch 0: one mass and 19 lag pretrained configurations. The original five nominal native models per regime are also reported, without fictitious calibration-campaign replication. All 100 fitted models and all curves remain archived.

The native procedure has no explicit physical-feature/add-back path and uses a different output parameterization and more active prediction modules than affine adaptation. Sharing an episode/epoch budget does not equate information preprocessing, compute or optimization conditioning. Comparisons are between specified procedures, not isolated causal effects of recurrence or physical-prior removal. No new model size, loss grid, scene, sensor noise or simulator setting was searched.

D29 independently checks all 20,200 selection candidates, strict reloads and validation reconstruction (maximum 3.67×10^{-8} m difference); CPU/GPU agreement is within 1.55×10^{-6} m. Preparation failures involved a missing recovery directory and evaluation-mode cuDNN backward in the preflight checker. They preceded fitting; exact failed preflight source and correction are retained. All 100 remote fits completed without a changed training protocol.

S11.3 Complete shared-state outcomes and failed screens

D30 adds 100 selected native models and 10 nominal references to the 190 existing C1 configurations. All 216 old snapshots, nine actions and three named continuations remain. Common action support is 98 mass/106 lag; contact uses 99/108 initially alive histories. Held-command prediction RMSE uses the same 98/106 all-nine-valid states. Native outputs are computed in float32, then world conversion and costs in float64. No forecast RMSE is computed against changed future-command continuations. All 11,880 new prediction rows and 35,640 state/mode rows remain, including exclusions, all tiers and all campaigns. No new inferential intervals or full-flight rollouts are added.

Table S45: All ten predictor groups on identical held-command support. RMSE in m; accuracy/contact in percent. Native-0: original nominal recurrent model, without target-regime adaptation. Values average seeds and, when applicable, campaigns. Contact uses initially alive histories.

Regime	Predictor	RMSE	Regret	Accuracy	Contact
Mass	Frozen	0.2091	0.03256	40.04	0.00
Mass	Scalar	0.2734	0.01474	75.92	0.00
Mass	Phys-feat	0.3241	0.03769	29.80	0.00
Mass	Raw	0.2326	0.02979	45.10	0.00
Mass	Raw-depth	0.3154	0.03042	45.31	0.00

Table S45 continued

Regime	Predictor	RMSE	Regret	Accuracy	Contact
Mass	Enc-rand	0.2982	0.03669	38.78	0.00
Mass	Enc-pre	0.2433	0.03336	38.41	0.00
Mass	Native-rand	0.4833	0.06329	22.33	0.48
Mass	Native-pre	0.4301	0.05633	24.57	0.16
Mass	Native-0	0.4282	0.05603	24.29	0.00
Lag	Frozen	0.2651	0.02895	50.64	0.00
Lag	Scalar	0.3056	0.01629	63.96	0.00
Lag	Phys-feat	0.3811	0.03992	35.28	0.00
Lag	Raw	0.3072	0.02943	49.43	0.00
Lag	Raw-depth	0.3512	0.03192	50.38	0.37
Lag	Enc-rand	0.3941	0.03559	44.26	0.00
Lag	Enc-pre	0.2922	0.02719	51.81	0.00
Lag	Native-rand	0.5362	0.10400	16.60	0.48
Lag	Native-pre	0.4658	0.10998	20.68	0.15
Lag	Native-0	0.4656	0.11037	20.19	0.19

Native-pre does not improve on scalar physics in either regime or any continuation. In lag, its pooled RMSE is 0.4658 m versus Native-rand 0.5362 m, while held regret is 0.10998 versus 0.10400. This pooled descriptive reversal is not consistent at campaign means: three lag campaigns improve both held RMSE and held regret, and two worsen both. Under shared nominal/observer continuation, pretrained regret is higher in all five lag campaigns. The complete campaign differences prevent treating a pooled result as campaign-uniform evidence. Native-0’s similar means and the epoch 0 selections further limit attribution to target-regime calibration.

Table S46: All native pretrained-minus-random campaign differences, averaging five matched seeds. RMSE uses held-command truth; the three regret columns name distinct continuations. Negative favors pretrained. No inferential intervals are introduced.

Regime	Campaign	Δ RMSE	Δ held	Δ nominal	Δ observer
Mass	0	-0.12524	-0.01160	-0.00413	-0.00436
Mass	1	+0.01868	+0.00834	+0.00107	+0.00113
Mass	2	+0.00066	-0.00194	-0.00062	-0.00040
Mass	3	-0.02711	-0.01014	-0.00406	-0.00422
Mass	4	-0.13316	-0.01946	-0.00352	-0.00322
Lag	0	-0.23025	-0.00336	+0.00281	+0.00493
Lag	1	+0.07861	+0.02180	+0.00309	+0.00285
Lag	2	-0.18391	-0.00870	+0.00216	+0.00417
Lag	3	+0.03313	+0.02420	+0.00417	+0.00408
Lag	4	-0.04935	-0.00404	+0.00119	+0.00208

Table S47: All predictors under the two accepted shared feedback continuations. The initially selected action is unchanged. Regret/accuracy use common support; contact uses initially alive histories. Accuracy/contact in percent. These are short branches, not full-flight controller results.

Regime	Predictor	Continuation	Regret	Accuracy	Contact
Mass	Frozen	Nominal	0.00787	31.59	0.00
Mass	Frozen	Observer	0.01036	23.76	0.00
Mass	Scalar	Nominal	0.00390	64.08	0.00
Mass	Scalar	Observer	0.00551	50.82	0.00
Mass	Phys-feat	Nominal	0.00729	31.02	0.00
Mass	Phys-feat	Observer	0.00784	31.22	0.00
Mass	Raw	Nominal	0.00660	40.41	0.00
Mass	Raw	Observer	0.00836	34.29	0.00
Mass	Raw-depth	Nominal	0.00599	46.33	0.00
Mass	Raw-depth	Observer	0.00673	42.24	0.00
Mass	Enc-rand	Nominal	0.00680	38.73	0.00
Mass	Enc-rand	Observer	0.00744	36.24	0.00

Table S47 continued

Regime	Predictor	Continuation	Regret	Accuracy	Contact
Mass	Enc-pre	Nominal	0.00767	33.14	0.00
Mass	Enc-pre	Observer	0.00955	27.92	0.00
Mass	Native-rand	Nominal	0.01017	23.76	0.00
Mass	Native-rand	Observer	0.01000	24.86	0.36
Mass	Native-pre	Nominal	0.00792	28.82	0.00
Mass	Native-pre	Observer	0.00778	34.16	0.08
Mass	Native-0	Nominal	0.00814	28.37	0.00
Mass	Native-0	Observer	0.00830	32.65	0.00
Lag	Frozen	Nominal	0.00612	42.30	0.00
Lag	Frozen	Observer	0.00889	31.51	0.00
Lag	Scalar	Nominal	0.00242	67.17	0.00
Lag	Scalar	Observer	0.00290	65.66	0.00
Lag	Phys-feat	Nominal	0.00522	36.79	0.00
Lag	Phys-feat	Observer	0.00522	39.81	0.00
Lag	Raw	Nominal	0.00528	44.72	0.00
Lag	Raw	Observer	0.00704	35.85	0.00
Lag	Raw-depth	Nominal	0.00504	47.55	0.00
Lag	Raw-depth	Observer	0.00597	43.96	0.00
Lag	Enc-rand	Nominal	0.00503	43.43	0.00
Lag	Enc-rand	Observer	0.00588	40.45	0.00
Lag	Enc-pre	Nominal	0.00552	43.06	0.00
Lag	Enc-pre	Observer	0.00794	33.77	0.00
Lag	Native-rand	Nominal	0.01281	21.47	0.00
Lag	Native-rand	Observer	0.01302	20.87	0.00
Lag	Native-pre	Nominal	0.01549	18.08	0.00
Lag	Native-pre	Observer	0.01664	28.04	0.00
Lag	Native-0	Nominal	0.01553	17.17	0.00
Lag	Native-0	Observer	0.01664	28.30	0.00

Before outcomes, the descriptive screen required Native-pre to reduce pooled regret by at least 10% versus each of Scalar, Phys-feat, Raw, Raw-depth and Frozen in all three continuations, with no worse pooled accuracy/contact and lower regret in at least 4/5 campaigns. It additionally required no pooled regret/accuracy/contact harm versus Native-rand. Accuracy/contact equality is checked with exact rational counts. All 36 individual comparisons follow. Only the three mass comparisons against Native-rand pass; neither regime passes its complete 18-comparison screen. Mass/observer continuation improves regret against Frozen, but introduces selected contact, and therefore fails. Negative and partially favorable results are retained.

Table S48: All 36 descriptive screen comparisons for Native-pre. Gain: percent regret reduction; k : campaigns with lower regret. R: required regret gain; K: campaign rule; A/C: no accuracy/contact harm; Pass: conjunction. Native-rand requires no pooled harm and has no campaign threshold. Y/N: passed/failed.

Regime	Mode	Against	Gain	k	R	K	A	C	Pass
Mass	Held	Scalar	-282.2	0	N	N	N	N	N
Mass	Held	Phys-feat	-49.4	0	N	N	N	N	N
Mass	Held	Raw	-89.1	0	N	N	N	N	N
Mass	Held	Raw-depth	-85.2	0	N	N	N	N	N
Mass	Held	Frozen	-73.0	0	N	N	N	N	N
Mass	Held	Native-rand	+11.0	4	Y	Y	Y	Y	Y
Mass	Nominal	Scalar	-103.0	0	N	N	N	Y	N
Mass	Nominal	Phys-feat	-8.7	2	N	N	N	Y	N
Mass	Nominal	Raw	-20.1	2	N	N	N	Y	N
Mass	Nominal	Raw-depth	-32.2	2	N	N	N	Y	N
Mass	Nominal	Frozen	-0.6	2	N	N	N	Y	N
Mass	Nominal	Native-rand	+22.2	4	Y	Y	Y	Y	Y
Mass	Observer	Scalar	-41.4	0	N	N	N	N	N
Mass	Observer	Phys-feat	+0.8	2	N	N	Y	N	N
Mass	Observer	Raw	+6.9	3	N	N	N	N	N

Table S48 continued

Regime	Mode	Against	Gain	k	R	K	A	C	Pass
Mass	Observer	Raw-depth	-15.7	2	N	N	N	N	N
Mass	Observer	Frozen	+24.8	5	Y	Y	Y	N	N
Mass	Observer	Native-rand	+22.1	4	Y	Y	Y	Y	Y
Lag	Held	Scalar	-575.3	0	N	N	N	N	N
Lag	Held	Phys-feat	-175.5	0	N	N	N	N	N
Lag	Held	Raw	-273.7	0	N	N	N	N	N
Lag	Held	Raw-depth	-244.5	0	N	N	N	Y	N
Lag	Held	Frozen	-280.0	0	N	N	N	N	N
Lag	Held	Native-rand	-5.8	3	N	Y	Y	Y	N
Lag	Nominal	Scalar	-540.1	0	N	N	N	Y	N
Lag	Nominal	Phys-feat	-196.6	0	N	N	N	Y	N
Lag	Nominal	Raw	-193.7	0	N	N	N	Y	N
Lag	Nominal	Raw-depth	-207.2	0	N	N	N	Y	N
Lag	Nominal	Frozen	-153.1	0	N	N	N	Y	N
Lag	Nominal	Native-rand	-21.0	0	N	Y	N	Y	N
Lag	Observer	Scalar	-473.3	0	N	N	N	Y	N
Lag	Observer	Phys-feat	-219.0	0	N	N	N	Y	N
Lag	Observer	Raw	-136.4	0	N	N	N	Y	N
Lag	Observer	Raw-depth	-178.5	0	N	N	N	Y	N
Lag	Observer	Frozen	-87.2	0	N	N	N	Y	N
Lag	Observer	Native-rand	-27.8	0	N	Y	Y	Y	N

D31 independently reconstructs world coordinates, costs, choices, validity, all summaries and screens from saved native forecasts and original branch traces. Maximum arithmetic difference is 2.14×10^{-14} ; all old table entries are unchanged. This checks saved numerical evidence, not an independent training implementation or aerodynamic model. The native comparator closes a scope gap but establishes neither a useful remedy nor a superior controller. No new full-flight or independent confirmation campaign follows from these failed screens. All 700 external binary files across the previous and native archives remain separately hash-listed; the compact bundle retains protocols, code, curves, outcomes and checks, rather than being a standalone binary reproduction package.

S12 Rotor-Drag Model-Form Sensitivity

S12.1 Fixed physical question and observer comparator

The S10 motor-state intervention retains thrust-scaled aerodynamic drag. A distinct development question is whether the lag observer boundary persists under nominal-hover-matched rotor-speed/hub-airflow drag. Protocol fd743df fixes this intervention before outcomes. The per-rotor formulation is checked against the archived RotorPy multirotor source (commit 40b0969; RotorPy paper). This uses a mapped force law within our simulator, not the full RotorPy simulator. Its stock coefficients are approximate or adjusted, not independent measurements for our vehicle.

Let T_i denote the original unscaled nonnegative thrust state, $T_h = m_{\text{nom}}g/4$, r_i the rotor position in body coordinates, and R the body-to-world rotation. With original body-frame drag coefficients k_{rot} , the intervention is

$$v_i = R^\top(v - w) + \omega \times r_i, \quad F_i = -\frac{1}{4} \text{diag}(k_{\text{rot}}) \sqrt{T_i/T_h} v_i, \quad M_{\text{rot}} = \sum_i r_i \times F_i.$$

The square root is a proportional-speed proxy, not measured RPM. Summed translational drag matches the original model only at nominal hover with zero angular velocity. Torque and thrust derivatives are not matched. Original first-order thrust dynamics, propulsion-gain randomization, quadratic body drag and angular damping remain; propulsion gain does not multiply this speed proxy. There is no added flapping term or fitted coefficient.

A zero-fit Speed-obs comparator changes the aggregate thrust factor in the original observer’s inversion and feedforward to $\sqrt{\hat{T}/(m_{\text{nom}}g)}$. It retains the same 12 observed state records, backward differences, nominal coefficients, last-six estimate average, clipping and timing. It receives no true wind, rotor states or randomized parameters. This equal-rotor approximation is not an exact inversion of hub forces. Both observers run on both plants.

S12.2 Complete retained outcomes and common-time support

All 30 prior paired-data controller configurations run on the hub law; four additional configurations evaluate Speed-obs on both plants/regimes. The 34 new configurations provide 1,224 flights. Reusing all 30 original configurations gives 64 total; no original performance run is repeated. All 36 tasks per configuration, costs, gains, warm-up, references, coefficients, checkpoints and activation timing remain fixed. Results condition on reused development tasks and one paired calibration campaign.

Table S49: All eight controller groups on both plants. RMSE (m) uses each flight’s own valid tracking time; success/contact are percentages; duration (s) includes warm-up. Available: flights with post-warm-up tracking records per configuration. Seeds and old scalar subsets are averaged within method. Hub: altered rotor-hub drag.

Plant	Regime	Controller	RMSE	Success	Contact	Duration	Available
Original	Mass	Feedback	1.1205	30.56	11.11	11.170	36/36
Original	Mass	Observer	1.0333	36.11	11.11	11.107	36/36
Original	Mass	Speed-obs	1.0333	36.11	11.11	11.109	36/36
Original	Mass	Scalar	1.1186	30.56	11.11	11.170	36/36
Original	Mass	Old-scalar	1.0969	33.33	11.11	11.195	36/36
Original	Mass	Phys-feat	1.0950	33.33	11.11	11.169	36/36
Original	Mass	Raw	1.1403	30.56	11.11	11.168	36/36
Original	Mass	Frozen	1.0785	36.11	8.33	11.292	36/36
Original	Lag	Feedback	0.5759	83.33	0.00	12.000	36/36
Original	Lag	Observer	0.4091	94.44	0.00	12.000	36/36
Original	Lag	Speed-obs	0.4087	94.44	0.00	12.000	36/36
Original	Lag	Scalar	0.5196	91.67	0.00	12.000	36/36
Original	Lag	Old-scalar	0.5260	88.33	0.00	12.000	36/36
Original	Lag	Phys-feat	0.5182	86.11	0.00	12.000	36/36
Original	Lag	Raw	0.5520	86.11	0.00	12.000	36/36
Original	Lag	Frozen	0.5060	87.78	0.00	12.000	36/36
Hub	Mass	Feedback	1.1217	30.56	11.11	11.180	36/36
Hub	Mass	Observer	1.0283	36.11	11.11	11.120	36/36
Hub	Mass	Speed-obs	1.0289	36.11	11.11	11.123	36/36
Hub	Mass	Scalar	1.1199	30.56	11.11	11.180	36/36

Table S49 continued

Plant	Regime	Controller	RMSE	Success	Contact	Duration	Available
Hub	Mass	Old-scalar	1.0946	35.00	10.00	11.244	36/36
Hub	Mass	Phys-feat	1.0946	36.11	11.11	11.180	36/36
Hub	Mass	Raw	1.1401	30.56	11.11	11.177	36/36
Hub	Mass	Frozen	1.0733	37.78	8.33	11.302	36/36
Hub	Lag	Feedback	0.5830	83.33	0.00	12.000	36/36
Hub	Lag	Observer	0.4045	94.44	0.00	12.000	36/36
Hub	Lag	Speed-obs	0.4042	94.44	0.00	12.000	36/36
Hub	Lag	Scalar	0.5251	88.89	0.00	12.000	36/36
Hub	Lag	Old-scalar	0.5315	87.22	0.00	12.000	36/36
Hub	Lag	Phys-feat	0.5241	86.11	0.00	12.000	36/36
Hub	Lag	Raw	0.5563	83.33	0.00	12.000	36/36
Hub	Lag	Frozen	0.5087	86.11	0.00	12.000	36/36

Table S50 reports every declared observer comparison for common-time RMSE. Two-arm differences intersect both valid-flight masks; gap changes intersect all four masks before computing each RMSE. All seed/subset cells retain 36 tasks, with unequal valid durations allowed. The 10,000 paired bootstrap draws use scene clusters stratified by joint-extrapolation geometry and five model seeds; old scalar subsets are fixed-averaged. No draw has empty support. All 88 unadjusted intervals across own-valid RMSE, common-time RMSE, success and contact remain in the complete JSON and readable source tables. They are not a new confirmatory family.

Table S50: All 12 observer common-time contrasts (m), including two-arm procedure differences and four-arm gap changes. Gap change: (Frozen minus named observer) on hub minus original plant, using four-way common time. Intervals are unadjusted exploratory 95% intervals within the complete 88-comparison family. Negative two-arm differences favor the first named controller.

Regime	Comparison	Difference	95% interval
Mass	Hub: Frozen–Observer	-0.001437	[-0.025501, +0.022393]
Mass	Hub: Frozen–Speed-obs	-0.001014	[-0.024807, +0.022652]
Mass	Original: Speed-obs–Observer	-0.000896	[-0.001758, -0.000112]
Mass	Hub: Speed-obs–Observer	-0.000730	[-0.001512, -0.000011]
Mass	Gap change: Observer	+0.000230	[-0.008758, +0.007255]
Mass	Gap change: Speed-obs	+0.000072	[-0.008885, +0.007014]
Lag	Hub: Frozen–Observer	+0.104203	[+0.062393, +0.147119]
Lag	Hub: Frozen–Speed-obs	+0.104532	[+0.062725, +0.147464]
Lag	Original: Speed-obs–Observer	-0.000447	[-0.000870, -0.000028]
Lag	Hub: Speed-obs–Observer	-0.000329	[-0.000714, +0.000055]
Lag	Gap change: Observer	+0.007334	[+0.002417, +0.011558]
Lag	Gap change: Speed-obs	+0.007216	[+0.002317, +0.011433]

In lag, Frozen remains about 0.104 m worse than either observer, with positive common-time intervals and zero contacts. The four-arm gaps increase by about 0.007 m relative to the original law. In mass, Frozen’s own-valid RMSE is higher but it has fewer contacts (3/36 per seed versus 4/36); both common-time difference intervals include zero. Own-flight and common-time orderings must not be conflated. The small Speed-obs comparisons establish no new controller contribution or equivalence.

S12.3 Failed complete screens and numerical acceptance

The fixed screen requires at least 10% common-time RMSE improvement against each of seven controls, improvement in at least four of five seeds, and no worse success/contact. Only lag versus ordinary feedback passes one of 14 components; both complete regime screens fail. Partial favorable results remain visible. These outcomes do not justify fresh confirmation or another coefficient, gain or scenario search.

Table S51: Every hub-plant Frozen-minus-control screen component. Gain: relative common-time RMSE reduction in percent. k : improved seeds out of five. Success/contact differences are exact fractions of task outcomes, averaged over seeds or fixed old subsets. Pass requires gain at least 10%, $k \geq 4$, success difference at least zero and contact difference at most zero.

Regime	Against	Gain	k	$\Delta\text{success}$	$\Delta\text{contact}$	Pass
Mass	Feedback	+3.81	5	13/180	-1/36	N
Mass	Observer	+0.14	5	1/60	-1/36	N
Mass	Speed-obs	+0.10	4	1/60	-1/36	N
Mass	Scalar	+3.60	5	13/180	-1/36	N
Mass	Old-scalar	+1.45	5	1/36	-1/60	N
Mass	Phys-feat	+1.52	5	1/60	-1/36	N
Mass	Raw	+5.36	5	13/180	-1/36	N
Lag	Feedback	+12.74	5	1/36	0	Y
Lag	Observer	-25.76	0	-1/12	0	N
Lag	Speed-obs	-25.86	0	-1/12	0	N
Lag	Scalar	+3.12	5	-1/36	0	N
Lag	Old-scalar	+4.29	5	-1/90	0	N
Lag	Phys-feat	+2.94	5	0	0	N
Lag	Raw	+8.56	5	1/36	0	N

D32 checks all 9,792 saved force cases (maximum force discrepancy 1.30×10^{-6} N; moment 5.55×10^{-9} N m), all 43,200 observer estimates (wind 9.53×10^{-7} m/s), and all physics positions/contact times, with zero threshold ambiguities. Tracking reconstruction differs by at most 2.67×10^{-15} m. D34 independently checks all D33 points, support counts, 256 configuration/tier rows, 128 method/tier means and 14 screens (maximum 4.44×10^{-16}). Synthetic cluster/seed draws were independently expanded before performance; actual interval quantiles were not recomputed by a second implementation. The linked trace archive retains all 238 new configuration files, including all force, observer and position traces. This closes a specific drag-assumption objection while leaving shared coefficients, clean-state inputs, external physical validity and demonstrated task utility unresolved.

S13 Prospective Model-Selection Validation

S13.1 A prospective deployment-selection question

This experiment asks whether a decision diagnostic helps choose an already calibrated predictor for subsequent complete flights. It keeps all earlier results and adds new geometry instances rather than reusing the development flight matrix. It does not introduce a new controller or independently repeat model training. The candidate pool comprises eight fixed predictors from the paired calibration campaign in S6–S7: scalar physics (index 0), physical features (1), raw history (2), and frozen supervised context seeds 0–4 (indices 3–7). Their weights are unchanged. In particular, the affine readouts used trajectory fitting loss but calibration-validation regret to select the ridge penalty; the pool is not exclusively RMSE-tuned. Nominal feedback (8) and the causal observer (9) are test references and cannot be selected.

S13.2 New instances, unchanged plant and interface

Five existing procedural geometry families each supply one excluded pilot instance (seed 900), three validation instances (seeds 1000–1002) and four test instances (2000–2003). Each validation panel contains one instance per family. Each instance has winds at 3, 6 and 8 m/s, with a shared cardinal inlet direction fixed by the family and instance indices. Thus there are 15 validation tasks per panel and 60 test tasks per regime. Full box geometry and rasterized occupancy are checked for reuse against earlier scenes and across new splits. These are unseen instances from the same geometry families and simulator, not unseen obstacle types or independent aerodynamic validation.

Wind preparation retains the planar lattice-Boltzmann solver, 0.25 m cells, 8,000 solver steps, the final 2,000-step average and relaxation time 0.56. Waypoint and plant seeds, batching, 12 s duration, two-second warm-up, command limits, timing and controller gains are fixed. Actual mass and actuator time constant remain fixed within each flight. Across candidate models, task identities, randomized plant parameters and the complete warm-up are paired. No true wind, scene identity or randomized plant parameter enters a predictor.

S13.3 Three fixed model-selection rules

At 3, 6 and 9 s of each nominal-feedback validation parent, the simulator and controller are cloned. All nine candidates are executed both as 1.5 s held commands and as 100 ms commands after the common 20 ms activation delay, followed by nominal feedback from 120 ms. Each predictor retains its original held-command forecast and chosen action; continuation truth evaluates that choice under specified subsequent feedback. The original surrogate forecast does not exactly simulate the activation delay or the continuation policy.

R selects the predictor with minimum held-command position RMSE. D selects minimum realized regret under nominal continuation, using the original MPC cost and choice. Both receive the same complete parent/branch library, and both use the same initially alive snapshots for which all nine held and all nine continuation branches remain valid. Scores average snapshots within a task and then weight supported tasks equally. A panel needs at least 15 common snapshots spanning at least three families; otherwise both rules use a declared scalar fallback and remain in the primary comparison. Contact counts outside this shared support are retained. C runs all eight predictors on the 15 complete validation tasks and selects the minimum failure-inclusive loss below. Ties within 10^{-12} use the smaller model index.

Per validation task, the common R/D library costs $12 + 3 \times 9 \times 1.5 \times 2 = 93$ simulated seconds. C costs $8 \times 12 = 96$ seconds. Both meet a declared 96 s/task cap, but costs are not exactly equal. Each standalone R/D strategy is charged the full shared library. Pilot work, duplicate branches and replay checks are separate QA cost; test flights are evaluation cost. This accounting does not establish superior computational efficiency.

S13.4 Fixed-horizon loss and inference

The primary complete-flight endpoint is

$$L_c = \frac{1}{200} \sum_{t \in \{2.00, 2.05, \dots, 11.95\}} \begin{cases} \min(\|p_t - r_t\|^2, c), & \text{while alive,} \\ c, & \text{after contact,} \end{cases} \quad c = 4 \text{ m}^2.$$

This is failure-inclusive capped tracking loss, not RMSE or a safety probability. A flight ending before warm-up receives c . Terminal contact, including the final unsampled interval, is reported separately. Prespecified sensitivities use $c = 1$

and 9 m^2 . Other secondary outcomes retain success, contact, flight exposure, valid-observation RMSE, terminal error and paired common-flight-time RMSE. Improved loss with worse contact cannot be described as a general safety improvement.

The two primary contrasts are D-minus-R test loss at $c = 4$, one per regime, averaged over the three fixed panel selections. All panels use the same test matrix. Twenty thousand paired bootstrap draws resample four test scenes within each of the five families, preserving the three winds, all models and all panel selections together. The two-sided 97.5% intervals (1.25/98.75 percentiles) adjust the two-comparison family. The fixed RNG seed is 2026091207 and the resampling indices are saved. Intervals condition on the fixed trained models and three fixed selection panels; panel repetition does not create independent training campaigns. Secondary contrasts use exploratory 95% intervals and are not promoted to primary tests after observing results.

S13.5 Integrity and reproducibility

An excluded pilot checks duplicate branches, parent replay, no-intervention equivalence, input isolation, action activation, state preservation and raw loss reconstruction. A cross-machine absolute-path error stopped the first six pilot entries before simulation; relative paths corrected deployment without changing wind arrays or scientific rules. The replacement six pilot configurations completed. The protocol, source, configurations and candidate weights are hashed before formal collection. Validation data are fully accepted and an immutable selection receipt is written before any test flight. Every test candidate, failed flight and unsupported panel is retained. Raw trajectories, forecast arrays, selection scores, source hashes, execution records and bootstrap indices accompany the research code; the Overleaf project itself only requires the typesetting assets.

The complete validation acceptance reconstructs 90 parent flights, 4,860 candidate branches, 270 snapshots (263 initially alive and 258 common-valid), 2,160 model/snapshot records and 720 validation flights. Test acceptance reconstructs all 1,200 flights. A separate actual-predictor replay confirms all 2,160 saved validation choices exactly, including near-tie cases; maximum cost discrepancy is 5.33×10^{-15} . Independent statistical reconstruction checks every primary and secondary interval against the saved draws, with maximum numerical discrepancy 3.56×10^{-15} . All 55 synthetic tests pass. These are numerical and protocol checks, not independent aerodynamic validation.

The two primary D-minus-R loss contrasts are 0.0174 m^2 with adjusted interval $[0.0060, 0.0294]$ in mass, and -0.0084 m^2 with $[-0.0185, 0.0008]$ in lag. Every panel meets the diagnostic support threshold: fallback is not responsible for either result. The mass disadvantage also appears in common-time RMSE and both secondary caps. Lag’s favorable secondary cap/common-time results remain exploratory and do not replace its unconfirmed primary. Direct-flight selection has lower loss than prediction-error selection in the secondary comparisons, while the observer’s lower lag loss accompanies one additional contact. The experiment therefore narrows the use of this particular local diagnostic; it does not validate it as a generally superior selector.

S13.6 Independent validation of model selection

R selects by held-command prediction RMSE; D selects by the regret of its original held-forecast choice when nominal feedback resumes after 120 ms; C selects by complete-flight capped loss. The eight candidate predictors remain fixed. Indices 0–2 identify scalar physics, physical-feature and raw-history readouts; 3–7 identify supervised-context seeds 0–4. Indices 8–9 are feedback and observer test references, excluded from selection. Display panels 1–3 correspond to recorded indices 0–2.

All three fixed panel selections remain in the test means. An unsupported R or D panel uses scalar index 0 and cannot be interpreted as a successful diagnostic selection. Each selector’s validation cost is charged separately: R/D use 93 simulated seconds per parent; C uses 96. QA and independent test flights are separate costs. The primary loss averages 200 observations at 2.00–11.95 s: $\min(\|p - r\|^2, 4)$ while alive and 4 after contact. The cap defines an engineering loss, not RMSE or safety probability. Caps 1 and 9 are prespecified secondary sensitivities.

Table S52: All validation panels. Valid denotes the common all-nine-valid held/continuation snapshot support. Parents and families count that support; R/D require at least 15 snapshots and three families. R, D and C columns give chosen model indices.

Regime	Panel	Valid	Parents	Families	R	D	C	R/D enabled
Mass	1	42/45	14	5	5	0	4	yes
Mass	2	38/45	13	5	2	2	3	yes
Mass	3	45/45	15	5	6	0	0	yes
Lag	1	45/45	15	5	6	0	7	yes
Lag	2	43/45	15	5	3	0	7	yes
Lag	3	45/45	15	5	2	0	4	yes

Table S53: Every frozen candidate's validation scores. RMSE (m), held/continuation regret (m^2 plus the declared command penalties), and C's complete-flight loss are parent-weighted. H/F contact percentages use all initially alive histories, beyond common valid support. Missing supported scores are shown as -; no candidate is removed.

Regime	Panel	Model	RMSE	F regret	H regret	C loss	H hit (%)	F hit (%)
Mass	1	0	0.6523	0.0005	0.0088	1.0273	6.7	6.7
Mass	1	1	0.1416	0.0008	0.0029	1.0194	6.7	6.7
Mass	1	2	0.1002	0.0015	0.0033	1.0424	6.7	6.7
Mass	1	3	0.0906	0.0025	0.0045	1.0074	6.7	6.7
Mass	1	4	0.0848	0.0028	0.0058	1.0036	6.7	6.7
Mass	1	5	0.0799	0.0025	0.0045	1.0054	6.7	6.7
Mass	1	6	0.0879	0.0025	0.0063	1.0111	6.7	6.7
Mass	1	7	0.0948	0.0024	0.0053	1.0059	6.7	6.7
Mass	2	0	0.4799	0.0005	0.0027	0.8999	0.0	0.0
Mass	2	1	0.0912	0.0005	0.0031	0.8934	7.1	0.0
Mass	2	2	0.0562	0.0004	0.0008	0.9075	7.1	0.0
Mass	2	3	0.0663	0.0014	0.0039	0.8895	7.1	0.0
Mass	2	4	0.0620	0.0013	0.0036	0.8896	7.1	0.0
Mass	2	5	0.0660	0.0019	0.0046	0.8908	7.1	0.0
Mass	2	6	0.0588	0.0014	0.0039	0.8915	7.1	0.0
Mass	2	7	0.0648	0.0014	0.0039	0.8905	7.1	0.0
Mass	3	0	0.7184	0.0011	0.0098	1.1483	0.0	0.0
Mass	3	1	0.2243	0.0017	0.0099	1.1539	0.0	0.0
Mass	3	2	0.3052	0.0040	0.0169	1.1857	0.0	0.0
Mass	3	3	0.1587	0.0027	0.0136	1.1508	0.0	0.0
Mass	3	4	0.1644	0.0031	0.0138	1.1512	0.0	0.0
Mass	3	5	0.1693	0.0027	0.0136	1.1529	0.0	0.0
Mass	3	6	0.1570	0.0032	0.0145	1.1557	0.0	0.0
Mass	3	7	0.1744	0.0028	0.0128	1.1532	0.0	0.0
Lag	1	0	0.1380	0.0005	0.0024	0.0702	0.0	0.0
Lag	1	1	0.1628	0.0021	0.0115	0.0840	0.0	0.0
Lag	1	2	0.1212	0.0011	0.0069	0.0696	0.0	0.0
Lag	1	3	0.1095	0.0018	0.0082	0.0674	0.0	0.0
Lag	1	4	0.0995	0.0011	0.0049	0.0650	0.0	0.0
Lag	1	5	0.1007	0.0013	0.0054	0.0652	0.0	0.0
Lag	1	6	0.0916	0.0010	0.0044	0.0699	0.0	0.0
Lag	1	7	0.0951	0.0008	0.0040	0.0645	0.0	0.0
Lag	2	0	0.1697	0.0003	0.0039	0.0941	0.0	0.0
Lag	2	1	0.1718	0.0007	0.0077	0.1002	0.0	0.0
Lag	2	2	0.1177	0.0005	0.0031	0.0931	0.0	0.0
Lag	2	3	0.0968	0.0012	0.0059	0.0910	0.0	0.0
Lag	2	4	0.1077	0.0010	0.0046	0.0891	0.0	0.0
Lag	2	5	0.0995	0.0013	0.0038	0.0907	0.0	0.0
Lag	2	6	0.1014	0.0005	0.0026	0.0929	0.0	0.0
Lag	2	7	0.1068	0.0005	0.0026	0.0869	0.0	0.0
Lag	3	0	0.2280	0.0005	0.0025	0.4351	0.0	0.0
Lag	3	1	0.2571	0.0009	0.0079	0.4755	0.0	0.0
Lag	3	2	0.2120	0.0018	0.0084	0.4874	0.0	0.0

Table S53 (continued)

Regime	Panel	Model	RMSE	F regret	H regret	C loss	H hit (%)	F hit (%)
Lag	3	3	0.2273	0.0010	0.0056	0.4184	0.0	0.0
Lag	3	4	0.2346	0.0010	0.0048	0.4162	0.0	0.0
Lag	3	5	0.2664	0.0010	0.0045	0.4226	0.0	0.0
Lag	3	6	0.2362	0.0022	0.0121	0.4412	0.0	0.0
Lag	3	7	0.2361	0.0007	0.0032	0.4364	0.0	0.0

Table S54: All fixed-panel selections on the same independent test matrix. L_k denotes the failure-inclusive tracking loss capped at k m². Panel points are selection outcomes, not confidence intervals or independent test repetitions.

Regime	Panel	Rule	Model	L_4	L_1	L_9	Hit (%)	Success (%)
Mass	1	R	5	0.8898	0.4896	1.2425	6.7	45.0
Mass	2	R	2	0.9324	0.5090	1.2959	6.7	41.7
Mass	3	R	6	0.8879	0.4887	1.2409	6.7	45.0
Mass	1	D	0	0.9149	0.5017	1.2642	6.7	45.0
Mass	2	D	2	0.9324	0.5090	1.2959	6.7	41.7
Mass	3	D	0	0.9149	0.5017	1.2642	6.7	45.0
Mass	1	C	4	0.8856	0.4880	1.2366	6.7	45.0
Mass	2	C	3	0.8883	0.4878	1.2418	6.7	45.0
Mass	3	C	0	0.9149	0.5017	1.2642	6.7	45.0
Lag	1	R	6	0.3824	0.1668	0.5980	1.7	83.3
Lag	2	R	3	0.3870	0.1667	0.6088	1.7	81.7
Lag	3	R	2	0.3747	0.1636	0.5896	1.7	83.3
Lag	1	D	0	0.3729	0.1599	0.5884	1.7	85.0
Lag	2	D	0	0.3729	0.1599	0.5884	1.7	85.0
Lag	3	D	0	0.3729	0.1599	0.5884	1.7	85.0
Lag	1	C	7	0.3708	0.1588	0.5861	1.7	83.3
Lag	2	C	7	0.3708	0.1588	0.5861	1.7	83.3
Lag	3	C	4	0.3795	0.1620	0.5993	1.7	81.7

Table S55: The complete 10-model test matrix: 60 tasks per model and regime, including both feedback references. Own-support RMSE (m) is accompanied by its task count n , flight exposure (s), terminal error (m), contact and success. It must not be read as failure-inclusive utility.

Regime	Model	L_4	L_1	L_9	Hit (%)	Succ. (%)	RMSE	n	Time	Terminal
Mass	0	0.9149	0.5017	1.2642	6.7	45.0	0.9091	60	11.436	1.1098
Mass	1	0.8977	0.4943	1.2497	6.7	45.0	0.8979	60	11.434	1.0965
Mass	2	0.9324	0.5090	1.2959	6.7	41.7	0.9236	60	11.436	1.1415
Mass	3	0.8883	0.4878	1.2418	6.7	45.0	0.8929	60	11.434	1.0977
Mass	4	0.8856	0.4880	1.2366	6.7	45.0	0.8914	60	11.434	1.0952
Mass	5	0.8898	0.4896	1.2425	6.7	45.0	0.8939	60	11.432	1.0960
Mass	6	0.8879	0.4887	1.2409	6.7	45.0	0.8928	60	11.433	1.0965
Mass	7	0.8902	0.4897	1.2439	6.7	45.0	0.8947	60	11.434	1.0987
Mass	8	0.9221	0.5051	1.2725	6.7	43.3	0.9138	60	11.436	1.1198
Mass	9	0.8428	0.4731	1.1946	6.7	45.0	0.8661	60	11.423	1.0408
Lag	0	0.3729	0.1599	0.5884	1.7	85.0	0.6843	60	11.874	0.8795
Lag	1	0.3813	0.1710	0.5943	1.7	83.3	0.6684	60	11.869	0.8519
Lag	2	0.3747	0.1636	0.5896	1.7	83.3	0.7049	60	11.940	0.7981
Lag	3	0.3870	0.1667	0.6088	1.7	81.7	0.7000	60	11.870	0.9084
Lag	4	0.3795	0.1620	0.5993	1.7	81.7	0.6882	60	11.870	0.8995
Lag	5	0.3785	0.1606	0.5979	1.7	85.0	0.6892	60	11.869	0.8958
Lag	6	0.3824	0.1668	0.5980	1.7	83.3	0.6976	60	11.869	0.9174
Lag	7	0.3708	0.1588	0.5861	1.7	83.3	0.6767	60	11.870	0.8797
Lag	8	0.4114	0.1886	0.6277	1.7	81.7	0.7372	60	11.933	0.8230
Lag	9	0.2370	0.0961	0.4524	3.3	91.7	0.5212	60	11.792	0.7029

Table S56: All two primary D-minus-R contrasts in capped loss at 4 m². Two-sided 97.5% percentile intervals adjust the two-regime family. Lower favors D. Null intervals do not establish equivalence.

Regime	$D - R$	97.5% interval	Draws
Mass	0.0174	[0.0060, 0.0294]	20000
Lag	-0.0084	[-0.0185, 0.0008]	20000

Bootstrap resampling retains all three winds within each scene and pairs every selector and fixed panel. Four scenes are sampled within each of five families. The 20,000 draws use seed 2026091207. Intervals condition on existing model weights and three fixed validation panels; model seeds and panels are not resampled.

Table S57: All prespecified secondary comparisons, with unadjusted exploratory 95% intervals. Percentage-point units are used for contact/success. For common-time RMSE, zero-support draws are not estimable; the number of valid draws is reported. These comparisons do not create additional primary claims.

Regime	Contrast	Metric	Difference	95% interval	Valid draws
Mass	D-C	L_4	0.0245	[0.0149, 0.0345]	20000
Mass	R-C	L_4	0.0071	[0.0032, 0.0111]	20000
Mass	D-R	L_1	0.0084	[0.0041, 0.0131]	20000
Mass	D-C	L_1	0.0116	[0.0076, 0.0161]	20000
Mass	R-C	L_1	0.0033	[0.0024, 0.0042]	20000
Mass	D-R	L_9	0.0150	[0.0020, 0.0280]	20000
Mass	D-C	L_9	0.0272	[0.0158, 0.0401]	20000
Mass	R-C	L_9	0.0122	[0.0052, 0.0205]	20000
Mass	D-R	Contact (pp)	0.0000	[0.0000, 0.0000]	20000
Mass	D-C	Contact (pp)	0.0000	[0.0000, 0.0000]	20000
Mass	R-C	Contact (pp)	0.0000	[0.0000, 0.0000]	20000
Mass	D-R	Success (pp)	0.0000	[0.0000, 0.0000]	20000
Mass	D-C	Success (pp)	-1.1111	[-2.7778, 0.0000]	20000
Mass	R-C	Success (pp)	-1.1111	[-2.7778, 0.0000]	20000
Mass	D-R	Exposure (s)	0.0019	[-0.0004, 0.0060]	20000
Mass	D-C	Exposure (s)	0.0012	[-0.0001, 0.0039]	20000
Mass	R-C	Exposure (s)	-0.0006	[-0.0022, 0.0003]	20000
Mass	D-R	Common RMSE (m)	0.0094	[0.0037, 0.0155]	20000
Mass	D-C	Common RMSE (m)	0.0155	[0.0096, 0.0219]	20000
Mass	R-C	Common RMSE (m)	0.0061	[0.0038, 0.0085]	20000
Lag	D-C	L_4	-0.0008	[-0.0116, 0.0095]	20000
Lag	R-C	L_4	0.0077	[0.0034, 0.0118]	20000
Lag	D-R	L_1	-0.0058	[-0.0102, -0.0019]	20000
Lag	D-C	L_1	0.0000	[-0.0056, 0.0053]	20000
Lag	R-C	L_1	0.0058	[0.0031, 0.0088]	20000
Lag	D-R	L_9	-0.0104	[-0.0212, -0.0009]	20000
Lag	D-C	L_9	-0.0021	[-0.0146, 0.0091]	20000
Lag	R-C	L_9	0.0083	[0.0037, 0.0130]	20000
Lag	D-R	Contact (pp)	0.0000	[0.0000, 0.0000]	20000
Lag	D-C	Contact (pp)	0.0000	[0.0000, 0.0000]	20000
Lag	R-C	Contact (pp)	0.0000	[0.0000, 0.0000]	20000
Lag	D-R	Success (pp)	2.2222	[0.0000, 5.5556]	20000
Lag	D-C	Success (pp)	2.2222	[0.0000, 5.5556]	20000
Lag	R-C	Success (pp)	0.0000	[-2.2222, 2.2222]	20000
Lag	D-R	Exposure (s)	-0.0192	[-0.0576, 0.0000]	20000
Lag	D-C	Exposure (s)	0.0040	[0.0000, 0.0119]	20000
Lag	R-C	Exposure (s)	0.0232	[0.0000, 0.0695]	20000
Lag	D-R	Common RMSE (m)	-0.0134	[-0.0202, -0.0067]	20000
Lag	D-C	Common RMSE (m)	-0.0021	[-0.0111, 0.0065]	20000
Lag	R-C	Common RMSE (m)	0.0113	[0.0024, 0.0190]	20000

Table S58: Pair-specific common-flight-time RMSE (m) for every panel and selector pair. Left/right model indices, eligible tasks and common observations are explicit. Empty supports remain missing; they do not become zero-error flights.

Regime	Panel	Pair	Models	n	Left	Right	Difference	Obs.
Mass	1	D-R	0/5	60	0.9067	0.8934	0.0132	11320
Mass	2	D-R	2/2	60	0.9236	0.9236	0.0000	11325
Mass	3	D-R	0/6	60	0.9073	0.8923	0.0149	11321
Mass	1	D-C	0/4	60	0.9079	0.8910	0.0169	11322
Mass	2	D-C	2/3	60	0.9224	0.8929	0.0295	11323
Mass	3	D-C	0/0	60	0.9091	0.9091	0.0000	11324
Mass	1	R-C	5/4	60	0.8939	0.8902	0.0037	11321
Mass	2	R-C	2/3	60	0.9224	0.8929	0.0295	11323
Mass	3	R-C	6/0	60	0.8923	0.9073	-0.0149	11321
Lag	1	D-R	0/6	60	0.6780	0.6976	-0.0195	11844
Lag	2	D-R	0/3	60	0.6780	0.7000	-0.0219	11844
Lag	3	D-R	0/2	60	0.6843	0.6830	0.0013	11849
Lag	1	D-C	0/7	60	0.6780	0.6767	0.0013	11844
Lag	2	D-C	0/7	60	0.6780	0.6767	0.0013	11844
Lag	3	D-C	0/4	60	0.6793	0.6882	-0.0089	11845
Lag	1	R-C	6/7	60	0.6976	0.6767	0.0209	11844
Lag	2	R-C	3/7	60	0.7000	0.6767	0.0233	11844
Lag	3	R-C	2/4	60	0.6781	0.6882	-0.0101	11845

Table S59: Descriptive paired common-time RMSE for every candidate against feedback (8) and observer (9). These support-specific comparisons are secondary and do not imply general control superiority.

Regime	Model	Ref.	n	Model RMSE	Ref. RMSE	Difference	Obs.
Mass	0	8	60	0.9091	0.9133	-0.0042	11324
Mass	0	9	60	0.9017	0.8661	0.0356	11310
Mass	1	8	60	0.8979	0.9127	-0.0148	11323
Mass	1	9	60	0.8912	0.8661	0.0251	11310
Mass	2	8	60	0.9236	0.9138	0.0098	11325
Mass	2	9	60	0.9159	0.8661	0.0498	11310
Mass	3	8	60	0.8929	0.9125	-0.0196	11323
Mass	3	9	60	0.8864	0.8661	0.0202	11310
Mass	4	8	60	0.8914	0.9125	-0.0211	11323
Mass	4	9	60	0.8848	0.8661	0.0186	11310
Mass	5	8	60	0.8939	0.9114	-0.0175	11321
Mass	5	9	60	0.8883	0.8661	0.0222	11310
Mass	6	8	60	0.8928	0.9119	-0.0192	11322
Mass	6	9	60	0.8867	0.8661	0.0205	11310
Mass	7	8	60	0.8947	0.9119	-0.0172	11322
Mass	7	9	60	0.8886	0.8661	0.0225	11310
Lag	0	8	60	0.6843	0.7198	-0.0355	11849
Lag	0	9	60	0.6126	0.5212	0.0914	11752
Lag	1	8	60	0.6684	0.7126	-0.0442	11843
Lag	1	9	60	0.6101	0.5212	0.0890	11752
Lag	2	8	60	0.7066	0.7372	-0.0306	11921
Lag	2	9	60	0.6148	0.5212	0.0937	11752
Lag	3	8	60	0.7000	0.7138	-0.0139	11844
Lag	3	9	60	0.6327	0.5212	0.1115	11752
Lag	4	8	60	0.6882	0.7150	-0.0268	11845
Lag	4	9	60	0.6230	0.5212	0.1018	11752
Lag	5	8	60	0.6892	0.7138	-0.0246	11844
Lag	5	9	60	0.6232	0.5212	0.1020	11752
Lag	6	8	60	0.6976	0.7138	-0.0163	11844
Lag	6	9	60	0.6294	0.5212	0.1082	11752
Lag	7	8	60	0.6767	0.7138	-0.0371	11844
Lag	7	9	60	0.6099	0.5212	0.0887	11752

S13.6.1 Source and scope record

All numerical entries above are generated from `runs/icra_selection_validation_20260912/selection.json` and `runs/icra_selection_validation_20260912/analysis/summary.json`. The former retains raw validation source hashes, per-model provenance and support counts; the latter retains complete test-result source hashes. `analysis/bootstrap.npz` retains actual resampling indices and comparison draws within the same experiment directory. Raw parent, held/continuation branch and complete-flight arrays remain in the research evidence package, not LaTeX compilation dependencies.

Protocol SHA-256: `c2ee6bbd852b13c372645c14d6e8419626397df46ecade6e8b8f62a497464327`.

Selection receipt SHA-256: `548a35af4eb1350f9dcf10999d90349876d3b0b4031f6d8b69813a8f1b9bb965`.

Bootstrap archive SHA-256: `9b96ba9f88a5bcb8b7c59c084095e618e4f7199c524504fc345d27ccdbf4c743`.

This experiment tests conditional selection utility on new instances of the same geometry families and simulated aerodynamic model. It does not independently repeat model training, establish a new controller, or prove that the affine structure causes every observed control difference. A reduction in capped loss with increased contact is not a general safety improvement.

S14 Frozen-Encoder Nonlinear Readout Sensitivity

This follow-up addresses the readout scope of S9 using the same C1 lag calibration sources and the same development controller states as main Figure 3 and S4. Its protocol was written before the new readouts were fit, after the earlier development evidence had been inspected. It is not a new independent confirmation, an additional simulator setting, or a complete-flight test. The prospective experiment in S13 and its adverse/null primary outcomes remain unchanged.

S14.1 Source-matched calibration and fixed fitting procedures

The five original C1 lag calibration caches were restored byte-for-byte and verified against their recorded SHA-256 values. Each campaign uses the original budget-32, subset-901 split: 24 FIT and eight VAL episodes. The valid windows and episode grouping follow C1; paired-action calibration data are not substituted. FIT/VAL window counts are respectively 316/100, 330/112, 322/112, 336/112 and 326/103 for campaigns 0–4. Future commands in these calibration windows are the recorded commands.

All five original supervised encoder checkpoints remain frozen, including their normalization buffers. As in the main paper, the representation was pretrained with a 256-unit history GRU, three-layer depth CNN and 128-dimensional fused context; its supervised pretraining objective is smooth-L1 on normalized future physical-state targets. The new heads use the same 338-dimensional feature vector

$$[z_t, U^a, d_{\text{nom}}^a] \in \mathbb{R}^{128+120+90}.$$

FIT windows alone determine head feature normalization, with standard deviations bounded below by 10^{-5} . The target is the 90-dimensional yaw-frame displacement residual around nominal response time $\tau = 0.3\text{s}$. This head-fitting MSE objective differs from encoder pretraining. The affine control refits the complete original closed-form ridge grid and verifies its selected fit and saved predictions before new-head evaluation; the scalar control remains the original campaign-specific C1 scalar fit.

Let $q^a = [U^a, d_{\text{nom}}^a]$. The additive MLP predicts $g(z_t) + h(q^a)$ with a width-106 hidden SiLU layer in each branch and 90 outputs, for 55,300 parameters. Its shared context term still cancels in candidate differences. The joint MLP predicts $f([z_t, q^a])$ using one width-128 SiLU hidden layer and 90 outputs, for 55,002 parameters; it permits context–candidate interactions. Their capacities are similar, not mathematically identical. They are direct readouts, not recurrent latent world models, and their difference cannot be attributed uniquely to interactions.

For each of five campaigns and five encoders, both MLP families retain head initializations 0, 1 and 2. Every candidate uses float64 head arithmetic and 1,000 full-batch Adam updates, learning rate 0.003, betas (0.9, 0.999) and epsilon 10^{-8} . There is no early stopping, scheduler, dropout, clipping or augmentation. The objective is mean residual MSE over all windows and 90 outputs plus $\lambda \sum W^2/90$, excluding biases. The seven penalties are $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$; candidates within an initialization start from identical weights. Only final-iterate VAL position RMSE selects the penalty, with exact ties resolved by grid order. All 1,050 neural candidates and 150 selected heads are retained; no head seed is selected for favorable outcomes. MLP optimization and effective regularization differ from closed-form ridge, so the comparison concerns these specified procedures.

S14.2 Shared support, continuation and uncertainty

Evaluation reuses all 108 lag snapshots from 36 tasks, three times per task, each with nine commands and a 1.5s horizon. The common all-nine-valid intersection across held, nominal-feedback and observer-feedback continuations contains 106 states in 21 scene clusters. Model choices remain fixed across modes. Held-command forecasts are scored against held-command truth; they are not scored as forecasts of changed feedback commands. As in S4, the continuation intervention activates the selected command after 20ms, holds it for 100ms, and resumes the named common state-feedback law at 120ms. The feedback law is shared; the resulting commands can differ as branch states diverge. Existing low-level stabilization remains active throughout.

The matrix contains five scalar controls, 25 affine heads, 75 additive MLP heads and 75 joint MLP heads: 180 configurations. All 108 records and initially-alive selected-contact accounting are retained in the machine-readable evidence. Pooled metrics use the same common support, with each configuration’s position MSE square-rooted before averaging RMSE across encoders and head seeds. Candidate-centered MSE subtracts the candidate mean from prediction and truth within a state before scoring response differences. Three head seeds are averaged within campaign/encoder cells, not treated as independent environments or combined into an ensemble forecast.

The fixed family has four directions (joint-affine, additive-affine, joint-additive and joint-scalar) across forecast RMSE, centered MSE and three regrets. All 20 contrasts use 20,000 paired scene-cluster/campaign/encoder bootstrap draws, seed 2026091401, with scene strata joint-extrapolation versus other. Repeated snapshots and winds stay within their scene cluster; paired methods use the same draw. The intervals are exploratory, unadjusted 95%, conditional on three fixed head seeds and reused development data.

Table S60: Lag readout sensitivity on the same 106 common-valid development states. RMSE is in metres; centered MSE and regrets are in squared metres. Means retain all campaigns, encoder seeds and selected head seeds. These are the unrounded-source counterparts of the main-paper table.

Readout	RMSE	Centered MSE	Held	Nominal	Observer
Scalar physics	0.305595	0.004078	0.016285	0.002421	0.002903
Affine	0.265143	0.002136	0.028945	0.006121	0.008892
Additive MLP	0.298978	0.002460	0.028273	0.005721	0.008223
Joint MLP	0.287426	0.002446	0.028550	0.005838	0.008274

Table S61: All five calibration campaigns, with encoder and head-seed metrics averaged within each campaign. Campaign identifiers are zero-based. The scalar has no fictitious encoder or head-seed replication.

Campaign	Readout	RMSE	Centered MSE	Held	Nominal	Observer
0	Scalar physics	0.246007	0.009489	0.012563	0.002421	0.003685
1	Scalar physics	0.332236	0.002227	0.018042	0.002494	0.002677
2	Scalar physics	0.332236	0.002227	0.018042	0.002494	0.002677
3	Scalar physics	0.285258	0.004221	0.014738	0.002198	0.002798
4	Scalar physics	0.332236	0.002227	0.018042	0.002494	0.002677
0	Affine	0.281734	0.002238	0.029342	0.006622	0.009686
1	Affine	0.254675	0.002113	0.034138	0.007337	0.010214
2	Affine	0.260674	0.002242	0.031226	0.005420	0.007867
3	Affine	0.267067	0.002087	0.023315	0.005784	0.008807
4	Affine	0.261565	0.001999	0.026704	0.005443	0.007886
0	Additive MLP	0.308003	0.003037	0.022104	0.004936	0.007546
1	Additive MLP	0.338424	0.002419	0.036335	0.007095	0.009883
2	Additive MLP	0.327934	0.002216	0.038681	0.006714	0.008710
3	Additive MLP	0.268955	0.002503	0.023553	0.005931	0.008984
4	Additive MLP	0.251574	0.002125	0.020691	0.003929	0.005993
0	Joint MLP	0.280535	0.002533	0.024620	0.005047	0.007327
1	Joint MLP	0.326418	0.002917	0.035393	0.007285	0.010118
2	Joint MLP	0.314970	0.002309	0.040671	0.007235	0.009437
3	Joint MLP	0.276036	0.002350	0.020802	0.005296	0.008078
4	Joint MLP	0.239171	0.002119	0.021263	0.004326	0.006412

S14.3 Results and limits

Joint-MLP pooled forecast RMSE is 0.2874m versus scalar physics' 0.3056m; their exploratory difference interval $[-0.0894, 0.0420]$ includes zero. Joint-minus-scalar held regret is 0.01226m^2 , with interval $[0.00450, 0.02091]$, and its regret is higher in all five campaigns. The disadvantage persists under nominal and observer continuation. Joint-minus-affine regret intervals include zero in all three modes; centered MSE increases from 0.002136 to 0.002446m^2 . This procedure therefore extends the observed disadvantage beyond the fixed affine readout without establishing a sole cause, universal failure of nonlinear models, or a complete-flight gain.

Table S62: Complete 20-contrast family. Differences are left minus right, and lower values are favorable for every metric. Intervals are unadjusted 95% paired bootstrap intervals on reused development data. They are not an independent confirmation family.

Metric	Contrast	Difference	Interval
Forecast RMSE	joint – affine	+0.022283	$[-0.018937, 0.064287]$
Forecast RMSE	additive – affine	+0.033835	$[-0.019796, 0.085792]$
Forecast RMSE	joint – additive	-0.011552	$[-0.032817, 0.005441]$
Forecast RMSE	joint – scalar	-0.018169	$[-0.089366, 0.041968]$
Centered MSE	joint – affine	+0.000310	$[0.000051, 0.000699]$
Centered MSE	additive – affine	+0.000324	$[0.000060, 0.000617]$
Centered MSE	joint – additive	-0.000014	$[-0.000371, 0.000388]$
Centered MSE	joint – scalar	-0.001632	$[-0.004472, 0.000328]$

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Metric	Contrast	Difference	Interval
Held regret	joint – affine	-0.000395	[−0.006673, 0.006972]
Held regret	additive – affine	-0.000672	[−0.006874, 0.006385]
Held regret	joint – additive	+0.000277	[−0.003444, 0.004750]
Held regret	joint – scalar	+0.012264	[0.004498, 0.020908]
Nominal regret	joint – affine	-0.000283	[−0.001497, 0.001146]
Nominal regret	additive – affine	-0.000400	[−0.001575, 0.000886]
Nominal regret	joint – additive	+0.000117	[−0.000665, 0.001012]
Nominal regret	joint – scalar	+0.003418	[0.001636, 0.005453]
Observer regret	joint – affine	-0.000617	[−0.002019, 0.000959]
Observer regret	additive – affine	-0.000669	[−0.001969, 0.000640]
Observer regret	joint – additive	+0.000051	[−0.000866, 0.001110]
Observer regret	joint – scalar	+0.005372	[0.002876, 0.008348]

S14.4 Numerical acceptance and complete source record

Independent acceptance rescored all 1,050 neural candidates on FIT/VAL, verified all 150 selected forecasts and the 180-configuration matrix, and reconstructed raw costs, supports, model selection, stage order, summary means and all 20 intervals. It passed 31,628 checks while verifying the 446 earlier source files were unchanged. These checks concern implementation and recorded data, not independent aerodynamic validation.

The self-contained typesetting source includes `nonlinear_readout/summary.json`, retaining 720 configuration/tier rows, 80 campaign/tier rows, 16 family/tier means and all 20 contrasts. It also contains the written protocol, exact main-table source receipt and independent acceptance receipts. `SOURCE_COVERAGE.json` records original repository-relative paths and SHA-256 hashes, distinguishes unchanged S1–S13 material from this addition, and identifies the main-paper version used. Large original calibration caches, checkpoints and per-candidate prediction arrays are indexed by the research evidence records; they are not required to compile this supplement and are not all contained in its typesetting source.